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Preface

These are the notes I took for a second semester analysis course.

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Chapter 1

Vector Spaces, Algebras, and Normed Spaces

For the entirety of this chapter assume F to be $\mathbb{R}$ or $\mathbb{C}$.

§ 1.1. Vector Spaces

Definition 1.1.1. A *vector space* (or *linear space*) over F is a nonempty set V equipped with two operations:

$$\begin{aligned} V \times V &\xrightarrow{+} V \text{ defined by } (v, w) \mapsto v + w \\ F \times V &\rightarrow V \text{ defined by } (\alpha, v) \mapsto \alpha v \end{aligned}$$

satisfying:

- (1) $(V, +)$ is an abelian group:
 - (i) $u + (v + w) = (u + v) + w$ for all $u, v, w \in V$;
 - (ii) there exists 0_V such that $v + 0_V = 0_V + v = v$ for all $v \in V$;
 - (iii) for all $v \in V$, there exists $w \in V$ satisfying $v + w = w + v = 0_V$;
 - (iv) $v + w = w + v$ for all $v, w \in V$;
- (2) $(\alpha + \beta)v = \alpha v + \beta v$ for all $\alpha, \beta \in F, v \in V$;
- (3) $\alpha(\beta v) = (\alpha\beta)v$ for all $\alpha, \beta \in F, v \in V$;
- (4) $\alpha(v + w) = \alpha v + \alpha w$ for all $\alpha \in F, v, w \in V$;
- (5) $1_F v = v$ for all $v \in V$.

It can be shown that the vector 0_V is unique, the additive inverse in (iii) is unique (which we denote as $-v$), that $0v = 0_V$, and $(-1)v = -v$.

Exercise 1.1.1. Show (iv) follows from the other axioms.

Exercise 1.1.2. Show $nv = \underbrace{v + v + \dots + v}_{n \text{ times}}$ for $n \in \mathbb{Z}_{\geq 1}$.

It can be shown that a subspace is a vector space in its own right.

Example 1.1.1. Let $\{W_i\}_{i \in I}$ be a family of vector spaces. Then $\bigcap_{i \in I} W_i$ is also a vector space.

Example 1.1.2. Planes and lines through the origin are subspaces of $\mathbb{R}^3$.

Definition 1.1.2. Let V be a vector space and $S \subseteq V$ a subset.

- (1) A *linear combination* from S is a finite sum $\sum_{j=1}^n \alpha_j v_j$ with $\alpha_j \in F, v_j \in S$.
- (2) The *linear span* of S is:

$$\text{span}(S) := \left\{ \sum_{j=1}^n \alpha_j v_j \mid n \in \mathbb{N}, \alpha_j \in F, v_j \in S \right\}.$$

Exercise 1.1.3. Show that $\text{span}(S) \subseteq V$ is a subspace and:

$$\text{span}(S) = \bigcap \{W \mid S \subseteq W, W \text{ is a subspace}\},$$

that is, $\text{span}(S)$ is the smallest subspace of V containing S .

Definition 1.1.3. Let V be a vector space and $S \subseteq V$ a subset.

- (1) S is *spanning* for V if $\text{span}(S) = V$.
- (2) S is *independent* if, given $n \in \mathbb{N}, \alpha_1, \dots, \alpha_n \in F, v_1, \dots, v_n \in S$, then $\sum_{j=1}^n \alpha_j v_j = 0$ implies $\alpha_j = 0$ for all j .

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Our goal is to show that every vector space admits a basis. As such, recall the following definitions from a standard course in Real Analysis.

Definition 1.1.4. An *ordering* on a set X is a relation $R \subseteq X \times X$ on X that is reflexive, transitive, and antisymmetric. We write xRy as $x \leq_R y$. The pair $(X, \leq_R)$ is called an *ordered set*. An ordering $\leq$ on X is called *total* (or *linear*) if for all $x, y \in X$, $x \leq y$ or $y \leq x$.

Note that if $(X, \leq)$ is an ordered set and $Y \subseteq X$ is a subset, then $(Y, \leq)$ is an ordered set as well.

Definition 1.1.5. Let $(X, \leq)$ be an ordered set and $Y \subseteq X$. An *upper bound* for Y is an element $u \in X$ with $u \geq y$ for all $y \in Y$. An element $m \in X$ is called *maximal* if $x \in X, x \geq m$ implies $x = m$.

Lemma 1.1.1 (Zorn's Lemma). *Let $(X, \leq_X)$ be an ordered set. Suppose every subset $Y \subseteq X$ for which $(Y, \leq_X)$ is totally ordered has an upper bound in X . Then X admits a maximal element.*

The proof of Zorn's Lemma is outside the interest of this text.

Theorem 1.1.2. *Every vector space admits a basis. Moreover, every independent set is contained in a basis.*

Proof. Let $S \subseteq V$ be linearly independent. Define:

$$\mathfrak{T}(S) = \{T \subseteq V \mid S \subseteq T, T \text{ linearly independent}\}.$$

Let $\mathfrak{C} \subseteq \mathfrak{T}(S)$ be a totally ordered subset. Set $R = \bigcup_{T \in \mathfrak{C}} T$. Clearly $R \supseteq S$. Assume $\sum_{j=1}^n \alpha_j v_j = 0$, where $\alpha_j \in F$ and $v_j \in R$. Since $\mathfrak{C}$ is totally ordered, there exists $T_0 \in \mathfrak{C}$ with $v_j \in T_0$ for all $j = 1, \dots, n$. Since T_0 is independent, $\alpha_j = 0$ for all $j = 1, \dots, n$. Thus R is independent as well. Whence R is an upper bound for $\mathfrak{C}$. By Zorn's Lemma, $\mathfrak{T}(S)$ admits a maximal element, call it B .

Claim: B is a basis for V . Suppose towards contradiction it's not, then there exists $v_0 \in V \setminus \text{span}(B)$. Consider $B \cup \{v_0\}$ and let $\alpha_0 v_0 + \sum_{j=1}^n \alpha_j v_j = 0_V$. If $\alpha_0 \neq 0$, then $\sum_{j=1}^n \alpha_j v_j = -\alpha_0 v_0$, giving $v_0 \in \text{span}(B)$ which is a contradiction. If $\alpha_0 = 0$, then $\sum_{j=1}^n \alpha_j v_j = 0_V$. Since B is independent, $\alpha_j = 0$ for all $j = 1, \dots, n$. Thus $B \cup \{v_0\}$ is independent, contradicting the maximality of B . Whence B is a basis for V . $\square$

Theorem 1.1.3. *If B_1 and B_2 are bases for V , then $\text{card}(B_1) = \text{card}(B_2)$.*

Definition 1.1.6. If V is a vector space, its *dimension* is the cardinality of any of its bases.

Corollary 1.1.4. *If B is a basis for V , then every $v \in V$ can be written $v = \sum_{j=1}^n \alpha_j \beta_j$, $\alpha_j \in F$, $\beta_j \in B$ in a unique way.*

Theorem 1.1.5. *Let V be a linear space and $B \subseteq V$ a subset. The following are equivalent:*

- (1) B is a basis for V ;
- (2) B is a maximal element in $\mathfrak{T} = \{T \subseteq V \mid T \text{ independent}\}$;
- (3) B is a minimal element in $\mathfrak{S} = \{S \subseteq V \mid S \text{ spans } V\}$;

Definition 1.1.7. Let $\{V_i\}_{i \in I}$ be a family of vector spaces over a field F .

- (1) The *product* of $\{V_i\}_{i \in I}$ is denoted:

$$\prod_{i \in I} V_i := \{(v_i)_{i \in I} \mid v_i \in V_i\}.$$

- (2) The *co-product* (or *sum*) is denoted

$$\bigoplus_{i \in I} V_i := \{(v_i)_{i \in I} \mid v_i \in V_i, \text{supp}((v_i)_{i \in I}) < \infty\}.$$

Exercise 1.1.4.

- (1) Show that $\prod_{i \in I} V_i$ equipped with pointwise operations:

$$\begin{aligned} (v_i)_{i \in I} + (w_i)_{i \in I} &= (v_i + w_i)_{i \in I} \\ \alpha(v_i)_{i \in I} &= (\alpha v_i)_{i \in I} \end{aligned}$$

is a linear space.

- (2) Show that $\bigoplus_{i \in I} V_i$ is a subspace of $\prod_{i \in I} V_i$.

Proposition 1.1.6. *Let V be a vector space over F and $W \subseteq V$. The (additive, abelian) quotient group V/W can be made into a vector space by defining multiplication by scalars as $\alpha(v + W) = \alpha v + W$ for all $\alpha \in F, v + W \in V/W$.*

Example 1.1.3.

- (1) The set $F^n = \{(x_1, \dots, x_n) \mid x_j \in F\}$ with component-wise operations is a vector space.
- (2) The set $M_{n,m}(F) = \{(a_{ij}) \mid a_{ij} \in F\}$ with linear operations is a vector space.
- (3) Let Ω be a nonempty set. Then $\mathcal{F}(\Omega, F) = \{f \mid f : \Omega \rightarrow F\}$ with pointwise operations is a vector space.
- (4) The set $\ell_\infty(\Omega, F) = \{f \in \mathcal{F}(\Omega, F) \mid \|f\|_\infty < \infty\}$ with pointwise operations is a vector space.

Exercise 1.1.5. Show $\ell_\infty(\Omega, F) \subseteq \mathcal{F}(\Omega, F)$ is a subspace.

- (5) Let $f : [a, b] \rightarrow \mathbb{R}$ be any function. Let $\mathcal{P} = \{a = x_0 < x_1 < \dots < x_{n-1} < x_n = b\}$ be a partition of $[a, b]$. The *variation of f on $\mathcal{P}$* is defined as:

$$\text{Var}(f; \mathcal{P}) = \sum_{k=1}^n |f(x_k) - f(x_{k-1})|.$$

We say f is a *bounded variation* if:

$$\text{Var}(f) := \sup_{\mathcal{P}} \text{Var}(f; \mathcal{P}) < \infty.$$

The set of all functions of bounded variation is defined:

$$\text{BV}([a, b]) = \{f : [a, b] \rightarrow \mathbb{R} \mid \text{Var}(f) < \infty\}.$$

This is a vector space by defining addition and scalar multiplication componentwise.

Exercise 1.1.6. Show that $\text{BV}([a, b]) \subseteq \ell_\infty([a, b], \mathbb{R})$ is a subspace.

- (6) Let $K \subseteq V$ be a convex subset of a vector space V , that is, for all $v, w \in K$ and $t \in [0, 1]$, then $(1 - t)v + tw \in K$. A function $f : K \rightarrow F$ is said to be *affine* if $x, y \in K$ and $t \in [0, 1]$ implies $f((1 - t)x + ty) = (1 - t)f(x) + tf(y)$. The set $\text{Aff}(K, F) = \{f \in \mathcal{F}(K, F) \mid f \text{ affine}\}$ with pointwise operations is a vector space.

Exercise 1.1.7. Show $\text{Aff}(\Omega, F) \subseteq \mathcal{F}(\Omega, F)$ is a subspace.

- (7) The set $C([a, b], F) = \{f : [a, b] \rightarrow F \mid f \text{ continuous}\}$ with pointwise operations is a vector space.

Exercise 1.1.8. Explain why $C([a, b], F) \subseteq \ell_\infty([a, b], F)$ is a subspace.

- (8) Consider the following sequence spaces:

- $s = \{(a_k)_k \mid a_k \in F\} = \mathcal{F}(\mathbb{N}, F)$;
- $\ell_\infty = \ell_\infty(\mathbb{N}, F) = \{(a_k)_k \mid \sup_{k \geq 1} |a_k| < \infty\}$;
- $c = \{(a_k)_k \mid (a_k)_k \text{ converges}\}$;
- $c_0 = \{(a_k)_k \mid (a_k)_k \rightarrow 0\}$;
- $c_{00} = \{(a_k)_k \mid \text{supp}((a_k)_k) < \infty\}$;
- $\ell_1 = \{(a_k)_k \mid \sum_{k=1}^{\infty} |a_k| < \infty\}$.

These are all vector spaces with pointwise operations. In fact, $c_{00} \subseteq c_0 \subseteq c \subseteq \ell_\infty \subseteq s$ are all subspaces.

Exercise 1.1.9. Show that $\ell_1 \subseteq c_0$ is a subspace.

(9) Consider the following continuous function spaces on $\mathbb{R}$:

- $C(\mathbb{R}) = \{f : \mathbb{R} \rightarrow F \mid f \text{ continuous}\}$;
- $C_b(\mathbb{R}) = C(\mathbb{R}) \cap \ell_\infty(\mathbb{R})$;
- $C_0(\mathbb{R}) = \{f \in C(\mathbb{R}) \mid \lim_{x \rightarrow \pm\infty} f(x) = 0\}$;
- Recall that a function is *compactly supported* if for all $\epsilon > 0$, there exists $\alpha > 0$ such that $|x| \geq \alpha$ implies $f(x) = 0$. The set of compactly supported functions is denoted $C_c(\mathbb{R}) = \{f \in C(\mathbb{R}) \mid f \text{ compactly supported}\}$.

These are all vector spaces with pointwise operations, and $C_c(\mathbb{R}) \subseteq C_0(\mathbb{R}) \subseteq C_b(\mathbb{R}) \subseteq C(\mathbb{R})$ are all subspace inclusion.

Definition 1.1.8. If V and W are linear spaces over a common field F , a map $T : V \rightarrow W$ is called *linear* if $T(v_1 + \alpha v_2) = T(v_1) + \alpha T(v_2)$ for all $v_1, v_2 \in V$ and $\alpha \in F$.

Example 1.1.4. Let $A \in M_{m,n}(F)$. Then $T_A : F^n \rightarrow F^m$ defined by $T_A(v) = Av$ is linear. Let $\{e_1, \dots, e_n\}$ be a basis for F^n . If $T : F^n \rightarrow F^m$ is linear, set:

$$[T] = \left(T(e_1) \mid T(e_2) \mid \dots \mid T(e_n) \right).$$

This gives $T(v) = [T]v$ for all $v \in F^n$. In fact, we also have $[T_A] = A$ and $T_{[T]} = T$.

Example 1.1.5. The *canonical projection* is linear:

$$\pi_j : \prod_{i \in I} V_i \rightarrow V_j \text{ defined by } \pi_j((v_i)_i) = v_j.$$

We also have that the *coordinate exclusions* are linear:

$$\iota_j : V_j \hookrightarrow \bigoplus_{i \in I} V_i \text{ defined by } \iota_j(v) = (v_i)_i, \text{ where } v_i = \begin{cases} 0_v, & i \neq j \\ v_j, & \text{otherwise.} \end{cases}$$

The *evaluation map* is linear as well. For $s \in S$, consider:

$$e_s : \mathcal{F}(S, F) \rightarrow F \text{ defined by } e_s(f) = f(s).$$

Proposition 1.1.7. *Let V be a vector space with basis B . Let W be a vector space and suppose $\varphi : B \rightarrow W$ is a map. Then there exists a unique linear map $T_\varphi : V \rightarrow W$ with $T_\varphi(b) = \varphi(b)$ for all $b \in B$. We have the following diagram.*

$$\begin{array}{ccc} B & \xhookrightarrow{\iota} & V \\ & \searrow \varphi & \downarrow \text{---} T_\varphi \\ & & W \end{array}$$

Proof. Define $T_\varphi : V \rightarrow W$ by:

$$\begin{aligned} T_\varphi(v) &= T_\varphi\left(\sum_{j=1}^n \alpha_j b_j\right) \\ &= \sum_{j=1}^n \alpha_j \varphi(b_j). \end{aligned}$$

Let $v_1, v_2 \in V$ and $c \in F$. We have that:

$$\begin{aligned} T_\varphi(v_1 + cv_2) &= T_\varphi\left(\sum_{j=1}^n \alpha_j b_j + c \sum_{j=1}^n \beta_j b_j\right) \\ &= T_\varphi\left(\sum_{j=1}^n (\alpha_j + c\beta_j) b_j\right) \\ &= \sum_{j=1}^n (\alpha_j + c\beta_j) \varphi(b_j) \\ &= \sum_{j=1}^n \alpha_j \varphi(b_j) + c \sum_{j=1}^n \beta_j \varphi(b_j) \\ &= T_\varphi(v_1) + cT_\varphi(v_2). \end{aligned}$$

Thus T_φ is linear. Chasing the above diagram makes it clear that $T_\varphi(b) = \varphi(b)$. It remains to show that T_φ is unique. Let T be another linear transformation satisfying $T(b) = \varphi(b)$ for all $b \in B$. Then:

$$\begin{aligned} T(v) &= T\left(\sum_{j=1}^n \alpha_j b_j\right) \\ &= \sum_{j=1}^n \alpha_j \varphi(b_j) \\ &= T_\varphi\left(\sum_{j=1}^n \alpha_j b_j\right) \\ &= T_\varphi(v). \end{aligned}$$

Thus T_φ is unique. □

Proposition 1.1.8. *Let $T : V \rightarrow W$ be linear.*

- (1) $\ker(T) = \{v \in V \mid T(v) = 0_W\}$ is a linear subspace of V .
- (2) $\text{im}(T) = \{T(v) \mid v \in V\}$ is a linear subspace of W .
- (3) $\ker(T) = \{0_V\}$ if and only if T is injective.
- (4) $\text{im}(T) = W$ if and only if T is surjective.

Proof. (1) Let $v_1, v_2 \in \ker(T)$ and $\alpha \in F$. Observe that:

$$\begin{aligned} T(v_1 + cv_2) &= T(v_1) + cT(v_2) \\ &= 0. \end{aligned}$$

Thus $v_1 + cv_2 \in \ker(T)$, giving $\ker(T)$ as a linear subspace of V .

(2) Let $w_1, w_2 \in \text{im}(T)$. Then there exists $v_1, v_2 \in V$ with $T(v_1) = w_1$ and $T(v_2) = w_2$. We have:

$$\begin{aligned} w_1 + cw_2 &= T(v_1) + cT(v_2) \\ &= T(v_1 + cv_2). \end{aligned}$$

Whence $w_1 + cw_2 \in \text{im}(T)$, giving $\text{im}(T)$ as a linear subspace of W .

(3) Let $\ker(T) = \{0\}$. Suppose $T(v_1) = T(v_2)$. Then $T(v_1) - T(v_2) = T(v_1 - v_2) = 0_W$. It must be that $v_1 - v_2 = 0_W$, giving $v_1 = v_2$. Thus T is injective. Conversely, suppose T is injective and let $v \in \ker(T)$. Then $T(v) = 0_W = T(0_V)$. Hence $v = 0_V$, establishing $\ker(T) = \{0\}$.

(4) This is by definition of surjectivity. $\square$

Proposition 1.1.9. *If $T : V \rightarrow W$ is linear and bijective, then the inverse map $T^{-1} : W \rightarrow V$ is linear.*

Proof. We have that:

$$T(T^{-1}(w_1) + \alpha T^{-1}(w_2)) = w_1 + \alpha w_2 = T \circ T^{-1}(w_1 + \alpha w_2).$$

Applying T^{-1} to both sides gives the desired result. $\square$

Proposition 1.1.10 (Vector Spaces are Injective). *Let U, V, W be vector spaces and $0 \rightarrow U \xrightarrow{j} V$ be exact (that is, j is injective). Let $\varphi : U \rightarrow W$ be linear. There exists a linear map $\Psi : V \rightarrow W$ such that $\varphi = \Psi \circ j$; i.e., the following diagram commutes:*

$$\begin{array}{ccccc} 0 & \longrightarrow & U & \xrightarrow{j} & V \\ & & \downarrow \varphi & \searrow \Psi & \\ & & W & & \end{array}$$

Proof. Let $\{u_i\}_{i \in I}$ be a basis for U . We must first show that $\{j(u_i)\}_{i \in I}$ is linearly independent. Notice that:

$$\begin{aligned} 0_V &= \sum_{i \in I} \alpha_i j(u_i) \\ &= j \left(\sum_{i \in I} \alpha_i u_i \right). \end{aligned}$$

By the injectivity of j , we have that $\sum_{i \in I} \alpha_i u_i = 0_U$. Thus $\alpha_i = 0$ for all $i \in I$, giving $\{j(u_i)\}_{i \in I}$ as linearly independent.

Since $\{j(u_i)\}_{i \in I}$ is linearly independent in V , we can extend it to a basis $B = \{v_i\}_{i \in J}$ where $I \subseteq J$ and $v_i = j(u_i)$ whenever $i \in I$. Now define $\psi : B \rightarrow W$ by:

$$\psi(v_i) = \begin{cases} \varphi(u_i), & i \in I \\ w, & i \in J \setminus I, \end{cases}$$

where $w \in W$ is arbitrary. Since this is a map of basis elements, there exists a unique linear map $\Psi : V \rightarrow W$ with $\Psi(v_i) = \psi(v_i)$ for all $v_i \in B$. We can finally see that:

$$\begin{aligned} \varphi(u_i) &= \psi(v_i) \\ &= \Psi(v_i) \\ &= \Psi(j(u_i)). \end{aligned}$$

This establishes that $\varphi = \Psi \circ j$. □

Proposition 1.1.11 (Vector Spaces are Projective). *Let U, V, W be vector spaces and $V \xrightarrow{\pi} U \rightarrow 0$ be exact (that is, π is onto). Let $\varphi : W \rightarrow U$ be linear. There exists a linear map $\Psi : V \rightarrow W$ such that $\varphi = \pi \circ \Psi$; i.e., the following diagram commutes:*

$$\begin{array}{ccc} & W & \\ \Psi \swarrow & \downarrow \varphi & \\ V & \xrightarrow{\pi} & U \longrightarrow 0 \end{array}$$

Proof. Let $B = \{w_i\}_{i \in I}$ be a basis for W . Define $\psi : B \rightarrow V$ by $\psi(w_i) = \pi^{-1}(\varphi(w_i))$. Since this is a map of basis elements, it extends to a unique (dependent on π^{-1}) linear map $\Psi : W \rightarrow V$ with $\Psi(w_i) = \psi(w_i)$ for all $w_i \in B$. Moreover, we have that:

$$\begin{aligned} (\pi \circ \Psi)(w_i) &= (\pi \circ \psi)(w_i) \\ &= (\pi \circ (\pi^{-1} \circ \varphi))(w_i) \\ &= \varphi(w_i). \end{aligned}$$
□

Definition 1.1.9. Let V and W be vector spaces over F . A *linear isomorphism* between V and W is a bijective linear map $T : V \rightarrow W$. If such a T exists, we say V and W are *linearly isomorphic*, and write $V \cong W$.

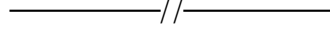
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Finite dimensional vector spaces are boring. This is illustrated through the following theorem.

Theorem 1.1.12. *Let V and W be finite-dimensional vector spaces over F . Then $V \cong W$ if and only if $\dim(V) = \dim(W)$.*

Proof. Suppose $V \cong W$. Then there is an isomorphism taking basis of V to a basis of W . Therefore they have the same dimension.

Conversely, if $\dim(V) = \dim(W) = n$, then they are each isomorphic to F^n , giving that they are isomorphic to each other. □



Example 1.1.6. Let V be a vector space, $W \subseteq V$ a subspace. The *natural projection*:

$$\pi : V \rightarrow V/W \text{ defined by } \pi(v) = v + W$$

is a linear surjective map.

Theorem 1.1.13 (First Isomorphism Theorem for Vector Spaces). *Let $T : V \rightarrow V'$ be a linear map and $W \subseteq V$ a subspace.*

- (1) *If T "kills" W (that is, $W \subseteq \ker(T)$), then there exists a linear map $\tilde{T} : V/W \rightarrow V'$ with $\tilde{T} \circ \pi = T$; i.e., the following diagram commutes.*

$$\begin{array}{ccc} V & \xrightarrow{T} & V' \\ & \searrow \pi & \uparrow \tilde{T} \\ & & V/W \end{array}$$

- (2) *If $\ker(T) = W$, then $\tilde{T}$ is injective.*
 (3) *If $\ker(T) = W$ and $\text{im}(T) = V'$, then $V/W \cong V'$.*

Proof. (1) As stipulated, define $\tilde{T}(v + W) = T(v)$. We must show that $\tilde{T}$ is well-defined: suppose $v_1 + W = v_2 + W$ for some $v_1, v_2 \in V$. Then $v_1 = v_2 + w$ for some $w \in W$. This gives:

$$\begin{aligned} \tilde{T}(v_1 + W) &= \tilde{T}(v_2 + w + W) \\ &= \tilde{T}(v_2 + W). \end{aligned}$$

Whence $\tilde{T}$ is well-defined. Now given $v_1 + W, v_2 + W \in V/W$ and $\alpha \in F$, observe that:

$$\begin{aligned} \tilde{T}((v_1 + W) + \alpha(v_2 + W)) &= \tilde{T}((v_1 + \alpha v_2) + W) \\ &= T(v_1 + \alpha v_2) \\ &= T(v_1) + \alpha T(v_2) \\ &= \tilde{T}(v_1 + W) + \alpha \tilde{T}(v_2 + W). \end{aligned}$$

Thus $\tilde{T}$ is linear.

- (2) If $\ker(T) = W$, then:

$$\begin{aligned} \ker(\tilde{T}) &= \{v + W \mid \tilde{T}(v + W) = 0_{V'}\} \\ &= \{v + W \mid T(v) = 0_{V'}\} \\ &= \{v + W \mid v \in \ker(T)\} \\ &= \{v + W \mid v \in W\} \\ &= \{0\}. \end{aligned}$$

Thus $\tilde{T}$ is injective.

(3) It remains to show that $\text{im}(T) = V'$ implies $\tilde{T}$ is surjective. Observe that:

$$\begin{aligned} \text{im}(\tilde{T}) &= \{\tilde{T}(v + W) \mid v + W \in V/W\} \\ &= \{\tilde{T}(\pi(v)) \mid v \in V\} \\ &= \{T(v) \mid v \in V\} \\ &= \text{im}(T) \\ &= V'. \end{aligned}$$

Thus $\tilde{T}$ is surjective, which establishes it as a bijection. This gives $V/W \cong V'$. $\square$

Definition 1.1.10. Let S be a nonempty set. The *free vector space* of S is:

$$\mathbb{F}(S) = \{f : S \rightarrow F \mid \text{supp}(f) < \infty\}.$$

Exercise 1.1.10. Show $\mathbb{F}(S) \subseteq \mathcal{F}(S, F)$ is a subspace.

Proposition 1.1.14. The set $\{\delta_s \mid s \in S\}$ is a basis for $\mathbb{F}(S)$, where $\delta_s : S \rightarrow F$ is defined by:

$$\delta_s(t) = \begin{cases} 1, & t = s \\ 0, & \text{otherwise.} \end{cases}$$

Proof. If $f \in \mathbb{F}(S)$ with $\text{supp}(f) = \{s_1, \dots, s_n\}$, then $f = \sum_{k=1}^n f(s_k)\delta_{s_k}$. If $\sum_{k=1}^n \alpha_k \delta_{s_k} = 0$, then for $j = 1, \dots, n$ we have $0 = (\sum_{k=1}^n \alpha_k \delta_{s_k})(s_j) = \alpha_j$. $\square$

Theorem 1.1.15. Given any vector space V and a map (of sets) $\varphi : S \rightarrow V$, there exists a unique linear map $T_\varphi : \mathbb{F}(S) \rightarrow V$ with $T_\varphi \circ \iota = \varphi$, where $\iota : S \rightarrow \mathbb{F}(S)$ is defined by $\iota(s) = \delta_s$ for all $s \in S$. The following diagram commutes:

$$\begin{array}{ccc} S & \xrightarrow{\iota} & \mathbb{F}(S) \\ & \searrow \varphi & \downarrow T_\varphi \\ & & V \end{array}$$

Proof. By the previous proposition, we have that $B = \{\delta_s \mid s \in S\}$ is a basis for $\mathbb{F}(S)$. Define $T : B \rightarrow V$ by $T(\delta_s) = \varphi(s)$. Since this is a map of basis elements, there exists a unique linear map $T_\varphi : \mathbb{F}(S) \rightarrow V$ with $T_\varphi(\delta_s) = T(\delta_s)$ for all $\delta_s \in B$. The diagram commutes because:

$$\begin{aligned} \varphi(s) &= T(\delta_s) \\ &= T_\varphi(\delta_s) \\ &= T_\varphi(\iota(s)). \end{aligned}$$

Moreover, if T' satisfies $\varphi = T' \circ \iota$, then:

$$\begin{aligned} T'(\delta_s) &= T'(\iota(s)) \\ &= \varphi(s) \\ &= T_\varphi(\iota(s)) \\ &= T_\varphi(\delta_s). \end{aligned}$$

Thus T_φ is unique. $\square$

Definition 1.1.11. Let V and W be vector spaces. The set of linear transformations between V and W is $\mathcal{L}(V, W) = \{T \mid T : V \rightarrow W \text{ linear}\}$. The set of linear functionals is $V' := \mathcal{L}(V, F)$.

Exercise 1.1.11. Show $\mathcal{L}(V, W)$ is a vector space.

Exercise 1.1.12. Show $M_{m,n}(F) \cong \mathcal{L}(F^m, F^n)$ by $a \mapsto T_a : (v \mapsto av)$.

§ 1.2. Algebras

Definition 1.2.1. An *algebra* over F is a linear space A over F equipped with a multiplication operation:

$$A \times A \rightarrow A \text{ defined by } (a, b) \mapsto ab$$

satisfying:

- (1) $(ab)c = a(bc)$ for all $a, b, c \in A$;
- (2) $(\alpha a)b = \alpha(ab) = a(\alpha b)$ for all $a, b \in A, \alpha \in F$;
- (3) $a(b + c) = ab + ac$ for all $a, b, c \in A$;
- (4) $(a + b)c = ac + bc$ for all $a, b, c \in A$.

If $ab = ba$ for all $a, b \in A$ we say that A is *commutative*. If there exists $1_A \in A$ with $1_A a = a 1_A = a$ for all $a \in A$ we say A is *unital*.

Example 1.2.1.

- (1) $M_n(F)$ is a noncommutative unital algebra over F under the usual matrix multiplication.
- (2) If V is a vector space over F , $\mathcal{L}(V)$ is a unital algebra over F . It is noncommutative provided $\dim(V) > 1$.
- (3) $\mathcal{F}(S, F)$ is a unital commutative algebra over F .

Definition 1.2.2. Let B be a (unital) algebra over F .

- (1) A (unital) *subalgebra* of B is a subspace $A \subseteq B$ ($1_B \in A$) satisfying the property that if $a, a' \in A$, then $aa' \in A$.
- (2) An *ideal* of B is a subspace $I \subseteq B$ with $b \in B, a \in I$ implying $ba, ab \in I$.

Example 1.2.2.

- (1) $\ell_\infty(\Omega, F) \subseteq \mathcal{F}(\Omega, F)$ is a unital subalgebra.
- (2) $c_{00} \subseteq c_0 \subseteq c \subseteq \ell_\infty \subseteq s$ are all subalgebras. In particular, $c_0 \subseteq \ell_\infty$ and $c_{00} \subseteq s$ are ideals.
- (3) $C([a, b]) \subseteq \ell_\infty([a, b])$ is a unital subalgebra.
- (4) $C_c(\mathbb{R}) \subseteq C_0(\mathbb{R}) \subseteq C_b(\mathbb{R}) \subseteq \ell_\infty(\mathbb{R})$ are all subalgebras. In fact, $C_b(\mathbb{R}) \subseteq C(\mathbb{R})$ and $C_b(\mathbb{R}) \subseteq \ell_\infty(\mathbb{R})$ are unital, whereas $C_0(\mathbb{R}) \subseteq C_b(\mathbb{R})$ and $C_c(\mathbb{R}) \subseteq C(\mathbb{R})$ are ideals.
- (5) The set $T_n(F) = \{(a_{ij}) \in M_n(F) \mid a_{ij} = 0, i > j\}$ is a unital subalgebra of $M_n(F)$.

Example 1.2.3 (Group Algebra). Let Γ denote a group (not necessarily abelian). Take the free vector space $\mathbb{F}(\Gamma)$ and define multiplication as *convolution*: given $f, g \in \mathbb{F}(\Gamma)$ let:

$$(f * g)(r) = \sum_{\substack{\{(s,t) \mid \\ s \in \text{supp}(f), \\ t \in \text{supp}(g), \\ st=r\}}} f(s)g(t).$$

Since $\text{supp}(f)$ and $\text{supp}(g)$ are finite, this is a finite sum. We often suppress this notation and write $(f * g)(r) = \sum_{st=r} f(s)g(t)$.

We can also make substitutions:

$$\begin{aligned} (f * g)(r) &= \sum_{st=r} f(s)g(t) \\ &= \sum_{t \in \Gamma} f(rt^{-1})g(t) \\ &= \sum_{s \in \Gamma} f(s)g(s^{-1}r). \end{aligned}$$

It is clear that:

$$\begin{aligned} (f + g) * h &= f * h + g * h \\ g * (f + h) &= f * g + g * h \\ \alpha(f * g) &= (\alpha f) * g = f * (\alpha g) \end{aligned}$$

for $f, g, h \in \mathbb{F}(\Gamma)$, $\alpha \in \mathbb{F}$. Associativity can be similarly shown using the above definition. Rather, we will prove associativity by first show that $\delta_s * \delta_t = \delta_{st}$. Given:

$$(\delta_s * \delta_t)(r) = \sum_{q \in \Gamma} \delta_s(rq^{-1})\delta_t(q),$$

notice that:

$$\delta_s(rt^{-1}) = \begin{cases} 1, & s = rt^{-1} \\ 0, & \text{otherwise} \end{cases} = \begin{cases} 1, & r = st \\ 0, & \text{otherwise} \end{cases} = \delta_{st}(r).$$

Since $\{\delta_t \mid t \in \Gamma\}$ is a basis for $\mathbb{F}(\Gamma)$, every $f \in \mathbb{F}(\Gamma)$ looks like:

$$f = \sum_{t \in J} \alpha_t \delta_t, \quad J \subseteq \Gamma \text{ finite.}$$

Using distributivity we get:

$$\begin{aligned} \delta_r * (\delta_s * \delta_t) &= \delta_r * \delta_{st} \\ &= \delta_{rst} \\ &= \delta_{rs} * \delta_t \\ &= (\delta_r * \delta_s) * \delta_t. \end{aligned}$$

Whence convolution is associative.

Exercise 1.2.1. Let $\{A_i\}_{i \in I}$ be a family of algebras over F .

- (1) $\prod_{i \in I} A_i$ is an algebra under $(a_i)_i(b_i)_i = (a_i b_i)_i$.
- (2) $\bigoplus_{i \in I} A_i \subseteq \prod_{i \in I} A_i$ is an ideal.

Exercise 1.2.2. Let A be an algebra over F and $I \subseteq A$ an ideal. Then A/I is an algebra under $(a + I)(b + I) = ab + I$.

§ 1.3. Normed Vector Spaces

To each vector v in a vector space V , we want to assign a "length", denoted $\|v\|$.

Definition 1.3.1. A *norm* on a vector space V is a map:

$$\|\cdot\| : V \rightarrow [0, \infty), \quad v \mapsto \|v\|$$

satisfying:

- (1) $\|\alpha v\| = |\alpha| \|v\|$ for all $\alpha \in F, v \in V$ (homogeneity);
- (2) $\|v + w\| \leq \|v\| + \|w\|$ (triangle inequality);
- (3) If $\|v\| = 0$, then $v = 0_V$ (positive-definite).

If $\|\cdot\|$ satisfies (1) and (2), it is called a *seminorm*. The pair $(V, \|\cdot\|)$ is called a *normed space*.

Definition 1.3.2. Two norms $\|\cdot\|$ and $\|\cdot\|'$ on a vector space V are called *equivalent* if there exists $c_1 \geq 0$ and $c_2 \geq 0$ with $\|v\| \leq c_1 \|v\|'$ and $\|v\|' \leq c_2 \|v\|$ for all $v \in V$.

Exercise 1.3.1. If p is a seminorm on V , show that $|p(v) - p(w)| \leq p(v - w)$.

Definition 1.3.3. Let $(V, \|v\|)$ be a normed space.

- (1) The *closed unit ball* is denoted $B_V = \{v \in V \mid \|v\| \leq 1\}$.
- (2) The *open unit ball* is denoted $U_V = \{v \in V \mid \|v\| < 1\}$.
- (3) The *unit sphere* is denoted $S_V = \{v \in V \mid \|v\| = 1\}$.

Example 1.3.1. Let $V = F^n$ and $x = (x_1, \dots, x_n)$. We define:

$$\begin{aligned} \|x\|_1 &= \sum_{j=1}^n |x_j|; \\ \|x\|_\infty &= \max_{j=1}^n |x_j|; \\ \|x\|_2 &= \left(\sum_{j=1}^n |x_j|^2 \right)^{\frac{1}{2}}. \end{aligned}$$

For $p \geq 1$:

$$\|x\|_p = \left(\sum_{j=1}^n |x_j|^p \right)^{\frac{1}{p}}.$$

Exercise 1.3.2. Show that $\|\cdot\|_1$ and $\|\cdot\|_\infty$ are norms

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We aim to show that $\|\cdot\|_p$ is a norm for $p \in [0, \infty]$.

Lemma 1.3.1. Let $p, q \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = 1$. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be defined by $f(t) = \frac{1}{p}t^p - t + \frac{1}{q}$. Then $f(t) \geq 0$ for $t \geq 0$.

Proof. Note that $f'(t) = t^{p-1} - 1$. Since:

$$\begin{aligned} f'(1) &= 0 \\ f'(t) &> 0 \quad \text{for } t > 1 \\ f'(t) &< 0 \quad \text{for } 0 \leq t < 1, \end{aligned}$$

we can see that $f(t) \geq 0$ for all $t \geq 0$. □

Lemma 1.3.2 (Young's Inequality). Let $p, q \in [0, \infty]$ with $\frac{1}{p} + \frac{1}{q} = 1$. If $x, y \geq 0$, then $xy \leq \frac{1}{p}x^p + \frac{1}{q}y^q$.

Proof. By Lemma 1.3.1, $t \leq \frac{1}{p}t^p + \frac{1}{q}$. Multiplying both sides by y^q gives:

$$ty^q \leq \frac{1}{p}t^p y^q + \frac{1}{q}y^q.$$

Let $t = xy^{1-q}$. Then:

$$xy^{1-q}y^q \leq \frac{1}{p}x^p y^{p-pq}y^q + \frac{1}{q}y^q.$$

Since $\frac{1}{p} + \frac{1}{q} = 1$, we have that $p - pq = -q$. Whence:

$$xy \leq \frac{1}{p}x^p + \frac{1}{q}y^q.$$

□

Lemma 1.3.3 (Hölder's Inequality). Let $p, q \in [0, \infty]$ with $\frac{1}{p} + \frac{1}{q} = 1$. Then for $x, y \in F^n$:

$$\left| \sum_{j=1}^n x_j y_j \right| \leq \|x\|_p \|y\|_q.$$

Proof. We proceed by cases.

Case 1: $p = 1$. Then:

$$\begin{aligned} \left| \sum_{j=1}^n x_j y_j \right| &\leq \sum_{j=1}^n |x_j| |y_j| \\ &\leq \sum_{j=1}^n |x_j| \|y\|_\infty \\ &= \|x\|_1 \|y\|_\infty. \end{aligned}$$

Case 2: $p = \infty$. This follows similarly to Case 1.

Case 3: $1 < p < \infty$. Suppose $\|x\|_p = \|y\|_q = 1$. Then:

$$\begin{aligned}
 \left| \sum_{j=1}^n x_j y_j \right| &\leq \sum_{j=1}^n |x_j| |y_j| \\
 &\leq \sum_{j=1}^n \left(\frac{1}{p} |x_j|^p + \frac{1}{q} |y_j|^q \right) \\
 &= \frac{1}{p} \left(\sum_{j=1}^n |x_j|^p \right) + \frac{1}{q} \left(\sum_{j=1}^n |y_j|^q \right) \\
 &= \frac{1}{p} + \frac{1}{q} \\
 &= 1.
 \end{aligned}$$

Whence the inequality holds. Now suppose $\|x\|_p = 0$ or $\|y\|_q = 0$. Then $x = 0_{F^n}$ or $y = 0_{F^n}$, whence the inequality holds. Suppose $\|x\|_p \neq 0$ and $\|y\|_q \neq 0$. Set:

$$\begin{aligned}
 x' &= \frac{x}{\|x\|_p} \\
 y' &= \frac{y}{\|y\|_q}.
 \end{aligned}$$

Then $\|x'\|_p = 1 = \|y'\|_q$. Observe that:

$$\begin{aligned}
 1 &\geq \left| \sum_{j=1}^n x'_j y'_j \right| \\
 &= \left| \sum_{j=1}^n \frac{x_j}{\|x\|_p} \frac{y_j}{\|y\|_q} \right|.
 \end{aligned}$$

Multiplying both sides by $\|x\|_p \|y\|_q$ gives the desired result. $\square$

Lemma 1.3.4 (Minkowski's Inequality). *Let $p, q \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = 1$. For $x, y \in F^n$:*

$$\|x + y\|_p \leq \|x\|_p + \|y\|_p.$$

Proof. The only nontrivial case is for $1 < p < \infty$. Observe that:

$$\begin{aligned}
(\|x + y\|_p)^p &= \sum_{j=1}^n |x_j + y_j|^p \\
&= \sum_{j=1}^n |x_j + y_j| |x_j + y_j|^{p-1} \\
&\leq \sum_{j=1}^n |x_j| |x_j + y_j|^{p-1} + |y_j| |x_j + y_j|^{p-1} \\
&\leq \left(\sum_{j=1}^n |x_j|^p \right)^{\frac{1}{p}} \left(\sum_{j=1}^n |x_j + y_j|^{(p-1)q} \right)^{\frac{1}{q}} + \left(\sum_{j=1}^n |y_j|^p \right)^{\frac{1}{p}} \left(\sum_{j=1}^n |x_j + y_j|^{(p-1)q} \right)^{\frac{1}{q}} \\
&= \left(\sum_{j=1}^n |x_j|^p \right)^{\frac{1}{p}} \left(\sum_{j=1}^n |x_j + y_j|^{p-1 \left(\frac{p}{p-1} \right)} \right)^{1-\frac{1}{p}} + \left(\sum_{j=1}^n |y_j|^p \right)^{\frac{1}{p}} \left(\sum_{j=1}^n |x_j + y_j|^{p-1 \left(\frac{p}{p-1} \right)} \right)^{1-\frac{1}{p}} \\
&= \left(\sum_{j=1}^n |x_j|^p \right)^{\frac{1}{p}} \left(\sum_{j=1}^n |x_j + y_j|^p \right)^{1-\frac{1}{p}} + \left(\sum_{j=1}^n |y_j|^p \right)^{\frac{1}{p}} \left(\sum_{j=1}^n |x_j + y_j|^p \right)^{1-\frac{1}{p}} \\
&= (\|x\|_p + \|y\|_p) \frac{\|x + y\|_p^p}{\|x + y\|_p}.
\end{aligned}$$

Multiplying both sides by $\frac{\|x+y\|_p}{\|x+y\|_p^p}$ gives the desired inequality. $\square$

Theorem 1.3.5. Let $V = F^n$. Then $(F^n, \|\cdot\|_p)$ is a normed space.

Proof. Let $x = (x_1, \dots, x_n) \in F^n$ and $\alpha \in F$. Observe that:

$$\begin{aligned}
\|\alpha x\|_p &= \left(\sum_{j=1}^n |\alpha x_j|^p \right)^{\frac{1}{p}} \\
&= \left(\sum_{j=1}^n |\alpha|^p |x_j|^p \right)^{\frac{1}{p}} \\
&= |\alpha| \|x\|_p.
\end{aligned}$$

This satisfies homogeneity. Moreover, Minkowski's Inequality satisfies the triangle inequality. It remains to show that $\|\cdot\|_p$ is positive-definite. If $\|x\|_p = 0$, then $x_j = 0$ for all $1 \leq j \leq n$. Thus $x = 0_{F^n}$. $\square$

Corollary 1.3.6. Let $p \in [1, \infty]$. Then $\ell_p = \{(a_k)_k \mid \sum_{k=1}^{\infty} |a_k|^p < \infty\}$ with norm $\|(a_k)_k\|_p = \left(\sum_{k=1}^{\infty} |a_k|^p \right)^{\frac{1}{p}}$ is a normed space.

Proof. Homogeneity and positive-definiteness are trivial to prove. Let $(x_k)_k, (y_k)_k \in \ell_p$. It

is clear that:

$$\begin{aligned} \left(\sum_{k=1}^n |x_k + y_k|^p \right)^{\frac{1}{p}} &\leq \left(\sum_{k=1}^n |x_k|^p \right)^{\frac{1}{p}} + \left(\sum_{k=1}^n |y_k|^p \right)^{\frac{1}{p}} \\ &\leq \left(\sum_{k=1}^{\infty} |x_k|^p \right)^{\frac{1}{p}} + \left(\sum_{k=1}^{\infty} |y_k|^p \right)^{\frac{1}{p}} \\ &= \|(x_k)_k\|_p + \|(y_k)_k\|_p. \end{aligned}$$

We have that $\sum_{k=1}^n |x_k + y_k|^p$ is increasing and bounded above by $(\|(x_k)_k\|_p + \|(y_k)_k\|_p)^p$. By the Monotone Convergence Theorem $\lim_{n \rightarrow \infty} \sum_{k=1}^n |x_k + y_k|^p = \sum_{k=1}^{\infty} |x_k + y_k|^p$ exists. Whence $(\sum_{k=1}^{\infty} |x_k + y_k|^p)^{\frac{1}{p}} = \|(x_k)_k + (y_k)_k\|_p \leq \|(x_k)_k\|_p + \|(y_k)_k\|_p$ $\square$

Theorem 1.3.7. *All p -norms on ℓ_p^n are equivalent for $1 \leq p \leq \infty$.*

Proof. Let $x \in \ell_p^n$. We have that:

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}} \leq \left(\sum_{i=1}^n \left(\max_{i=1}^n |x_i| \right)^p \right)^{\frac{1}{p}} = \left(\sum_{i=1}^n \|x\|_{\infty}^p \right)^{\frac{1}{p}} = n^{\frac{1}{p}} \|x\|_{\infty}.$$

$$\|x\|_{\infty} = \left(\left(\max_{i=1}^n |x_i| \right)^p \right)^{\frac{1}{p}} \leq \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}} = \|x\|_p.$$

$$\|x\|_{\infty} = \max_{i=1}^n |x_i| \leq \sum_{i=1}^n |x_i| = \|x\|_1.$$

$$\|x\|_1 = \sum_{i=1}^n |x_i| \leq \sum_{i=1}^n \max_{i=1}^n |x_i| = n \max_{i=1}^n |x_i| = n \|x\|_{\infty}.$$

By transitivity, all norms on ℓ_p^n are equivalent. $\square$

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Example 1.3.2.

- (1) $(\ell_{\infty}(\Omega, F), \|\cdot\|_{\infty})$ where $\|f\|_{\infty} = \sup_{x \in \Omega} |f(x)|$ is a normed space. This includes its subspaces, such as $C([a, b], F) \subseteq \ell_{\infty}([a, b], F)$ and $C_c(\mathbb{R}) \subseteq C_0(\mathbb{R}) \subseteq \ell_{\infty}(\mathbb{R})$, all with $\|\cdot\|_{\infty}$.
- (2) Take $\Omega = \mathbb{N}$ in the previous example. Then $(\ell_{\infty}, \|\cdot\|_{\infty})$ is a normed space. This includes its subspaces $c_{00} \subseteq c_0 \subseteq \ell_{\infty}$ with $\|\cdot\|_{\infty}$.
- (3) $(\ell_1, \|\cdot\|_1)$ is a normed space.

- (4) $(C([a, b]), \|\cdot\|_1)$ with $\|f\|_1 = \int_a^b |f(t)| dt$ is a normed space.
- (5) $(BV([a, b]), \|\cdot\|_{BV})$ where $\|f\|_{BV} = |f(a)| + \text{Var}(f)$ is a normed space.
- (6) Let $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ be normed spaces. Then $(B(V, W), \|\cdot\|_{\text{op}})$ is a normed space, where $B(V, W) = \{T \in \mathcal{L}(V, W) \mid \|T\|_{\text{op}} < \infty\}$ is the set of bounded linear maps and $\|T\|_{\text{op}} = \sup_{v \in B_V} \|T(v)\|_W$. Intuitively, $\|T\|_{\text{op}}$ measures the radius of the smallest ball which contains B_V .

Exercise 1.3.3. Show that $V^* := B(V, F)$ is a subspace of V' .

- (7) Let S be a nonempty set. Both $(F(S), \|\cdot\|_1)$ and $(F(S), \|\cdot\|_p)$ are normed spaces, where $\|f\|_1 = \sum_{s \in S} |f(s)|$ and $\|f\|_p = (\sum_{s \in S} |f(s)|^p)^{\frac{1}{p}}$. Note that since $f(s) \neq 0$ for finitely many $s \in S$, both $\|\cdot\|_1$ and $\|\cdot\|_p$ are well-defined.

Exercise 1.3.4. Show that $\|f\|_\infty := \sup_{s \in S} |f(s)|$ is a norm on $F(S)$.

§ 1.4. Inner Product Spaces

Definition 1.4.1. Let V be a vector space over F and $\varphi : V \times V \rightarrow F$ a map.

- (1) The map φ is said to be a *bilinear form* if it is linear in the first and second variable separately; i.e., for all $v_1, v_2, v \in V$ and $c \in F$ we have:
- (i) $\varphi(cv_1 + v_2, v) = c\varphi(v_1, v) + \varphi(v_2, v)$
 - (ii) $\varphi(v, cv_1 + v_2) = c\varphi(v, v_1) + \varphi(v, v_2)$.
- (2) The map φ is said to be a *sesquilinear form* if it is linear in the first variable and conjugate linear in the second variable; i.e., for all $v_1, v_2, v \in V$ and $c \in F$ we have:
- (i) $\varphi(cv_1 + v_2, v) = c\varphi(v_1, v) + \varphi(v_2, v)$
 - (ii) $\varphi(v, cv_1 + v_2) = \bar{c}\varphi(v, v_1) + \varphi(v, v_2)$.

If we wish to keep track of a bilinear form on V we write (V, φ) .

Definition 1.4.2. Let V be a vector space over F .

- (1) A bilinear form φ on V is said to be *symmetric* if $\varphi(v, w) = \varphi(w, v)$ for all $v, w \in V$.
- (2) A sesquilinear form φ on V is said to be *Hermitian* if $\varphi(v, w) = \overline{\varphi(w, v)}$ for all $v, w \in V$.

Definition 1.4.3. Let (V, φ) be a vector space over F such that if φ is symmetric, then $F = \mathbb{R}$ or if φ is Hermitian, then $F = \mathbb{C}$. We say φ is *positive-definite* if for all nonzero $v \in V$ we have $\varphi(v, v) \neq 0$.

Definition 1.4.4. Let (V, φ) be a vector space over $\mathbb{R}$ with φ a positive-definite symmetric bilinear form or over $\mathbb{C}$ with φ a positive-definite Hermitian sesquilinear form. Then we say φ is an *inner product* on V and write φ as $\langle \cdot, \cdot \rangle$. We say $(V, \langle \cdot, \cdot \rangle)$ is an *inner product space*.

—————//—————

Definition 1.4.5. If V is an inner product space we define $\|v\|_2 = \langle v, v \rangle^{\frac{1}{2}}$.

Definition 1.4.6. Let V be an inner product space. Two vectors $v, w \in V$ are *orthogonal* if $\langle u, v \rangle = 0$. We denote this as $u \perp v$.

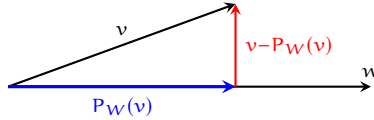
Theorem 1.4.1 (Pythagorean Theorem). *Let $v_1, \dots, v_n$ be mutually orthogonal. Then $\sum_{j=1}^n \|v_j\|_2^2 = \|\sum_{j=1}^n v_j\|_2^2$.*

Proof. Because $v_i \perp v_j$ for $1 \leq i, j \leq n$, we have $\langle v_i, v_j \rangle = 0$. Observe that:

$$\begin{aligned} \left\| \sum_{j=1}^n v_j \right\|_2^2 &= \left\langle \sum_{j=1}^n v_j, \sum_{j=1}^n v_j \right\rangle \\ &= \sum_{j=1}^n \left(\sum_{i=1}^n \langle v_j, v_i \rangle \right) \\ &= \sum_{j=1}^n \langle v_j, v_j \rangle \\ &= \sum_{j=1}^n \|v_j\|_2^2. \end{aligned}$$

□

Definition 1.4.7. Let V be an inner product space and $w \in V$ nonzero. The *projection* of a vector $v \in V$ onto w is a map $P_W : V \rightarrow V$ defined by $P_W(v) = \frac{\langle v, w \rangle}{\langle w, w \rangle} w$.



Proposition 1.4.2. *** Let V be an inner product space and $w \in V$ a nonzero vector. Then $P_W(v) \perp v - P_W(v)$.

Proof.

□

Corollary 1.4.3. *** Let V be an inner product space and $w \in W$ a nonzero vector. Then $\|v\|_2^2 = \|P_W(v)\|_2^2 + \|v - P_W(v)\|_2^2$.

Proof.

□

Lemma 1.4.4 (Cauchy-Schwartz Inequality). *Let V be an inner product space and $v, w \in V$. Then $|\langle v, w \rangle| \leq \|v\|_2 \|w\|_2$.*

Proof. The previous corollary gives $\|v\|_2 \geq \|P_w(v)\|_2$. We have that:

$$\begin{aligned} \|v\|_2 &\geq \left\| \frac{\langle v, w \rangle}{\langle w, w \rangle} w \right\|_2 \\ &= \frac{|\langle v, w \rangle|}{\|w\|_2^2} \|w\|_2 \\ &= \frac{\langle v, w \rangle}{\|w\|_2}. \end{aligned}$$

Multiplying both sides by $\|w\|_2$ gives the desired result. $\square$

Theorem 1.4.5. *Let V be an inner product space. Then $(V, \|\cdot\|_2)$ is a normed space.*

Proof. Let $v, w \in V$ and $\alpha \in \mathbb{F}$. We have that:

$$\begin{aligned} \|\alpha v\|_2 &= \langle \alpha v, \alpha v \rangle^{\frac{1}{2}} \\ &= (\alpha \bar{\alpha} \langle v, v \rangle)^{\frac{1}{2}} \\ &= (|\alpha|^2 \langle v, v \rangle)^{\frac{1}{2}} \\ &= |\alpha| \|v\|_2. \end{aligned}$$

Thus $\|\cdot\|_2$ satisfies homogeneity. It follows from the Cauchy-Schwartz Inequality that:

$$\begin{aligned} \|v + w\|_2^2 &= \langle v + w, v + w \rangle \\ &= \langle v, v \rangle + \langle v, w \rangle + \langle w, v \rangle + \langle w, w \rangle \\ &= \|v\|_2^2 + \langle v, w \rangle + \overline{\langle v, w \rangle} + \|w\|_2^2 \\ &= \|v\|_2^2 + 2\Re(\langle v, w \rangle) + \|w\|_2^2 \\ &\leq \|v\|_2^2 + 2|\langle v, w \rangle| + \|w\|_2^2 \\ &\leq \|v\|_2^2 + 2\|v\|_2 \|w\|_2 + \|w\|_2^2 \\ &= (\|v\|_2 + \|w\|_2)^2, \end{aligned}$$

where we used the fact that $2\Re(\langle v, w \rangle) = 2|\langle v, w \rangle|$. Squaring both sides proves that $\|\cdot\|_2$ satisfies the triangle inequality. It remains to show positive-definiteness. Suppose $\|v\|_2 = 0$. Then $\langle v, v \rangle = 0$, but since the inner-product is by definition positive-definite, we get that $v = 0_V$. $\square$

—————//—————

Example 1.4.1.

- (1) $\ell_2^n = \mathbb{F}^n$ is an inner product space where $\langle (x_1, \dots, x_n), (y_1, \dots, y_n) \rangle := \sum_{j=1}^n x_j \overline{y_j}$.

(2) ℓ_2 is an inner product space where $\langle (a_k)_k, (b_k)_k \rangle := \sum_{k=1}^{\infty} a_k \overline{b_k}$. Note that:

$$\begin{aligned} \sum_{k=1}^n |a_k \overline{b_k}| &= \sum_{k=1}^n |a_k| |b_k| \\ &\leq \left(\sum_{k=1}^n |a_k|^2 \right)^{\frac{1}{2}} \left(\sum_{k=1}^n |b_k|^2 \right)^{\frac{1}{2}} \\ &\leq \left(\sum_{k=1}^{\infty} |a_k|^2 \right)^{\frac{1}{2}} \left(\sum_{k=1}^{\infty} |b_k|^2 \right)^{\frac{1}{2}} \\ &= \|(a_k)_k\|_2 \|(b_k)_k\|_2 \\ &< \infty \quad (\text{Because } (a_k)_k, (b_k)_k \in \ell_2). \end{aligned}$$

Since $\sum_{k=1}^n |a_k \overline{b_k}|$ is increasing and bounded above, the Monotone Convergence Theorem says $\sum_{k=1}^{\infty} |a_k \overline{b_k}|$ exists and is finite. Whence $\langle (a_k)_k, (b_k)_k \rangle$ converges.

(3) Recall that $\text{Tr} : M_n(\mathbb{C}) \rightarrow \mathbb{C}$ is defined by $\text{Tr}(a_{ij}) = \sum_{i=1}^n a_{ii}$. Then $M_n(\mathbb{C})$ is an inner product space where $\langle a_{ij}, b_{ij} \rangle := \text{Tr}(b_{ij}^* a_{ij})$.

(4) $C([0, 1])$ is an inner product space where $\langle f, g \rangle := \int_0^1 f(x) \overline{g(x)} dx$.

§ 1.5. Normed Algebras

Definition 1.5.1. A *normed algebra* is an algebra A equipped with a norm $\|\cdot\|_A$ such that $\|ab\|_A \leq \|a\|_A \|b\|_A$. If A is unital, we require $\|1\|_A = 1$.

Example 1.5.1.

- (1) $\ell_{\infty}(\Omega)$ equipped with $\|\cdot\|_u$ is a normed algebra.
- (2) $C_c(\mathbb{R})$, $C_0(\mathbb{R})$, and $C([0, 1])$ are all normed algebras when equipped with $\|\cdot\|_u$.
- (3) $M_n(F)$ equipped with $\|\cdot\|_{\text{op}}$ is a normed algebra.
- (4) If V is a normed space, then $B(V, V)$ with $\|\cdot\|_{\text{op}}$ is a normed algebra: for $T, S \in B(V, V)$ and $v \in B_V$, we have that

$$\begin{aligned} \|(T \circ S)(v)\| &\leq \|T\|_{\text{op}} \|S(v)\| \\ &\leq \|T\|_{\text{op}} \|S\|_{\text{op}}. \end{aligned}$$

Taking the supremum over all $v \in B_V$ gives $\|T \circ S\|_{\text{op}} \leq \|T\|_{\text{op}} \|S\|_{\text{op}}$.

- (5) Let S be a group. Equip the algebra $\mathbb{F}(S)$ with $\|\cdot\|_1$. We get a normed algebra.

Exercise 1.5.1. For $a, b \in \ell_1(\mathbb{Z})$, define $a * b : \mathbb{Z} \rightarrow F$ by $(a * b)(n) = \sum_{k \in \mathbb{Z}} a(n - k)b(k)$. Show that $\ell_1(\mathbb{Z})$ with this multiplication is a normed algebra.

Chapter 2

Metric Spaces

§ 2.1. Basic Definitions and Examples

Definition 2.1.1. A *metric* on a nonempty set X is a map

$$d : X \times X \rightarrow [0, \infty)$$

satisfying for all $x, y, z \in X$:

- (1) $d(x, y) = d(y, x)$ (symmetry);
- (2) $d(x, z) \leq d(x, y) + d(y, z)$ (triangle inequality);
- (3) $d(x, x) = 0$;
- (4) if $d(x, y) = 0$ then $x = y$ (positivity).

If d satisfies all but (iv), then d is called a *semi-metric*. The pair (X, d) is called a *metric space*.

Definition 2.1.2. Two metrics d, ρ on X are called *equivalent* if there exists constants c, c' with $d(x, y) \leq c\rho(x, y)$ and $\rho(x, y) \leq c'd(x, y)$.

Definition 2.1.3. A family of metrics $\{d_k\}_{k=1}^{\infty}$ on X is *uniformly bounded* if, for all $x, y \in X$ and $k \in \mathbb{N}$, we have $d_k(x, y) \leq C$.

Example 2.1.1.

- (1) The *discrete metric* on $X \neq \emptyset$ is:

$$d(x, y) = \begin{cases} 1, & x \neq y \\ 0, & x = y. \end{cases}$$

- (2) The *hamming distance* between two bit strings of equal length: given $X = \{0, 1\}^n$, then $d_H : X \times X \rightarrow [0, \infty)$ is defined by $d_H((x_j)_{j=1}^n, (y_j)_{j=1}^n) = |\{j \mid x_j \neq y_j\}|$.
- (3) If $(V, \|\cdot\|)$ is any normed space, then $d(v, w) = \|v - w\|$ is a metric on V .

Exercise 2.1.1. If $\|\cdot\|$ and $\|\cdot\|'$ are norms on a linear space V , show they are equivalent if and only if their induced metrics are equivalent.

- (4) If (X, d) is a metric and $Y \subseteq X$ is a subset, then (Y, d) is a metric space.

- (5) Let (X, ρ) be a metric space. Fix $p \in X$. Then:

$$d(x, y) := \begin{cases} 0, & x = y \\ \rho(x, p) + \rho(p, y), & x \neq y \end{cases}$$

is a metric.

- (6) It is often beneficial to work with metrics that are bounded. Let ρ be a (semi)-metric of X . Set:

$$d(x, y) = \frac{\rho(x, y)}{1 + \rho(x, y)}.$$

Defining $d(x, y)$ as above gives $0 \leq d(x, y) \leq 1$. Although d and ρ are not equivalent metrics, they are topologically equivalent.

Clearly d is symmetric and $d(x, x) = 0$. Moreover, if $d(x, y) = 0$, then $\rho(x, y) = 0$, giving $x = y$ if ρ is a metric. For the triangle inequality, consider the function $g : [0, \infty) \rightarrow [0, 1)$ given by $g(t) = \frac{t}{1+t}$. We have that $g'(t) = \frac{(1+t)-t}{(1+t)^2} = \frac{1}{(1+t)^2} > 0$, whence g is strictly increasing. Since we know $\rho(x, z) \leq \rho(x, y) + \rho(y, z)$, observe that:

$$\begin{aligned} d(x, z) &= \frac{\rho(x, z)}{1 + \rho(x, z)} \\ &\leq \frac{\rho(x, y) + \rho(y, z)}{1 + \rho(x, y) + \rho(y, z)} \\ &= \frac{\rho(x, y)}{1 + \rho(x, y) + \rho(y, z)} + \frac{\rho(y, z)}{1 + \rho(x, y) + \rho(y, z)} \\ &\leq \frac{\rho(x, y)}{1 + \rho(x, y)} + \frac{\rho(y, z)}{1 + \rho(y, z)} \\ &= d(x, y) + d(y, z). \end{aligned}$$

- (7) If $d_1, \dots, d_n$ are metrics on X and $c_1, \dots, c_n > 0$, then:

$$d(x, y) = \sum_{i=1}^n c_i d_i(x, y)$$

is a metric on X .

- (8) Let $\{\rho_k\}_{k=1}^\infty$ be a family of semi-metrics on X . Assume that the family is *separating*: if $x, y \in X$ and $x \neq y$, then there exists k such that $\rho_k(x, y) \neq 0$. Let $d_k(x, y) = \frac{\rho_k(x, y)}{1 + \rho_k(x, y)}$. Then:

$$d(x, y) = \sum_{k=1}^{\infty} 2^{-k} d_k(x, y)$$

is a metric on X . Since $0 \leq d_k(x, y) \leq 1$, by comparison $d(x, y)$ will converge.

- (9) Let $(X_k, \rho_k)_{k \geq 1}$ be a sequence of metric spaces. For each k let d_k be as above. Let $X = \prod_{k=1}^\infty X_k$. Then the map $D : X \times X \rightarrow [0, \infty)$ defined by

$$D(f, g) = \sum_{k=1}^{\infty} 2^{-k} d_k(f(k), g(k))$$

is a metric on X . Note that we need not make d_k from ρ_k if all the d_k are uniformly bounded.

- (10) Let $X = \{0, 2\}$ with the discrete metric. The *abstract Cantor set* is $\Delta := \prod_{k \in \mathbb{N}} X$. Then the map $D : \Delta \times \Delta \rightarrow [0, \infty)$ defined by

$$D(f, g) = \sum_{k=1}^{\infty} 3^{-k} |f(k) - g(k)|$$

is a metric on Δ .

- (11) Let $\langle \cdot, \cdot \rangle$ be the standard inner product on $\mathbb{R}^3$ (or $\mathbb{R}^n$). The unit sphere $S^2 = \{x \in \mathbb{R}^3 \mid \|x\|_2 = 1\}$ paired with $d(x, y) := \arccos(\langle x, y \rangle)$ is a metric space.

Exercise 2.1.2. Show that (S^2, d) defined as above is indeed a metric space.

Definition 2.1.4.

- (1) Let (X, d) be a metric space with $E \subseteq X$. The *diameter* of E is $\text{diam}(E) = \sup_{x, y \in E} d(x, y)$. We say E is *bounded* (metrically) if $\text{diam}(E) < \infty$.
- (2) If Ω is any set and (Y, d) is a metric space, $f : \Omega \rightarrow Y$ is *bounded* if $\text{diam}(f(\Omega)) < \infty$. The set of bounded functions is $\text{Bd}(\Omega, Y) := \{f : \Omega \rightarrow Y \mid f \text{ bounded}\}$.

Exercise 2.1.3. If $(V, \|\cdot\|)$ is a normed space and $E \subseteq V$, the following are equivalent:

- (1) E is bounded;
- (2) $\sup_{v \in E} \|v\| < \infty$;
- (3) there exists $r > 0$ such that $E \subseteq B(0, r)$.

Example 2.1.2. The set $\text{Bd}(\Omega, Y)$ is a metric space with:

$$D_u(f, g) := \sup_{x \in \Omega} d(f(x), g(x)).$$

Clearly $D_u(f, g) = D_u(g, f)$ and $D_u(f, f) = 0$. If $D_u(f, g) = 0$, then $f(x) = g(x)$ for all $x \in \Omega$, giving $f = g$. Moreover, for every $x \in \Omega$:

$$\begin{aligned} d(f(x), h(x)) &\leq d(f(x), g(x)) + d(g(x), h(x)) \\ &\leq D_u(f, g) + D_u(g, h). \end{aligned}$$

Whence $D_u(f, h) \leq D_u(f, g) + D_u(g, h)$. Note that if we take the normed space $(\ell_\infty(\Omega), \|\cdot\|_u)$, the induced metric is:

$$\begin{aligned} d(f, g) &= \|f - g\|_u \\ &= \sup_{x \in \Omega} |(f - g)(x)| \\ &= \sup_{x \in \Omega} |f(x) - g(x)| \\ &= D_u(f, g). \end{aligned}$$

So as metric spaces, $\ell_\infty(\Omega) \cong \text{Bd}(\Omega, F)$. Now consider the subset $E = \{f \in \text{Bd}(\Omega, F) \mid f(x) \in \{0, 1\}\}$. We get:

$$\begin{aligned} D_u(f, g) &= \sup_{x \in \Omega} |f(x) - g(x)| \\ &= \begin{cases} 0, & f = g \\ 1, & f \neq g. \end{cases} \end{aligned}$$

So (E, D_u) is discrete.

§ 2.2. Topology of Metric Spaces

Unless otherwise stated, let (X, d) be a metric space.

Definition 2.2.1. Let X be a set. A collection T of subsets of X is called a *topology on X* if they satisfy:

- (1) $\emptyset, X \in T$;
- (2) arbitrary unions of elements in T are in T ;
- (3) finite intersections of elements in T are in T .

Definition 2.2.2. Let (X, d) be a metric space.

- (1) Let $x_0 \in X$ and $\delta > 0$.
 - (i) The *open ball* centered at x_0 of radius δ is $U(x_0, \delta) = \{x \mid d(x, x_0) < \delta\}$.
 - (ii) The *closed ball* centered at x_0 of radius δ is $B(x_0, \delta) = \{x \mid d(x, x_0) \leq \delta\}$.
 - (iii) The *sphere* centered at x_0 of radius δ is $S(x_0, \delta) = \{x \mid d(x, x_0) = \delta\}$.
- (2) A subset $U \subseteq X$ is *open in X* if:

$$(\forall x \in U)(\exists \delta > 0) : U(x, \delta) \subseteq U.$$

The collection of open sets is denoted $\tau_X := \{U \subseteq X \mid U \text{ is open}\}$.

- (3) A subset $D \subseteq X$ is *closed in X* if $D^c \subseteq X$ is open in X .
- (4) If $x \in U \in \tau_X$, then U is called an *open neighborhood of x* . If $x \in U \in \tau_X$ and $U \subseteq N \subseteq X$, then N is called a *neighborhood of x* . The collection of neighborhoods of x is denoted $\mathcal{N}_x = \{N \mid N \text{ is a neighborhood of } x\}$.
- (5) Let $A \subseteq X$.

- (i) The *interior of A* is:

$$A^\circ := \bigcup \{V \in \tau_X \mid V \subseteq A\}.$$

- (ii) The *closure of A* is:

$$\bar{A} := \bigcap \{C \mid C \supseteq A, C \text{ closed}\}.$$

(iii) The *boundary* of A is $\partial A := \bar{A} \setminus A^\circ$.

Exercise 2.2.1. Show that $\overline{A^c} = (A^\circ)^c$ and $\bar{A}^c = (A^c)^\circ$.

Proposition 2.2.1. Let (X, d) be a metric space. The open sets τ_X form a topology.

Proof. Both $\emptyset$ and X are open by assumption. Let $\{V_i\}_{i \in I}$ be a family of open sets of X . Let $x \in \bigcup_{i \in I} V_i$. Then $x \in V_i$ for some i . Since V_i is open, there exists $\delta > 0$ with $B(x, \delta) \subseteq V_i \subseteq \bigcup_{i \in I} V_i$. Whence the arbitrary union of open sets is open.

Now let $\{V_k\}_{k=1}^n$ be a family of open sets. Let $x \in \bigcap_{k=1}^n V_k$. Then $x \in V_k$ for all k . Since V_k is open, there exists $\delta_k > 0$ with $B(x, \delta_k) \subseteq V_k$. Pick $\delta = \min\{\delta_1, \delta_2, \dots, \delta_n\}$. Then $B(x, \delta) \subseteq \bigcap_{k=1}^n V_k$. Whence $\bigcap_{k=1}^n V_k$ is open. $\square$

Note that an arbitrary intersection of open sets is not necessarily open. Consider the sequence of intervals $(I_n)_n = (-\frac{1}{n}, \frac{1}{n})$. Then $\bigcap_{n=1}^\infty I_n = \{0\}$, which is closed.

Exercise 2.2.2. Let (X, d) be a metric space and consider a collection $\mathcal{C}$ of subsets of X with $\emptyset, X \in \mathcal{C}$. Show that:

- (1) if $\{C_i\}_{i \in I}$ is a family of closed sets, then $\bigcap_{i \in I} C_i$ is closed;
- (2) if $\{C_i\}_{i=1}^n$ is a family of closed sets then $\bigcup_{i=1}^n C_i$ is closed.

Proposition 2.2.2. Let (X, d) be a metric space and $x \in X$.

- (1) $N \in \mathcal{N}_x$ if and only if there exists $\delta > 0$ such that $U(x, \delta) \subseteq N$.
- (2) If $N \in \mathcal{N}_x$ and $N \subseteq M$, then $M \in \mathcal{N}_x$.
- (3) If $N_1, N_2 \in \mathcal{N}_x$, then $N_1 \cap N_2 \in \mathcal{N}_x$.

Proof. (1) Let $N \in \mathcal{N}_x$. Then there is an open set U with $x \in U \subseteq N \subseteq X$. Since U is open, there exists $\delta > 0$ such that $U(x, \delta) \subseteq U \subseteq N$. Conversely, suppose there exists $\delta > 0$ such that $U(x, \delta) \subseteq N$. Clearly $U(x, \delta) \subseteq N \subseteq X$, whence $N \in \mathcal{N}_x$.

(2) If $N \in \mathcal{N}_x$, then there is an open set U with $x \in U$ and $U \subseteq N \subseteq X$. So $U \subseteq N \subseteq M \subseteq X$. Whence $M \in \mathcal{N}_x$.

(3) If $N_1, N_2 \in \mathcal{N}_x$, then there are open sets U_1, U_2 with $x \in U_1, x \in U_2$ and $U_1 \subseteq N_1 \subseteq X, U_2 \subseteq N_2 \subseteq X$. Whence $U_1 \cap U_2 \subseteq N_1 \cap N_2 \subseteq X$. $\square$

Proposition 2.2.3. Let $U \subseteq \mathbb{R}$ be open. Then:

$$U = \bigcup_{j \in J} I_j,$$

where J is countable and I_j are open intervals.

Proof. For each $x \in U$, define:

$$I_x := \bigcup \{I \mid x \in I \subseteq U, I \text{ open interval}\}.$$

Clearly $x \in I_x \subseteq U$. If $s, t \in I_x$ with $s < t$, then there exists open intervals I, I' with $x \in I \subseteq U, x \in I' \subseteq U$, and $s \in I, t \in I'$. Since $I \cap I' \neq \emptyset$, $I \cup I'$ is an open interval. Moreover, since $s, t \in I \cup I'$, we know $[s, t] \subseteq I \cup I' \subseteq I_x$. This shows I_x is an interval—in particular, since I_x is the union of open intervals, it must be open.

Now suppose $x, y \in U$ and $I_x \cap I_y \neq \emptyset$. Then there exists $z \in I_x \cap I_y$, but $z \in I_x$ implies $I_x \subseteq I_z$ and $z \in I_y$ implies $I_y \subseteq I_z$. But we also have $x \in I_x \subseteq I_z$ which gives $I_z \subseteq I_x$, and similarly $y \in I_y \subseteq I_z$ gives $I_z \subseteq I_y$. Together, we have $I_x = I_y$, which means for any $x, y \in U$, then $I_x \cap I_y = \emptyset$ or $I_x = I_y$. Thus there exists $J \subseteq U$ with $U = \bigsqcup_{j \in J} I_j$.

It remains to show that J is countable. Define $J \rightarrow \mathbb{Q}$ by $x \mapsto q_x$, where $q_x \in \mathbb{Q} \cap I_x$. This map is injective, establishing the proposition. $\square$

Proposition 2.2.4. *Let $A \subseteq X$.*

- (1) $x \in A^\circ$ if and only if there exists $\delta > 0$ such that $U(x, \delta) \subseteq A$.
- (2) $x \in \bar{A}$ if and only if for all $\delta > 0$, $U(x, \delta) \cap A \neq \emptyset$.
- (3) $x \in \partial A$ if and only if for all $\delta > 0$, $U(x, \delta) \cap A \neq \emptyset$ and $U(x, \delta) \cap A^c \neq \emptyset$.

Proof. (1) If $x \in A^\circ$, then by definition $x \in \bigcup \{V \in \tau_X \mid V \subseteq A\}$. So there exists some $V \in \tau_X$ such that $x \in V \subseteq A$. Since V is open, there exists $\delta > 0$ with $U(x, \delta) \subseteq V \subseteq A$. Conversely, if $U(x, \delta) \subseteq A$ for some $\delta > 0$, then clearly $x \in A^\circ$, as $U(x, \delta) \in \tau_X$.

(2) We prove the converse of this statement. Note that $x \notin \bar{A}$ if and only if $x \in (\bar{A})^c = (A^c)^\circ$. By (1) there exists $\delta > 0$ with $U(x, \delta) \subseteq (A^c)^\circ \subseteq A^c$. This is true if and only if $U(x, \delta) \cap A = \emptyset$, establishing the proposition.

(3) Note that:

$$\begin{aligned} \partial A &= \bar{A} \setminus A^\circ \\ &= \bar{A} \cap (A^\circ)^c \\ &= \bar{A} \cap \bar{A}^c. \end{aligned}$$

So $x \in \partial A$ if and only if $x \in \bar{A} \cap \bar{A}^c$. Applying (2) to $x \in \bar{A}$ and $x \in \bar{A}^c$ establishes (3). $\square$

Exercise 2.2.3. Show that open balls are open, closed balls are closed, and spheres are closed.

Proposition 2.2.5. *** *For any normed space:*

- (1) $\overline{U(x, \delta)} = B(x, \delta)$
- (2) $B(x, \delta)^\circ = U(x, \delta)$
- (3) $\partial U(x, \delta) = \partial B(x, \delta) = S(x, \delta)$.

Proof. $\square$

Proposition 2.2.6. *** *Let (X, d) be a metric space with $\{A_i\}_{i \in I}$ a family of subsets. Let $K \subseteq I$ be finite.*

- (1) $\bigcup_{i \in I} A_i^\circ \subseteq \left(\bigcup_{i \in I} A_i \right)^\circ$ (Inclusion may be strict).
- (2) $\overline{\bigcap_{i \in I} A_i} \subseteq \bigcap_{i \in I} \bar{A}_i$ (Inclusion may be strict).

$$(3) \bigcap_{i \in K} A_i^\circ = \left(\bigcap_{i \in K} A_i \right)^\circ$$

$$(4) \overline{\bigcup_{i \in K} A_i} = \bigcup_{i \in K} \overline{A_i}.$$

Proof.

□

Proposition 2.2.7. *Let $S \subseteq X$.*

- (1) $\partial S = \partial S^c$.
- (2) ∂S is closed.
- (3) $\bar{S} = S \cup \partial S$.
- (4) $S \setminus \partial S = S^\circ$.

Proof. (1) This follows from the characterization of ∂S . (2) We have $\partial S = \bar{S} \setminus S^\circ = \bar{S} \cap (S^\circ)^c$, which is closed. (3) Clearly $S \cup \partial S \subseteq \bar{S}$. Let $x \in \bar{S}$. If $x \in S$ we are done. Otherwise $x \in \bar{S} \setminus S \subseteq \bar{S} \setminus S^\circ = \partial S$. (4) Observe that:

$$\begin{aligned} S \setminus \partial S &= S \cap (\partial S)^c \\ &= S \cap (\bar{S} \setminus S^\circ)^c \\ &= S \cap (\bar{S} \cap (S^\circ)^c)^c \\ &= S \cap (\bar{S}^c \cup S^\circ) \\ &= S \cap \bar{S}^c \cup S \cap S^\circ \\ &= S^\circ. \end{aligned}$$

□

Definition 2.2.3. Let (X, d) be a metric space.

- (1) A subset $A \subseteq X$ is *d-dense* if $\bar{A} = X$.
- (2) A subset $N \subseteq X$ is *nowhere dense* if $(\bar{N})^\circ = \emptyset$.
- (3) The space (X, d) is *separable* if there exists a countable dense subset $D \subseteq X$.

Exercise 2.2.4. If $N \subseteq X$ is closed, then N is nowhere dense if and only if N^c is dense.

Proposition 2.2.8. *** Let $A \subseteq X$. The following are equivalent:

- (1) A is dense;
- (2) $(\forall U \in \tau_X), U \cap A \neq \emptyset$;
- (3) $(\forall x \in X)(\forall \epsilon > 0), U(x, \epsilon) \cap A \neq \emptyset$;
- (4) $(\forall x \in X)(\forall \epsilon > 0)(\exists a \in A) : d(a, x) < \epsilon$.

Proof.

□

Definition 2.2.4. Let (X, d) be a metric space.

(1) A *base* for τ_X is a family of open subsets $\mathcal{B} \subseteq \tau_X$ such that:

$$(\forall U \in \tau_X)(\forall x \in U)(\exists B \in \mathcal{B}) : x \in B \subseteq U.$$

Equivalently, for all $U \in \tau_X$, we can write $U = \bigcup_{i \in I} B_i$, where $\{B_i\}_{i \in I} \subseteq \tau_X$.

(2) X is *second countable* if it has a countable base.

Note that this definition can be generalized to any topological space. Clearly $\mathcal{B} = \{U(x, \epsilon) \mid x \in X, \epsilon > 0\}$ forms a base for any metric space.

Example 2.2.1. The set $\mathcal{B} = \{U(q, \frac{1}{n}) \mid n \geq 1, q \in \mathbb{Q}^d\}$ is a base for $\mathbb{R}^d$.

Proposition 2.2.9. Let (X, d) be a metric space. X is separable if and only if X is second countable.

Proof. Let $\mathcal{B} = \{U_n\}_{n=1}^\infty$ be a countable base. Choose any $a_n \in U_n$. Then $\{a_n\}_{n=1}^\infty$ is dense. Indeed, given any $x \in X$ and $\epsilon > 0$, there exists U_m with $x \in U_m \subseteq U(x, \epsilon)$ (since $U_m \in \mathcal{B}$). Whence $d(a_m, x) < \epsilon$.

Let $\{a_n\}_{n=1}^\infty$ be dense. Consider:

$$\mathcal{B} = \{U(a_n, \frac{1}{m}) \mid n \geq 1, m \geq 1\}.$$

Clearly $\mathcal{B}$ is countable—it remains to show that it is a base for X . Given $x \in V \in \tau_X$, find $\epsilon > 0$ such that $U(x, \epsilon) \subseteq V$. Then there exists $m \geq 1$ with $\epsilon > \frac{1}{m}$. Since $\{a_n\}_{n=1}^\infty$ is dense, there exists $a_j \in \{a_n\}_{n=1}^\infty$ such that $d(a_j, x) < \frac{1}{2m}$. Let $y \in U(a_j, \frac{1}{2m})$. Observe that:

$$\begin{aligned} d(x, y) &\leq d(x, a_j) + d(a_j, y) \\ &< \frac{1}{2m} + \frac{1}{2m} \\ &= \frac{1}{m} \\ &< \epsilon. \end{aligned}$$

So $y \in U(x, \epsilon)$. Thus $x \in U(a_j, \frac{1}{2m}) \subseteq U(x, \epsilon) \subseteq V$, establishing $\mathcal{B}$ as a base. $\square$

Example 2.2.2.

(1) The space $(\mathbb{R}^d, \|\cdot\|_p)$ is separable for any $1 \leq p \leq \infty$. Indeed, if $(r_1, \dots, r_d) \in \mathbb{R}^d$ and $\epsilon > 0$, find $q_j \in \mathbb{Q}$, $j = 1, \dots, d$ with:

$$|r_j - q_j| < \frac{\epsilon}{d}.$$

Then:

$$\|r - q\|_1 = \sum_{j=1}^d |r_j - q_j| < \epsilon.$$

So $\mathbb{Q}^d$ is $\|\cdot\|_1$ -dense in $\mathbb{R}^d$. For $1 \leq p \leq \infty$, let $C > 0$ be such that $\|\cdot\|_p \leq C \|\cdot\|_1$. So given $\epsilon > 0$, find $q \in \mathbb{Q}^d$ with $\|r - q\|_1 \leq \frac{\epsilon}{C}$. Then $\|r - q\|_p < \epsilon$.

(2) Similarly, $\mathbb{C}_Q^d \subseteq \mathbb{C}^d$ is $\|\cdot\|_p$ -dense, where:

$$\mathbb{C}_Q = \{a + bi \mid a, b \in \mathbb{Q}\}.$$

(3) Recall that $c_{00} = \{(z_k)_k \mid \text{supp}((z_k)_k) < \infty\}$. The space $(c_{00}, \|\cdot\|_u)$ is separable.

Note that $c_{00} = \mathbb{C}\text{-span}\{e_k \mid k \in \mathbb{N}\}$. This space is not countable—clearly $\mathbb{C}\text{-span}\{e_1\} = \{\alpha e_1 \mid \alpha \in \mathbb{C}\}$ is not countable, so it must be that c_{00} is also not countable.

Instead, consider:

$$\begin{aligned} \mathbb{C}_Q\text{-span}\{e_k \mid k \in \mathbb{N}\} &= \left\{ \sum_{k=1}^{\infty} t_k e_k \mid t_k \in \mathbb{C}_Q, \right\} \\ &= \bigcup_{k=1}^{\infty} \{C_k \mid C_k = \mathbb{C}_Q\text{-span}\{e_1, \dots, e_k\}\} \end{aligned}$$

Note that C_k is in bijection with $\mathbb{Q}^{2k}$, whence $\mathbb{C}_Q\text{-span}\{e_k \mid k \in \mathbb{N}\}$ is countable.

Given $z \in c_{00}$, let $z = \sum_{k=1}^N z_k e_k$ and $\epsilon > 0$. Find $t_k \in \mathbb{C}_Q$ with $|z_k - t_k| < \epsilon$. Then:

$$\begin{aligned} \|z - t\|_u &= \left\| \sum_{k=1}^N z_k e_k - \sum_{k=1}^N t_k e_k \right\|_u \\ &= \left\| \sum_{k=1}^N (z_k - t_k) e_k \right\|_u \\ &= \sup_{k=1}^N |z_k - t_k| \\ &< \epsilon. \end{aligned}$$

Thus $\mathbb{C}_Q\text{-span}\{e_k \mid k \in \mathbb{N}\}$ is dense in c_{00} , whence the space $(c_{00}, \|\cdot\|_u)$ is separable.

Proposition 2.2.10. *If (X, d) is a separable metric space and $Y \subseteq X$, then (Y, d) is separable.*

Proof. Let $A = \{a_k\}_{k=1}^{\infty}$ be dense in X . Let:

$$N = \{(m, n) \mid U(a_m, \frac{1}{n}) \cap Y \neq \emptyset\}.$$

For each $(m, n) \in N$, choose $b_{(m,n)} \in Y \cap U(a_m, \frac{1}{n})$. Claim: the set

$$\{b_{(m,n)} \mid (m, n) \in N\}$$

is dense in Y . Let $y \in Y$ and $\epsilon > 0$. Then there exists $n \geq 1$ with $\frac{\epsilon}{2} > \frac{1}{n}$. Since A is dense, $U(y, \frac{1}{n}) \cap A \neq \emptyset$ (this is because for all $U \in \tau_X$, we have $U \cap A \neq \emptyset$). So $d(a_m, y) < \frac{1}{n}$. Whence:

$$\begin{aligned} d(b_{(m,n)}, y) &\leq d(b_{(m,n)}, a_m) + d(a_m, y) \\ &< \frac{1}{n} + \frac{1}{n} \\ &< \epsilon. \end{aligned}$$

□

Example 2.2.3.

- (1) The space $(\ell_\infty, \|\cdot\|_u)$ is not separable. If it were, consider:

$$E = \{(x_k)_k \mid x_k \in \{0, 1\}\} \subseteq \ell_\infty.$$

This set is uncountable, and by the previous proposition $(E, \|\cdot\|_u)$ is also separable. Let $a, b \in E$. We have:

$$\begin{aligned} \|(a_k)_k - (b_k)_k\|_u &= \|(a_k - b_k)_k\|_u \\ &= \sup_{k \geq 1} |a_k - b_k| \\ &= \begin{cases} 0, & (a_k)_k = (b_k)_k \\ 1, & (a_k)_k \neq (b_k)_k. \end{cases} \end{aligned}$$

So $(E, \|\cdot\|_u)$ is discrete. Note that a metric space is discrete if and only if its singletons are open. Whence if $(E, \|\cdot\|_u)$ is separable, then $A = \overline{A} = E$, which contradicts the countability of A . It must be that $(\ell_\infty, \|\cdot\|_u)$ is not separable.

- (2) The space $(\ell_p, \|\cdot\|_p)$ is separable for $1 \leq p < \infty$. Let $a = (a_k)_k \in \ell_p$ and $\epsilon > 0$. Find N large so that $\sum_{k > N} |a_k|^p < \frac{\epsilon^p}{2}$. Find $b_k \in \mathbb{C}_Q$ with $|a_k - b_k| < \frac{\epsilon}{(2N)^{\frac{1}{p}}}$. Let $b = (b_1, b_2, \dots, b_{N-1}, b_N, 0, 0, \dots)$. We have:

$$\begin{aligned} \|a - b\|_p^p &= \sum_{k=1}^{\infty} |a_k - b_k|^p \\ &= \sum_{k=1}^N |a_k - b_k|^p + \sum_{k > N} |a_k|^p \\ &< N \cdot \frac{\epsilon^p}{2N} + \frac{\epsilon^p}{2} \\ &= \epsilon. \end{aligned}$$

Whence the set $\mathbb{C}_Q$ -span $\{e_k \mid k \in \mathbb{N}\}$ is $\|\cdot\|_u$ -dense, giving $(\ell_p, \|\cdot\|_p)$ as separable.

- (3) We will eventually show that the set of polynomial functions:

$$P([0, 1]) = \left\{ \sum_{k=0}^n a_k x^k \mid a_k \in \mathbb{F}, n \geq 0 \right\}$$

is $\|\cdot\|_u$ -dense in $C([0, 1])$ (note that this set is not countable). With this fact, we can show that $C([0, 1])$ is separable. Indeed, given $f \in C([0, 1])$ and $\epsilon > 0$, find $p \in P([0, 1])$ with $\|f - p\|_u < \frac{\epsilon}{2}$. Now let $p(x) = \sum_{k=0}^n a_k x^k$. Find $b_k \in \mathbb{C}_Q$ with

$|a_k - b_k| < \frac{\epsilon}{2(n+1)}$ and define $q(x) = \sum_{k=0}^n b_k x^k$. Observe that:

$$\begin{aligned} \|f - q\|_u &= \|f - p + p - q\|_u \\ &\leq \|f - p\|_u + \|p - q\|_u \\ &= \|f - p\|_u + \sum_{k=0}^n |a_k - b_k| \\ &< \frac{\epsilon}{2} + (n+1) \cdot \frac{\epsilon}{2(n+1)} \\ &= \epsilon. \end{aligned}$$

Thus the set $\mathbb{C}_Q\text{-span}\{x^k \mid k \in \mathbb{N}\}$ is $\|\cdot\|_u$ -dense in $C([0, 1])$. In particular, since it is countable, $C([0, 1])$ is separable.

—————//—————

We have seen that subsets of metric spaces are metric spaces in their own right. Then what are their open sets?

Proposition 2.2.11. *Let (X, d) be a metric space and let $Y \subseteq X$ be any subset. Then $V \subseteq Y$ is open in Y if and only if there exists an open set $U \subseteq X$ with $U \cap Y = V$. That is, $\tau_Y = \{U \cap Y \mid U \in \tau_X\}$.*

Proof. Let $V \subseteq Y$ be open. Then for every $x \in V$, there exists $\delta_x > 0$ with $U_y(x, \delta_x) \subseteq V$, where:

$$\begin{aligned} U_y(x, \delta_x) &= \{y \in Y \mid d(y, x) < \delta_x\} \\ &= U(x, \delta_x) \cap Y. \end{aligned}$$

Set $U = \bigcup_{x \in V} U(x, \delta_x)$. Then U is indeed open in X . Also:

$$\begin{aligned} U \cap Y &= \left(\bigcup_{x \in V} U(x, \delta_x) \right) \cap Y \\ &= \bigcup_{x \in V} (U(x, \delta_x) \cap Y) \\ &= \bigcup_{x \in V} U_y(x, \delta_x) \\ &= V. \end{aligned}$$

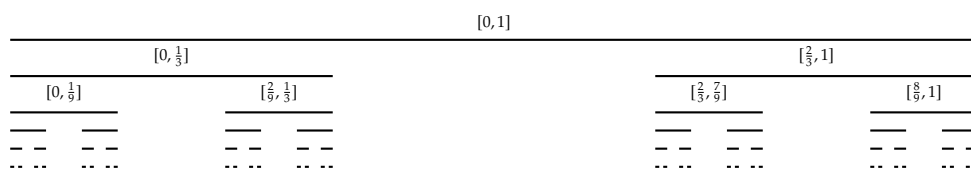
Conversely, suppose $V = U \cap Y$ for some open $U \in \tau_X$. Let $x \in V$. Since $x \in U$ and U is open, there exists $\delta > 0$ such that $U(x, \delta) \subseteq U$. So $U_y(x, \delta) = U(x, \delta) \cap Y \subseteq U \cap Y = V$. Thus V is open in Y . $\square$

Example 2.2.4.

- (1) $[0, \frac{1}{2})$ is not open in $\mathbb{R}$, but it is open in $[0, 1]$.
- (2) ℓ_∞ is not a discrete metric space, but $\{0, 1\}^{\mathbb{N}} \subseteq \ell_\infty$ is.

§ 2.3. The Cantor Set

Given the interval $[0, 1]$, start by deleting the open middle third $(\frac{1}{3}, \frac{2}{3})$, leaving two line segments $[0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$. Next, the open middle third of each of these remaining segments is deleted, leaving four line segments $[0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1]$.



We are interested in studying the topological properties of the points which are not deleted at any step of this infinite process. The set of these points have a special name and are defined below.

Definition 2.3.1. Let $C_0 := [0, 1]$ and $C_n = \frac{C_{n-1}}{3} \cup \left(\frac{2}{3} + \frac{C_{n-1}}{3}\right)$ for $n \geq 1$. The *Cantor set* is $\mathcal{C} = \bigcap_{n=0}^{\infty} C_n$.

Proposition 2.3.1. *The Cantor set is closed.*

Proof. Since the Cantor set is defined as the intersection of closed sets, it must be closed. $\square$

Proposition 2.3.2. *The Cantor set is nowhere dense.*

Proof. Suppose towards contradiction its not, that is, $\overline{\mathcal{C}}^o \neq \emptyset$. Then there is some $x \in \overline{\mathcal{C}}^o$. We can find an $\epsilon > 0$ with $(x - \epsilon, x + \epsilon) \subseteq \mathcal{C}$, in particular $(x - \epsilon, x + \epsilon) \subseteq C_n$ for all $n \geq 1$. Find m large so that $\epsilon > \frac{1}{3^m}$ and consider $(x - \epsilon, x + \epsilon) \subseteq C_m$. We have that $C_m = \bigsqcup_{j=1}^{2^m} C_{m,j}$ where $\text{length}(C_{m,j}) = \frac{1}{3^m}$. Since each $C_{m,j}$ is disjoint, it must be the case that $(x - \epsilon, x + \epsilon) \subseteq C_{m,j}$ for some $1 \leq j \leq 2^m$. But the length of $(x - \epsilon, x + \epsilon)$ is 2ϵ , which is impossible. It must be that $\mathcal{C}$ is nowhere dense. $\square$

Proposition 2.3.3. *The total length of the Cantor set is 0.*

Proof. The total length of the removed intervals is:

$$\begin{aligned} \frac{1}{3} + \frac{2}{9} + \frac{4}{27} + \dots &= \sum_{k=1}^{\infty} \frac{2^{k-1}}{3^k} \\ &= \frac{1}{2} \sum_{k=1}^{\infty} \left(\frac{2}{3}\right)^k \\ &= \frac{1}{2} \cdot \frac{\frac{2}{3}}{1 - \frac{2}{3}} \\ &= 1. \end{aligned}$$

Thus $\text{length}(\mathcal{C}) = 0$. □

Lemma 2.3.4. ***

Proof. □

Lemma 2.3.5. ***

Proof. □

Lemma 2.3.6. ***

Proof. □

Proposition 2.3.7. *** $\text{card}(\mathcal{C}) = \mathfrak{c}$.

Proof. □

§ 2.4. Convergent Sequences

Definition 2.4.1. Let (X, d) be a metric space.

- (1) A *sequence* in X is a map $x_{\bullet} : \mathbb{N} \rightarrow X$ defined by $n \mapsto x_n$. We denote a sequence as $(x_n)_{n \geq 1}$, $(x_n)_{n=1}^{\infty}$, or $(x_n)_n$.
- (2) A *natural sequence* is a sequence $(x_n)_n$ in X with $n_1 < n_2 < n_3 < \dots$
- (3) A *subsequence* of a sequence $(x_n)_n$ is a sequence $(x_{n_k})_k$, where $(n_k)_k$ is a natural sequence. This is equivalent to the composition of maps $\mathbb{N} \xrightarrow{k \mapsto n_k} \mathbb{N} \xrightarrow{n_k \mapsto x_{n_k}} X$.

Definition 2.4.2. A sequence $(x_n)_n$ *converges* to $x \in X$ if:

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N}) : (\forall n \in \mathbb{N})(n \geq N \implies d(x_n, x) < \epsilon).$$

We write $(x_n)_n \xrightarrow{d} x$ or $\lim x_n = x$.

Exercise 2.4.1. Show that a sequence can have at most one limit.

Proposition 2.4.1. Let $(x_n)_n$ be a sequence in X and $x \in X$. The following are equivalent:

- (1) $(x_n)_n \rightarrow x$;
- (2) $(d(x_n, x))_n \rightarrow 0$ in $\mathbb{R}$;
- (3) $(\forall V \in \mathcal{N}_x)(\exists N \in \mathbb{N}) : (\forall n \in \mathbb{N})(n \geq N \implies x_n \in V)$.

Proof. Exercise. □

Exercise 2.4.2. Let (X, d) be a metric space and $\rho(x, y) = \frac{d(x, y)}{1 + d(x, y)}$. Then $(x_n)_n \xrightarrow{d} x$ if and only if $(x_n)_n \xrightarrow{\rho} x$.

Proposition 2.4.2. *Convergent sequences are bounded.*

Proof. Suppose that $(x_n)_n \rightarrow x$ and let $\epsilon = 1$. Find N large so that for $n \geq N$ we have $d(x_n, x) < 1$. Then for all $m, n \geq N$, we have $d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < 2$. Set $C = \max_{1 \leq n, m \leq N} d(x_m, x_n)$. Now if $n \geq N$ and $m \leq N$, we have:

$$\begin{aligned} d(x_n, x_m) &\leq d(x_n, x_N) + d(x_N, x_m) \\ &\leq 1 + C. \end{aligned}$$

Let $K = \max\{2, 1 + C, C\}$. Then $\text{diam}(\{x_n\}_{n \geq 1}) = \sup_{m, n \geq 1} d(x_n, x_m) \leq K$. □

Definition 2.4.3. Let $(v_k)_k$ be a sequence in $(V, \|\cdot\|)$.

- (1) A sequence of partial sums $(s_n)_n$ is defined as $s_n = \sum_{k=1}^n v_k$.
- (2) If $(s_n)_n \rightarrow s$ in V we say the series $\sum_{k=1}^{\infty} v_k$ converges and write $\sum_{k=1}^{\infty} v_k = s$.
- (3) The series $\sum v_k$ converges *absolutely* if $\sum \|v_k\|$ converges.

—————//—————

Definition 2.4.4. Let Ω be a set, (X, d) a metric space, and $(f_n)_n$ a sequence of functions in X^Ω .

- (1) $(f_n)_n$ converges *pointwise* to $f \in X^\Omega$ if:

$$(\forall x \in \Omega)(\forall \epsilon > 0)(\exists N_{x, \epsilon} \in \mathbb{N}) : (\forall n \in \mathbb{N})(n \geq N \implies d(f_n(x), f(x)) < \epsilon).$$

- (2) $(f_n)_n$ converges *uniformly* to $f \in X^\Omega$ if:

$$\begin{aligned} (\forall \epsilon > 0)(\exists N_\epsilon \in \mathbb{N}) : (\forall n \in \mathbb{N})(\forall x \in \Omega)(n \geq N \implies d(f_n(x), f(x)) < \epsilon) \\ \equiv (\forall n \in \mathbb{N})(n \geq N \implies \sup_{x \in \Omega} d(f_n(x), f(x)) < \epsilon). \end{aligned}$$

Proposition 2.4.3. Let Ω be a set, (X, d) a metric space, and $(f_n)_n$ a sequence in X^Ω . The following are equivalent:

- (1) The sequence $(f_n)_n$ converges uniformly to f in X^Ω ;
- (2) The sequence $(D_u(f_n, f))_n$ converges to 0 in $\mathbb{R}$.

Proof. ($\Rightarrow$) Let $\epsilon > 0$. By assumption, find $N \in \mathbb{N}$ sufficiently large so that, for all $x \in X$, $n \geq N$ implies $d(f_n(x), f(x)) < \frac{\epsilon}{2}$. It follows then that:

$$\begin{aligned} |D_u(f_n, f)| &= \left| \sup_{x \in X} d(f_n(x), f(x)) \right| \\ &\leq \frac{\epsilon}{2} \\ &< \epsilon. \end{aligned}$$

Thus $(D_u(f_n, f))_n \rightarrow 0$.

($\Leftarrow$) Let $\epsilon > 0$. Find $N \in \mathbb{N}$ large so that $n \geq N$ implies $D_u(f_n, f) < \epsilon$. If $n \geq N$ and $z \in X$, then:

$$\begin{aligned} d(f_n(z), f(z)) &\leq \sup_{x \in X} d(f_n(x), f(x)) \\ &< \epsilon. \end{aligned}$$

Since z was arbitrary, $(f_n)_n \rightarrow f$ uniformly. $\square$

Example 2.4.1. Let (X, d) be a metric space and $(f_n)_n$ a sequence of functions in $(\text{Bd}(\Omega, X), D_u)$ which converges to $f \in \text{Bd}(\Omega, X)$. Proposition 2.4.1 says this is equivalent to $(D_u(f_n, f))_n \rightarrow 0$ in $\mathbb{R}$, and by Proposition 2.4.3 $(f_n)_n$ must converge uniformly to f . Thus convergence in $c_{00} \subseteq c_0 \subseteq c \subseteq \ell_\infty \subseteq \ell_\infty(\Omega) = \text{Bd}(\Omega, \mathbb{R})$ is uniform.

Proposition 2.4.4. Let $\{d_k\}_k$ be a separating family of semi-metrics which are uniformly bounded. Define:

$$d(x, y) := \sum_{k=1}^{\infty} 2^{-k} d_k(x, y).$$

Then $(x_n)_n \xrightarrow{d} x$ if and only if $(d_k(x_n, x))_n \rightarrow 0$ for all k .

Proof. ($\Rightarrow$) Let $k \geq 1$ be arbitrary. We have:

$$0 \leq 2^{-k} d_k(x_n, x) \leq d(x_n, x).$$

Multiplying all sides of the above equation by 2^k gives $0 \leq d_k(x_n, x) \leq 2^k d(x_n, x)$. Since $\lim_{n \rightarrow \infty} 2^k d(x_n, x) = 0$, by the Squeeze Theorem $(d_k(x_n, x))_n \rightarrow 0$.

($\Leftarrow$) Let $\epsilon > 0$. For all x, y, k , there exists $C > 0$ with $d_k(x, y) \leq C$. Since $\sum_{k=1}^{\infty} 2^{-k}$ converges (in particular, it converges to 1), find K large so:

$$\sum_{k > K} 2^{-k} < \frac{\epsilon}{2C}.$$

Note that $(d_k(x_n, x))_n \rightarrow 0$ for $k = 1, 2, \dots, K$. So there exists $N_1, N_2, \dots, N_K \in \mathbb{N}$ with $n \geq N_j$

implying $d_j(x_n, x) < \frac{\epsilon}{2}$ for $1 \leq j \leq K$. Let $N = \max_{j=1}^K N_j$. For $n \geq N$, we have:

$$\begin{aligned} d(x_n, x) &= \sum_{k=1}^K 2^{-k} d_k(x_n, x) + \sum_{k>K} 2^{-k} d_k(x_n, x) \\ &= \sum_{k=1}^K 2^{-k} d_k(x_n, x) + \sum_{k>K} 2^{-k} C \\ &< \sum_{k=1}^K 2^{-k} \frac{\epsilon}{2} + C \cdot \frac{\epsilon}{2C} \\ &\leq \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

□

Thus $(x_n)_n \xrightarrow{d} x$.

Example 2.4.2. Consider the space $C(\mathbb{R})$. How does one define a distance between two functions? Given $f, g \in C(\mathbb{R})$, note that the uniform metric:

$$d(f, g) := \sup_{x \in \Omega} |f(x) - g(x)|$$

does not guarantee $d(f, g) < \infty$. We can fix this as follows: define $\rho_k(f, g) := \sup_{x \in [-k, k]} |f(x) - g(x)|$. Note that the family of metrics $\{\rho_k\}_k$ is separating, but not uniformly bounded. Defining $d_k(f, g) := \frac{\rho_k(f, g)}{1 + \rho_k(f, g)}$ gives $\{d_k\}_k$ as a family of uniformly bounded semi-metrics. We can now define what is called the *Fréchet metric*:

$$d_F(f, g) := \sum_{k=1}^{\infty} 2^{-k} d_k(f, g).$$

By the comparison test, $d_F(f, g) < \infty$. In $(C(\mathbb{R}), d_F)$, observe that:

$$\begin{aligned} (f_n)_n \xrightarrow{d_F} f &\iff \forall k, (d_k(f_n, f))_n \rightarrow 0 \\ &\iff \forall k, (\rho_k(f_n, f))_n \rightarrow 0 \\ &\iff \forall k, \left(\sup_{x \in [-k, k]} |f_n(x) - f(x)| \right)_n \rightarrow 0 \\ &\iff \forall k, (f_n)_n \rightarrow f \text{ uniformly on } [-k, k]. \end{aligned}$$

We've obtained a new type of convergence called *compact convergence*.

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Proposition 2.4.5. Let (X, d) be a metric space and $A \subseteq X$. We have $x \in \overline{A}$ if and only if there exists a sequence $(a_n)_n$ in A with $(a_n)_n \rightarrow x$.

Proof. $(\Rightarrow)$ If $x \in \overline{A}$, then for each $n \geq 1$ we have $U(x, \frac{1}{n}) \cap A \neq \emptyset$. For each n choose $a_n \in U(x, \frac{1}{n}) \cap A$. Then $d(x, a_n) < \frac{1}{n}$, so $(a_n)_n \rightarrow x$. $(\Leftarrow)$ Given $\epsilon > 0$, find N large so $d(x, a_N) < \epsilon$. Then $a_N \in U(x, \epsilon) \cap A$. Since $U(x, \epsilon) \cap A \neq \emptyset$, we have $x \in \overline{A}$. □

Proposition 2.4.6. *Let (X, d) be a metric space and $A \subseteq X$. The following are equivalent:*

- (1) *A is closed;*
- (2) *If $(a_n)_n$ is a sequence in A which converges to $x \in X$, then $x \in A$.*

Proof. Let $(a_n)_n$ be a sequence in A which converges to $x \in X$. By Proposition 2.4.6 $x \in \overline{A}$. Since A is closed, $A = \overline{A}$. Thus $x \in A$.

We will show A is closed by proving $A = \overline{A}$. Clearly $A \subseteq \overline{A}$. Let $x \in \overline{A}$. Then there exists a sequence $(a_n)_n$ in A with $(a_n)_n \rightarrow x$. Thus $x \in A$, giving $\overline{A} \subseteq A$. $\square$

Exercise 2.4.3. Show $x \in \overline{A}$ if and only if $x \in A$ or there exists a sequence $(a_n)_n$ in $A \setminus \{x\}$ with $(a_n)_n \rightarrow x$.

Proposition 2.4.7. ***

- (1) The space $c_0 \subseteq \ell_\infty$ is closed.
 (2) $\overline{c_{00}} = c_0$.

Proof. (1) Let $(z_n)_n$ be a sequence in c_0 converging to $f \in \ell_\infty$. Let $\epsilon > 0$. Find N large so that:

$$\|z_N - f\|_\infty < \frac{\epsilon}{2}.$$

Since $z_N \in c_0$, we know $\lim_{k \rightarrow \infty} z_N(k) = 0$. Find K large so that for $k \geq K$:

$$|z_N(k)| < \frac{\epsilon}{2}.$$

Together, for $k \geq K$ we have:

$$\begin{aligned} |f(k)| &= |f(k) - z_N(k) + z_N(k)| \\ &\leq |f(k) - z_N(k)| + |z_N(k)| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

Thus $\lim_{k \rightarrow \infty} f(k) = 0$; i.e., $f \in c_0$. By Proposition 2.4.6, c_0 is closed in ℓ_∞ . □

Definition 2.4.5. Let (X, d) be a metric space and $A \subseteq X$. The *distance* of an element $x \in X$ to A is defined as the map $\text{dist}_A : X \rightarrow [0, \infty)$ given by $\text{dist}_A(x) = \inf_{a \in A} d(x, a)$.

Proposition 2.4.8. Let (X, d) be a metric space and $A \subseteq X$.

- (1) $\overline{A} = \{x \mid \text{dist}_A(x) = 0\}$.
- (2) $\text{dist}_A = \text{dist}_{\overline{A}}$.
- (3) $|\text{dist}_A(x) - \text{dist}_A(y)| \leq d(x, y)$.

Proof. (1) Let $D := \{x \mid \text{dist}_A(x) = 0\}$. By Proposition 2.4.5, $x \in \overline{A}$ if and only if there exists a sequence $(a_n)_n$ in A which converges to x . Equivalently, the sequence $(d(x, a_n))_n$ converges to 0 in $\mathbb{R}$. Since $\inf_{a \in A} d(x, a) \leq d(x, a_n)$, it must be the case that $\text{dist}_A(x) = 0$. Thus $\overline{A} \subseteq D$. Now suppose $x \in D$. Then $\inf_{a \in A} d(x, a) = 0$ implies the existence of a sequence $(a_n)_n$ in A with $(d(x, a_n))_n \rightarrow 0$. Thus $(a_n)_n \rightarrow x$, establishing that $x \in \overline{A}$ by Proposition 2.4.5. Thus $D = \overline{A}$.

(2) Let $x \in X$. Since $A \subseteq \overline{A}$, it is clear that:

$$\begin{aligned} \text{dist}_{\overline{A}}(x) &= \inf_{a \in \overline{A}} d(x, a) \\ &\leq \inf_{a \in A} d(x, a) \\ &= \text{dist}_A(x). \end{aligned}$$

Let $\epsilon > 0$ and $y \in \overline{A}$. From part (1), we have that $\text{dist}_A(y) = 0$. In particular, $\text{dist}_A(y) < \epsilon$. So there exists $a \in A$ such that $d(a, y) < \epsilon$. Observe that:

$$\begin{aligned} \text{dist}_A(x) &= \inf_{a \in A} d(x, a) \\ &\leq d(x, a) \\ &\leq d(x, y) + d(y, a) \\ &< d(x, y) + \epsilon. \end{aligned}$$

It follows that $\text{dist}_A(x) \leq d(x, y)$ for all $y \in \overline{A}$. Finally:

$$\begin{aligned} \text{dist}_A(x) &\leq \inf_{y \in \overline{A}} d(x, y) \\ &= \text{dist}_{\overline{A}}(x). \end{aligned}$$

Thus $\text{dist}_A = \text{dist}_{\overline{A}}$.

(3)

□

§ 2.5. Continuity

Definition 2.5.1. Let (X, d) and (Y, ρ) be metric spaces. A map $f : X \rightarrow Y$ is *continuous at* $x_0 \in X$ if any of the equivalent definitions are satisfied:

- (1) $(\forall \epsilon > 0)(\exists \delta > 0) : (\forall x \in X) \left(d(x, x_0) < \delta \implies \rho(f(x), f(x_0)) < \epsilon \right)$
- (2) $(\forall \epsilon > 0)(\exists \delta > 0) : (\forall x \in X) \left(x \in U_X(x_0, \delta) \implies f(x) \in U_Y(f(x_0), \epsilon) \right)$
- (3) $(\forall \epsilon > 0)(\exists \delta > 0) : f(U_X(x_0, \delta)) \subseteq U_Y(f(x_0), \epsilon)$

Proposition 2.5.1. Let $f : (X, d) \rightarrow (Y, \rho)$ be a map between metric spaces and $x_0 \in X$. The following are equivalent:

- (1) f is continuous at x_0 ;
- (2) $(\forall V \in \mathcal{N}_{f(x_0)})(\exists U \in \mathcal{N}_{x_0}) : f(U) \subseteq V$;
- (3) $(\forall (x_n)_n \in X^N)((x_n)_n \rightarrow x_0 \implies (f(x_n))_n \rightarrow f(x_0))$

Proof. (1) $\implies$ (2) follows from Definition 2.5.1.

(1) $\implies$ (3) Let $(x_n)_n \rightarrow x_0$. Let $\epsilon > 0$. Since f is continuous, find $\delta > 0$ so that $x \in U(x, \delta)$ implies $f(x) \in U(f(x_0), \epsilon)$. Whence $d(f(x_n), f(x_0)) < \epsilon$, establishing $(f(x_n))_n \rightarrow f(x_0)$.

(3) $\implies$ (1) We prove the contrapositive of this statement. If f is not continuous, choose $\epsilon_0 > 0$ so that $d(x_n, x_0) < \frac{1}{n}$ and $d(f(x_n), f(x_0)) \geq \epsilon_0$. Whence $(x_n)_n \rightarrow x_0$ and $(f(x_n))_n \not\rightarrow f(x_0)$. $\square$

Proposition 2.5.2. Let $f : (X, d) \rightarrow (Y, \rho)$ be a map of metric spaces. The following are equivalent:

- (1) f is continuous;
- (2) $(\forall V \in \tau_Y), f^{-1}(V) \in \tau_X$;

Proof. Let $V \subseteq Y$ be open. If $f^{-1}(V) = \emptyset$, we're done. If not, let $x \in f^{-1}(V)$. Then $f(x) \in V$. Since V is open, find $\epsilon > 0$ so that $U(f(x), \epsilon) \subseteq V$. Since f is continuous, find $\delta > 0$ so that $f(U(x, \delta)) \subseteq U(f(x), \epsilon) \subseteq V$. Whence $U(x, \delta) \subseteq f^{-1}(V)$.

Let $x \in X$ and $\epsilon > 0$. Since $U(f(x), \epsilon) \in \tau_Y$, we have $f^{-1}(U(f(x), \epsilon)) \in \tau_X$. Note that $x \in f^{-1}(U(f(x), \epsilon))$. Since this set is open, find $\delta > 0$ so that $U(x, \delta) \subseteq f^{-1}(U(f(x), \epsilon))$. Thus $f(U(x, \delta)) \subseteq U(f(x), \epsilon)$; i.e., f is continuous. $\square$

Proposition 2.5.3. *** Let $(X, d) \xrightarrow{f} (Y, \rho) \xrightarrow{g} (Z, \gamma)$ be maps of metric spaces. If f is continuous at $x \in X$ and g is continuous at $y = f(x)$, then $g \circ f$ is continuous at x .

Proof. Exercise. $\square$

Proposition 2.5.4. Let (X, d) be a metric space with $A \subseteq X$ dense. Let $f : X \rightarrow F$ be a continuous and bounded function. Then $\sup_{x \in A} f(x) = \sup_{x \in X} f(x)$.

Proof. Since $A \subseteq X$ we have $\sup_{x \in A} f(x) \leq \sup_{x \in X} f(x)$. Conversely, let $\epsilon > 0$. Find $x' \in X$ so that $\sup_{x \in X} f(x) - \frac{\epsilon}{2} < f(x')$. Since f is continuous, find $\delta > 0$ so that for all $x \in X$, $d(x, x') < \delta$ implies $|f(x) - f(x')| < \frac{\epsilon}{2}$. Since A is dense, find $a \in A$ so that $d(x', a) < \delta$. This implies

$|f(x') - f(a)| < \frac{\epsilon}{2}$, or equivalently $f(x') - \frac{\epsilon}{2} < f(a)$. This gives:

$$\begin{aligned} \sup_{x \in X} f(x) - \frac{\epsilon}{2} &< f(x') \\ &< f(a) + \frac{\epsilon}{2}. \end{aligned}$$

Upon simplifying, we have:

$$\sup_{x \in X} f(x) - \epsilon < f(a).$$

Hence for $\epsilon > 0$ we have:

$$\sup_{x \in X} f(x) < \sup_{x \in A} f(x) + \epsilon.$$

Taking $\epsilon \rightarrow 0$ gives $\sup_{x \in X} f(x) \leq \sup_{x \in A} f(x)$. □

Definition 2.5.2. Let $f : (X, d) \rightarrow (Y, \rho)$ be a map of metric spaces.

(1) We say f is *uniformly continuous* if:

$$(\forall \epsilon > 0)(\exists \delta > 0) : (\forall x, y \in X)(d(x, y) < \delta \implies \rho(f(x), f(y)) < \epsilon).$$

(2) We say f is *Lipschitz* if:

$$(\exists C > 0) : (\forall x, y \in X)(\rho(f(x), f(y)) \leq C d(x, y)).$$

If $C < 1$, we say f is *contractive*.

(3) We say f is an *isometry* if:

$$(\forall x, x' \in X)(\rho(f(x), f(x')) = d(x, x'))$$

Exercise 2.5.1. *** Show that Lipschitz implies uniform continuity. Show that uniform continuity implies continuity. Show that the converse direction fails in general.

Example 2.5.1. Let $(V, \|\cdot\|)$ be a normed space. Then $V \xrightarrow{\|\cdot\|} [0, \infty)$ is continuous. Indeed, we have $||\|v\| - \|w\|| \leq \|v - w\|$, so $\|\cdot\|$ is Lipschitz.

Example 2.5.2. Let (X, d) be a metric space and equip $X \times X$ with the product metric D_1 . Claim: $d : X \times X \rightarrow [0, \infty)$ is continuous. Indeed, given $(x, y), (x', y') \in X \times X$, then we have:

$$d(x, y) \leq d(x, x') + d(x', y') + d(y', y).$$

So we have $d(x, y) - d(x', y') \leq d(x, x') + d(y', y)$. But this is equivalent to $|d(x, y) - d(x', y')| \leq D_1((x, y), (x', y'))$. So d is Lipschitz.

Example 2.5.3. If (X, d) is a metric space and $A \subseteq X$, then $\text{dist}_A : X \rightarrow [0, \infty)$ is continuous. We've shown that $|\text{dist}_A(x) - \text{dist}_A(y)| \leq d(x, y)$, so dist_A is Lipschitz.

Definition 2.5.3. Let X be a topological space. We say X is *normal* (or T_4) if, given any disjoint closed sets $E, F \subseteq X$, there are neighbourhoods U

Definition 2.5.4. Let X be a topological space. We say X is *normal* (or T_4) if, for any $A, B \subseteq X$ closed satisfying $A \cap B = \emptyset$, then there exists $U, V \in \tau_X$ with $A \subseteq U$, $B \subseteq V$ satisfying $U \cap V = \emptyset$.

Proposition 2.5.5. *Metric spaces are normal.*

Proof. Let $A, B \subseteq (X, d)$ with $A \cap B = \emptyset$. Define $f : (X, d) \rightarrow \mathbb{R}$ by:

$$f(x) = \frac{\text{dist}_A(x)}{\text{dist}_A(x) + \text{dist}_B(x)}.$$

Then f is continuous. Moreover, define:

$$\begin{aligned} U &:= f^{-1}\left(\left(-\frac{1}{2}, \frac{1}{2}\right)\right) \\ V &:= f^{-1}\left(\left(-\frac{1}{2}, \frac{3}{2}\right)\right) \end{aligned}$$

Then U and V open with $U \cap V = \emptyset$. □

Proposition 2.5.6. *Let V and W be normed spaces and $T : V \rightarrow W$ linear. The following are equivalent:*

- (1) T is continuous at 0_V ;
- (2) T is continuous;
- (3) T is uniformly continuous;
- (4) T is Lipschitz;
- (5) there exists $C \geq 0$ such that $\|Tv\| \leq C\|v\|$ for all $v \in V$;
- (6) T is bounded.

Proof. We will show $(6) \Leftrightarrow (5) \Rightarrow (4) \Rightarrow (3) \Rightarrow (2) \Rightarrow (1) \Rightarrow (5)$. Let T be bounded. Given $v \in V$, $v \neq 0$, we have:

$$\begin{aligned} \|T\|_{\text{op}} &= \sup_{v \in B_V} \|Tv\| \\ &\geq \left\| T \frac{v}{\|v\|} \right\| \quad \text{for all } v \in V \\ &= \frac{1}{\|v\|} \|Tv\| \quad \text{for all } v \in V. \end{aligned}$$

Thus $\|Tv\| \leq \|T\|_{\text{op}} \|v\|$ for all $v \in V$. The converse is clear by inspection.

Suppose there exists $C \geq 0$ satisfying (5). We have that $\|Tv - Tv'\| \leq C\|v - v'\|$. Thus T is Lipschitz.

Let T be Lipschitz. Let $\epsilon > 0$ and find $\delta = \frac{\epsilon}{C}$. Then $\|v - v'\| < \delta$ implies:

$$\begin{aligned} \|Tv - Tv'\| &\leq C\|v - v'\| \\ &< C\frac{\epsilon}{C} \\ &= \epsilon. \end{aligned}$$

Thus T is uniformly continuous.

Suppose that T be uniformly continuous. Fix $x \in V$. Given $\epsilon > 0$, we can find $\delta > 0$ so that $\|v - x\| < \delta$ implies $\|Tv - Tx\| < \epsilon$. Thus T is continuous at $x \in V$. Since x was arbitrary, T is continuous. Moreover, T will be continuous at 0_V , establishing (1).

We will now show (1) implies (5). Let $\epsilon = 1$. We can find a $\delta > 0$ such that $T(U(0, \delta)) \subseteq U(0, 1)$. If $v \in V$, $v \neq 0$, then $\frac{\delta v}{2\|v\|} \in U(0, \delta)$. Since T is continuous at 0 , we have $T\frac{\delta v}{2\|v\|} \in U(0, 1)$ ¹. This gives $\left\|T\frac{\delta v}{2\|v\|}\right\| < 1$, which is equivalent to $\|Tv\| < \frac{2}{\delta}\|v\|$. This establishes (5). $\square$

Corollary 2.5.7. *Let V be a normed space with $\dim(V) = n$. If $T : \ell_p^n \rightarrow V$ is linear, then T is continuous.*

Proof. Let $\mathcal{B} = \{e_1, \dots, e_n\}$ be a basis for ℓ_p^n . Let $v \in V$. Then $v = \sum_{j=1}^n \alpha_j e_j$ where each $\alpha_j \in \ell_p$. Then:

$$\begin{aligned} \|Tv\| &= \left\|T\left(\sum_{j=1}^n \alpha_j e_j\right)\right\| \\ &= \left\|\sum_{j=1}^n \alpha_j Te_j\right\| \\ &\leq \sum_{j=1}^n |\alpha_j| \|Te_j\|. \end{aligned}$$

Let $c = \max_{j=1}^n \|Te_j\|$. We have:

$$\begin{aligned} \sum_{j=1}^n |\alpha_j| \|Te_j\| &\leq c \sum_{j=1}^n |\alpha_j| \\ &= c \left\|\sum_{j=1}^n \alpha_j e_j\right\|_1. \end{aligned}$$

We showed that all norms are equivalent in Theorem 1.3.7. So there exists $c' > 0$ such that $\|\cdot\|_1 \leq \|\cdot\|_p$. Thus:

$$\begin{aligned} \|Tv\| &\leq c \cdot c' \left\|\sum_{j=1}^n \alpha_j e_j\right\|_p \\ &= c \cdot c' \|v\|_p. \end{aligned}$$

By Proposition 2.5.6, we have that T is continuous. $\square$

Proposition 2.5.8. *Let (X, d) be a metric space with $A \subseteq X$ dense. If $f, g : X \rightarrow (Y, \rho)$ are continuous with $f(a) = g(a)$ for all $a \in A$, then $f = g$.*

¹Recall that $T(0) = 0$.

Proof. If $x \in X$, we can find a sequence $(a_n)_n$ in A with $(a_n)_n \xrightarrow{d} x$. Then:

$$\begin{aligned} (f(a_n))_n &\rightarrow f(x) \\ \parallel \\ (g(a_n))_n &\rightarrow g(x). \end{aligned}$$

Thus $f(x) = g(x)$. □

Definition 2.5.5. Let (X, d) and (Y, ρ) be metric spaces and $f : X \rightarrow Y$.

- (1) f is a *homeomorphism* if f is bijective with f and f^{-1} continuous. If such an f exists, we say $X \cong Y$ are *homeomorphic*.
- (2) f is a *uniformism* if f is bijective with f and f^{-1} uniformly continuous. If such an f exists, we say $X \cong Y$ are *uniformly isomorphic*.
- (3) f is an *metric isomorphism* if f is bijective with f and f^{-1} Lipschitz. If such an f exists, we say $X \cong Y$ are *metrically isomorphic*.
- (4) f is an *isometric isomorphism* if f and f^{-1} are isometries. We say $X \cong Y$ are *isometrically isomorphic*.

Example 2.5.4. $(0, 1) \cong \mathbb{R}$ are homeomorphic, but not uniformly isomorphic.

Example 2.5.5. *** Let $a = (a_k)_k \in \ell_1$. Define $\varphi_a : c_0 \rightarrow F$ by $\varphi_a(z) = \sum_{k \geq 1} a_k z_k$. This series converges since:

Definition 2.5.6. Let X be a set with two metrics d_1 and d_2 .

- (1) d_1 and d_2 are *metrically equivalent* if $\text{id} : (X, d_1) \rightarrow (X, d_2)$ and $\text{id}^{-1} : (X, d_2) \rightarrow (X, d_1)$ are Lipschitz.
- (2) d_1 and d_2 are *uniformly equivalent* if $\text{id} : (X, d_1) \rightarrow (X, d_2)$ and $\text{id}^{-1} : (X, d_2) \rightarrow (X, d_1)$ are uniformisms.
- (3) d_1 and d_2 are *topologically equivalent* if $\text{id} : (X, d_1) \rightarrow (X, d_2)$ and $\text{id}^{-1} : (X, d_2) \rightarrow (X, d_1)$ are homeomorphisms.

Example 2.5.6. ***

§ 2.6. Completeness

Definition 2.6.1. A sequence $(x_n)_n$ in a metric space (X, d) is *d-Cauchy* if:

$$(\forall \epsilon > 0)(\exists N \in \mathbb{N}) : (\forall p, q \in \mathbb{N})(p, q \geq N \implies d(x_p, x_q) < \epsilon).$$

Proposition 2.6.1. Let (x_n) be a sequence in (X, d) .

- (1) If $(x_n)_n$ converges, then $(x_n)_n$ is Cauchy.
- (2) If $(x_n)_n$ is Cauchy, then $(x_n)_n$ is bounded.

Proof. (1) Let $x \in X$ and suppose $(x_n)_n \rightarrow x$. Let $\epsilon > 0$. Find N large so that for $p \geq N$ we have $d(x_p, x) < \frac{\epsilon}{2}$. Then $p, q \geq N$ implies:

$$\begin{aligned} d(x_p, x_q) &\leq d(x_p, x) + d(x, x_q) \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

(2) Let $\epsilon = 1$. Find N large so $p, q \geq N$ implies $d(x_p, x_q) < 1$. Let $C = \max_{1 \leq p, q \leq N} d(x_p, x_q)$. Without loss of generality, if $p \geq N$ and $q \leq N$, then $d(x_p, x_q) \leq d(x_p, x_N) + d(x_N, x_q) < 1 + C$. Set $K = \max\{1, 1 + C\}$. Then $\text{diam}(\{x_n\}_{n \geq 1}) = \sup_{p, q \geq 1} d(x_p, x_q) < K$. $\square$

Proposition 2.6.2. *Let $(x_n)_n$ be a Cauchy sequence in X and suppose there exists a subsequence $(x_{n_k})_k$ converging to $x \in X$. Then $(x_n)_n$ converges to x .*

Proof. Let $\epsilon > 0$. Since $(x_n)_n$ is Cauchy, there exists N large so $n, n_k \geq N$ implies $d(x_n, x_{n_k}) < \frac{\epsilon}{2}$. This gives:

$$\begin{aligned} d(x_n, x) &= d(x_n, \lim_{k \rightarrow \infty} x_{n_k}) \\ &= \lim_{k \rightarrow \infty} d(x_n, x_{n_k}) \\ &\leq \frac{\epsilon}{2} \\ &< \epsilon. \end{aligned}$$

$\square$

Definition 2.6.2. A metric space is said to be *complete* if every Cauchy sequence converges. A complete normed space is called a *Banach space*. A complete inner product space is called a *Hilbert space*.

Lemma 2.6.3. *Let $f : (X, d) \rightarrow (Y, \rho)$ be uniformly continuous. If $(x_n)_n$ is d -Cauchy then $(f(x_n))_n$ is ρ -Cauchy.*

Proof. Let $\epsilon > 0$. Find $\delta > 0$ so that $d(x, x') < \delta$ implies $\rho(f(x), f(x')) < \epsilon$. Pick N sufficiently large so that $p, q \geq N$ implies $d(x_p, x_q) < \delta$. This gives $\rho(f(x_p), f(x_q)) < \epsilon$, whence $(f(x_n))_n$ is ρ -Cauchy. $\square$

Corollary 2.6.4. *If $f : (X, d) \rightarrow (Y, \rho)$ is a uniformism, then (X, d) is complete if and only if (Y, ρ) is complete.*

Proof. Let (X, d) be complete. If $(y_n)_n$ is ρ -Cauchy, then $(f^{-1}(y_n))_n$ is d -Cauchy in X . So we can find some $x \in X$ such that $(f^{-1}(y_n))_n \rightarrow x$. Then $(f(f^{-1}(y_n)))_n = (y_n)_n \rightarrow f(x)$. The converse follows similarly. $\square$

Corollary 2.6.5. *If d_1 and d_2 are uniformly equivalent metrics on a set X , then (X, d_1) is complete if and only if (X, d_2) is complete.*

Proof. Since the map $\text{id} : (X, d_1) \rightarrow (X, d_2)$ is a uniformism, the previous corollary gives that (X, d_1) is complete if and only if (X, d_2) is complete. $\square$

Proposition 2.6.6. ℓ_p^d is a Banach space for $1 \leq p \leq \infty$.

Proof. We only need to show this for ℓ_∞^d since all the p -norms are equivalent. Let $(x_n)_n$ be $\|\cdot\|_\infty$ -Cauchy in ℓ_∞^d . Let $\epsilon > 0$. Find N large so for $n, m \geq N$ we have $\|x_n - x_m\|_\infty < \epsilon$. Observe that:

$$\begin{aligned} |x_n(k) - x_m(k)| &\leq \max_{1 \leq k \leq d} |x_n(k) - x_m(k)| \\ &= \|x_n - x_m\|_\infty \\ &< \epsilon, \end{aligned}$$

where $x_n(k)$ is the k^{th} entry of the d -tuple x_n . So for each $k = 1, \dots, d$, we know $(x_n(k))_n$ is Cauchy in F . Set $\lim_{n \rightarrow \infty} x_n(k) = x(k)$ for $k = 1, \dots, d$. This gives:

$$\|x - x_n\|_\infty = \max_{1 \leq k \leq d} |x(k) - x_n(k)| \xrightarrow{n \rightarrow \infty} 0.$$

Whence $(x_n)_n \rightarrow x$ in ℓ_∞^d . Now set $y = (y_1, \dots, y_n)$. We have:

$$\begin{aligned} \|x_n - y\|_p &\leq c' \|x_n - y\|_1 \\ &= c' \sum_{j=1}^d |x_n(j) - y_j| \\ &\leq c' \max_{1 \leq j \leq d} |x_n(j) - y_j| \xrightarrow{n \rightarrow \infty} 0. \end{aligned}$$

Thus ℓ_p^d is complete. □

Proposition 2.6.7. ℓ_p is a Banach space for $1 \leq p \leq \infty$.

Proof. Suppose that $(f_n)_n$ is $\|\cdot\|_{\ell_p}$ -Cauchy. Observe that

$$\begin{aligned} |f_n(k) - f_m(k)|^p &\leq \sum_{j=1}^{\infty} |f_n(j) - f_m(j)|^p \\ &= \|f_n - f_m\|_{\ell_p}^p \end{aligned}$$

So $(f_n(k))_n$ is Cauchy in F . Since this space is complete, define $\lim_{n \rightarrow \infty} f_n(k) := f(k)$.

Our goal is to find some function $f : \mathbb{N} \rightarrow F$ satisfying $f \in \ell_p$ and $\|f_n - f\|_{\ell_p} \rightarrow 0$. The $f(k)$ we've just obtained will lead us to the most suitable candidate.

Since $(f_n)_n$ is $\|\cdot\|_{\ell_p}$ -Cauchy, it is bounded by some constant. Fix $K \geq 1$ and observe that:

$$\begin{aligned} \sum_{k=1}^K |f(k)|^p &= \sum_{k=1}^K \left| \lim_{n \rightarrow \infty} f_n(k) \right|^p \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^K |f_n(k)|^p \\ &\leq \sup_{n \geq 1} \|f_n\|_{\ell_p}^p \\ &:= C. \end{aligned}$$

Since the sequence $\left(\sum_{k=1}^K |f(k)|^p\right)_{K=1}^\infty$ is increasing and bounded above C , the Monotone Convergence Theorem says that its limit exists. We obtain:

$$\begin{aligned} \lim_{K \rightarrow \infty} \sum_{k=1}^K |f(k)|^p &= \sum_{k=1}^{\infty} |f(k)|^p \\ &= \|f\|_{\ell_p}^p \\ &< \infty. \end{aligned}$$

Thus $f \in \ell_p$.

It remains to show that $(f_n)_n$ converges to f . Given $\epsilon > 0$, find N large so that $n, m \geq N$ implies $\|f_n - f_m\|_p < \epsilon$. For every $n, m \geq N$ we have:

$$\begin{aligned} \sum_{k=1}^K |f_m(k) - f_n(k)|^p &\leq \|f_m - f_n\|_{\ell_p}^p \\ &< \epsilon^p. \end{aligned}$$

Taking the limit as $m \rightarrow \infty$ and considering all $n \geq N$ gives:

$$\sum_{k=1}^K |f(k) - f_n(k)|^p < \epsilon^p.$$

Finally, taking the limit as $K \rightarrow \infty$ and simplifying gives $\|f - f_n\|_{\ell_p} < \epsilon$. Thus ℓ_p is complete. $\square$

Proposition 2.6.8. *Let (Y, d) be a complete metric space. The set of bounded functions $\text{Bd}(\Omega, Y)$ with $\|\cdot\|_u$ is complete.*

Proof. Let $(f_n)_n$ be D_u -Cauchy. Fix $x, x' \in \Omega$ and let $\epsilon > 0$. Find N large so that $n, m \geq N$ implies:

$$d(f_n(x), f_m(x)) \leq D_u(f_n, f_m) < \epsilon.$$

Thus $(f_n(x))_n$ is Cauchy in Y . Since Y is complete, define $\lim_{n \rightarrow \infty} f_n(x) := f(x)$. Now find N_1 large so $n \geq N_1$ implies $d(f(x), f_n(x)) < \epsilon$. Find N_2 large so $n \geq N_2$ implies $d(f(x'), f_n(x')) < \epsilon$. Observe that:

$$\begin{aligned} d(f(x), f(x')) &\leq d(f(x), f_{N_1}(x)) + d(f_{N_1}(x), f_{N_2}(x')) + d(f_{N_2}(x'), f(x')) \\ &< 2\epsilon + d(f_{N_1}(x), f_{N_2}(x')) \\ &\leq 2\epsilon + d(f_{N_1}(x), f_{N_2}(x)) + d(f_{N_2}(x), f_{N_2}(x')) \\ &\leq 2\epsilon + \sup_{n, m \geq 1} d(f_n(x), f_m(x)) + \sup_{x, x' \in \Omega} d(f_{N_2}(x), f_{N_2}(x')) \\ &= 2\epsilon + \text{diam}(\{f_n(x)\}_{n \geq 1}) + \text{diam}(f_{N_2}(\Omega)). \end{aligned}$$

Note that the sequence $(f_n(x))_n$ is bounded because it is Cauchy —so there exists $C_1 \geq 0$ such that $\text{diam}(\{f_n(x)\}_{n \geq 1}) < C_1$. Moreover, since $f_{N_2} \in \text{Bd}(\Omega, Y)$, there exists $C_2 \geq 0$ such that $\text{diam}(f_{N_2}(\Omega)) < C_2$. This gives:

$$d(f(x), f(x')) < 2\epsilon + C_1 + C_2.$$

Since this inequality is independent of any n , and since $x, x' \in \Omega$ was arbitrary, we have that $\text{diam}(f(\Omega)) < \infty$; i.e., $f \in \text{Bd}(\Omega, Y)$. With the same ϵ as above, find N_3 large so that for all $n, m \geq N_3$ then $D_u(f_n, f_m) < \frac{\epsilon}{2}$. We know:

$$d(f_n(x), f_m(x)) \leq D_u(f_n, f_m) < \frac{\epsilon}{2}.$$

Taking $m \rightarrow \infty$ gives $d(f_n(x), f(x)) \leq \frac{\epsilon}{2}$ for all $n \geq N$. But note that N does not depend on our fixed $x \in \Omega$. It follows that $D_u(f_n, f) \leq \frac{\epsilon}{2} < \epsilon$. Thus $(f_n)_n \rightarrow f$, establishing $\text{Bd}(\Omega, Y)$ as complete. $\square$

Corollary 2.6.9. $\ell_\infty(\Omega)$ is complete.

Proposition 2.6.10. Let (X, d) be a complete metric space and $Y \subseteq X$. Y is complete if and only if Y is closed.

Proof. Let $(y_n)_n$ be a sequence in Y converging to $x \in X$. Then $(y_n)_n$ is sequence in X . Since X is complete, $(y_n)_n$ is Cauchy. Since Y is complete, $(y_n)_n$ must converge to some $y \in Y$. Since sequences can have at most one limit, it must be that $y = x$. Thus $x \in Y$; i.e., Y is closed.

Conversely, if $(y_n)_n$ is Cauchy in Y , then it is Cauchy in X . Since X is complete, there exists $x \in X$ with $(y_n)_n \rightarrow x$. Since Y is closed, $x \in Y$. Thus Y is complete. $\square$

Proposition 2.6.10 and Proposition 2.4.6 are extremely useful tools for showing a space is complete.

Corollary 2.6.11. Let (X, d) and (Y, ρ) be metric spaces.

- (1) $C_b(X, Y) := C(X, Y) \cap \text{Bd}(X, Y)$ is D_u -complete.
- (2) $C_b(X)$ is a $\|\cdot\|_u$ -Banach space.
- (3) $C_0(\mathbb{R})$ is a $\|\cdot\|_u$ -Banach space.

Proof. (1) Let $(f_n)_n$ be a sequence in $C_b(X, Y)$ converging to $f \in \text{Bd}(X, Y)$. Let $x \in X$ and $\epsilon > 0$. Find N large so that $D_u(f_N, f) < \frac{\epsilon}{3}$. Find $\delta > 0$ so that for all $x' \in X$, $d(x, x') < \delta$ implies $\rho(f_N(x), f_N(x')) < \frac{\epsilon}{3}$. For $d(x, x') < \delta$:

$$\begin{aligned} \rho(f(x), f(x')) &\leq \rho(f(x), f_N(x)) + \rho(f_N(x), f_N(x')) + \rho(f_N(x'), f(x')) \\ &\leq 2D_u(f_N, f) + \rho(f_N(x), f_N(x')) \\ &< \frac{2\epsilon}{3} + \frac{\epsilon}{3} \\ &= \epsilon. \end{aligned}$$

Thus $f \in C_b(X, Y)$ because it is bounded and continuous. Since $C_b(X, Y) \subseteq \text{Bd}(X, Y)$ is closed, it is complete.

(2) As we've just shown, the space $C_b(X)$ is complete. It only remains to show it is a vector space.

(3) Let $(f_n)_n$ be a sequence in $C_0(\mathbb{R})$ converging to $f \in C_b(\mathbb{R})$. Let $\epsilon > 0$ and find N large so that $\|f - f_N\|_u < \frac{\epsilon}{2}$. Since $f_N \in C_0(\mathbb{R})$, we know that $\lim_{x \rightarrow \pm\infty} f_N(x) = 0$. So there

exists $M > 0$ with $|x| \geq M$ implying $|f_N(x)| < \frac{\epsilon}{2}$. For $|x| \geq M$ observe that:

$$\begin{aligned} |f(x)| &\leq |f(x) - f_N(x)| + |f_N(x)| \\ &\leq \|f - f_N\|_{\infty} + |f_N(x)| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

Thus $\lim_{x \rightarrow \infty} f(x) = 0$; i.e., $f \in C_0(\mathbb{R})$. Since $C_0(\mathbb{R}) \subseteq C_b(\mathbb{R})$ is closed, it is complete. $\square$

Proposition 2.6.12. *Let V be a normed space and W a Banach space. $B(V, W)$ with $\|\cdot\|_{\text{op}}$ is a Banach space.*

Proof. Let $(T_n)_n$ be $\|\cdot\|_{\text{op}}$ -Cauchy. Let $v \in V$ and $\epsilon > 0$. Find N_1 large so that $n, m \geq N_1$ implies $\|T_n - T_m\|_{\text{op}} < \frac{\epsilon}{\|v\|}$. We can see:

$$\begin{aligned} \|T_n v - T_m v\|_W &= \|(T_n - T_m)v\|_W \\ &\leq \|T_n - T_m\|_{\text{op}} \|v\| \\ &< \frac{\epsilon}{\|v\|} \cdot \|v\| \\ &= \epsilon. \end{aligned}$$

So $(T_n v)_n$ is $\|\cdot\|_W$ -Cauchy. Since W is a Banach space, define $\lim_{n \rightarrow \infty} T_n v = Tv$. We must show that T is linear, bounded, and $\|T_n - T\|_{\text{op}} \rightarrow 0$. Given $v_1, v_2 \in V$ and $c \in F$ we can see:

$$\begin{aligned} T(v_1 + cv_2) &= \lim_{n \rightarrow \infty} T_n(v_1 + cv_2) \\ &= \lim_{n \rightarrow \infty} T_n v_1 + c T_n v_2 \\ &= \lim_{n \rightarrow \infty} T_n v_1 + c \lim_{n \rightarrow \infty} T_n v_2 \\ &= Tv_1 + cTv_2. \end{aligned}$$

Thus T is linear. Now since $(T_n)_n$ is $\|\cdot\|_{\text{op}}$ -Cauchy, it is bounded, so there exists $C > 0$ with $\|T_n\|_{\text{op}} \leq C$ for all $n \geq 1$. Using the fact norms are continuous, we have:

$$\begin{aligned} \|Tv\|_W &= \left\| \lim_{n \rightarrow \infty} T_n v \right\|_W \\ &= \lim_{n \rightarrow \infty} \|T_n v\|_W \\ &\leq \limsup_{n \rightarrow \infty} \|T_n\|_{\text{op}} \|v\| \\ &\leq C \|v\|. \end{aligned}$$

Thus $T \in B(V, W)$. With the same epsilon as before, find N_2 so that $n, m \geq N_2$ implies $\|T_n - T_m\|_{\text{op}} < \frac{\epsilon}{2}$. We can show:

$$\|T_n v - T_m v\|_W \leq \|T_n - T_m\|_{\text{op}} \|v\| < \frac{\epsilon}{2}.$$

Taking $m \rightarrow \infty$ gives:

$$\|T_n v - T v\|_W \leq \frac{\epsilon}{2}.$$

Taking the supremum over all $v \in B_V$ gives:

$$\|T_n - T\|_{\text{op}} \leq \frac{\epsilon}{2} < \epsilon.$$

Thus $B(V, W)$ is complete. $\square$

Proposition 2.6.13. *Let $(V, \|\cdot\|)$ be a normed space. The following are equivalent:*

- (1) V is a Banach space;
- (2) If $(v_k)_k$ is a sequence in V with $\sum_{k=1}^{\infty} \|v_k\|$ convergent, then $\sum_{k=1}^{\infty} v_k$ converges.

Proof. Suppose V is a Banach space. Let $s_n = \sum_{k=1}^n v_k$ and $t_n = \sum_{k=1}^n \|v_k\|$. For $p > q > 1$:

$$\begin{aligned} \|s_p - s_q\| &= \left\| \sum_{k=q+1}^p v_k \right\| \\ &\leq \sum_{k=q+1}^p \|v_k\| \\ &= |t_p - t_q|. \end{aligned}$$

Since $(t_n)_n$ is convergent, it is Cauchy. So $(s_n)_n$ is Cauchy, implying it is convergent. Thus $\sum_{k=1}^{\infty} v_k$ converges.

Now let $(v_n)_n$ be Cauchy. Find $n_1 \in \mathbb{N}$ such that $p, q \geq n_1$ implies $\|v_p - v_q\| < 2^{-1}$. Find $n_2 > n_1$ such that $p, q \geq n_2$ implies $\|v_p - v_q\| < 2^{-2}$. Inductively, find $n_k > n_{k-1}$ such that $p, q \geq n_k$ implies $\|v_p - v_q\| < 2^{-k}$. Consider the sequence $(v_{n_{k+1}} - v_{n_k})_k$. Then:

$$\sum_{k=1}^{\infty} \|v_{n_{k+1}} - v_{n_k}\| \leq \sum_{k=1}^{\infty} 2^{-k} = 1.$$

By our hypothesis, $\sum_{k=1}^{\infty} (v_{n_{k+1}} - v_{n_k})$ converges. Let $(w_m)_m$ denote $\sum_{k=1}^m (v_{n_{k+1}} - v_{n_k})$ sequence of partial sums. Then each w_m telescopes as follows:

$$\begin{aligned} w_m &= \sum_{k=1}^m v_{n_{k+1}} - v_{n_k} \\ &= v_{n_{m+1}} - v_{n_1} \end{aligned}$$

In particular, since $(v_n)_n$ also converges to some $w \in V$ as $m \rightarrow \infty$. However, notice:

$$\begin{aligned} (v_{n_m})_m &= (v_{n_m} - v_{n_1})_m + v_{n_1} \\ &\xrightarrow{m \rightarrow \infty} w + v_{n_1}. \end{aligned}$$

Since $(v_n)_n$ is a Cauchy sequence which admits a convergent subsequence, $(v_n)_n$ converges. Thus V is a Banach space. $\square$

Example 2.6.1. *** Let $\mathcal{H}$ be a Hilbert space. Suppose $(e_n)_n$ is an orthonormal sequence in $\mathcal{H}$ and $(t_k)_k \in \ell_2$. We will show $\sum_{k=1}^{\infty} t_k e_k$ converges in $\mathcal{H}$, but not absolutely in general.

Let $s_n = \sum_{k=1}^n t_k e_k$. For $n > m > 1$:

$$\begin{aligned} \|s_n - s_m\|^2 &= \left\| \sum_{k=m+1}^n t_k e_k \right\|^2 \\ &= \sum_{k=m+1}^n |t_k|^2. \end{aligned}$$

Since $(\sum_{k=1}^n |t_k|^2)_n$ is Cauchy, then $(s_n)_n$ is Cauchy, whence $\sum_{k=1}^{\infty} t_k e_k$ converges. But notice $\|s_n\|^2 = \sum_{k=1}^n |t_k|^2$. As $n \rightarrow \infty$, we see $\left\| \sum_{k=1}^{\infty} t_k e_k \right\|^2 = \sum_{k=1}^{\infty} |t_k|^2$.

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Recall that if $f : (X, d) \rightarrow (Y, \rho)$ is a uniformly continuous map between metric spaces and $(x_n)_n$ is Cauchy, then $(f(x_n))_n$ is Cauchy. Complete spaces have the unique property that, given a map from a dense subset $A \rightarrow Y$, we can define an extension of such function.

Theorem 2.6.14. *Let (X, d) and (Y, ρ) be metric spaces with Y complete. Suppose $A \subseteq X$ is dense and $f : A \rightarrow Y$ is uniformly continuous. There exists a unique uniformly continuous map $\tilde{f} : X \rightarrow Y$ with $\tilde{f}(x) = f(x)$ for all $x \in A$.*

Proof. Let $x \in X$. We know there exists a sequence $(a_n)_n$ in A with $(a_n)_n \rightarrow x$. Since $(a_n)_n$ is convergent, it is Cauchy, so $(f(a_n))_n$ is also Cauchy. By the completeness of Y , $(f(a_n))_n$ converges. Define $\tilde{f}(x) := \lim_{n \rightarrow \infty} f(a_n)$.

We must show this extension is well-defined. Suppose $(b_n)_n$ is another sequence in A with $(b_n)_n \rightarrow x$. Then the mixed sequence $(a_1, b_1, a_2, b_2, \dots)$ will converge to x . The same reasoning as above tells us $(f(a_1), f(b_1), f(a_2), f(b_2), \dots)$ converges in Y . The two subsequences $(f(a_n))_n$ and $(f(b_n))_n$ must then converge to the same limit.

We will now show that $\tilde{f}$ is uniformly continuous. Let $\epsilon > 0$. Find $\delta > 0$ so that for all $a, b \in A$, $d(a, b) < \delta$ implies $\rho(f(a), f(b)) < \frac{\epsilon}{2}$. Now let $x, x' \in X$ with $d(x, x') < \frac{\delta}{4}$. Find sequences $(a_n)_n$ and $(b_n)_n$ in A with $(a_n)_n \rightarrow x$ and $(b_n)_n \rightarrow x'$. Find N large so that $n \geq N$ implies $d(a_n, x) < \frac{\delta}{4}$ and $d(b_n, x') < \frac{\delta}{4}$. The triangle inequality gives $d(a_n, b_n) < \frac{3\delta}{4} < \delta$. So $\rho(f(a_n), f(b_n)) < \frac{\epsilon}{2}$ for all $n \geq N$. Observe that:

$$\begin{aligned} \rho(\tilde{f}(x), \tilde{f}(x')) &= \lim_{n \rightarrow \infty} \rho(f(a_n), f(b_n)) \\ &\leq \frac{\epsilon}{2} \\ &< \epsilon. \end{aligned}$$

Thus $\tilde{f}$ is uniformly continuous. It remains to show that $\tilde{f}$ is unique. Suppose $g : X \rightarrow Y$ is also a continuous extension of f . Then $g(x) = f(x) = \tilde{f}(x)$ for all $x \in A$. Since g and $\tilde{f}$ agree on elements of a dense set, we must have $g = \tilde{f}$. $\square$

Proposition 2.6.15. *Let (X, d) and (Y, ρ) be metric spaces with $A \subseteq X$ dense, Y complete, and $f : A \rightarrow Y$ an isometry. Then the continuous extension $\tilde{f} : X \rightarrow Y$ is an isometry.*

Proof. Let $x, x' \in X$. Let $(a_n)_n$ and $(b_n)_n$ be sequences in A with $(a_n)_n \rightarrow x$ and $(b_n)_n \rightarrow x'$. We have:

$$\begin{aligned} \rho(\tilde{f}(x), \tilde{f}(x')) &= \lim_{n \rightarrow \infty} \rho(f(a_n), f(b_n)) \\ &= \lim_{n \rightarrow \infty} d(a_n, b_n) \\ &= d(x, x'). \end{aligned} \quad \square$$

Corollary 2.6.16. *Let V be a normed space, W a Banach space, and $U \subseteq V$ a dense linear subspace. Let $T_0 \in B(U, W)$. There exists a unique $T \in B(V, W)$ with $\|T\|_{\text{op}} = \|T_0\|_{\text{op}}$. Moreover, if T_0 is isometric, then so is T .*

Proof. We only need to show T is linear and $\|T\|_{\text{op}} = \|T_0\|_{\text{op}}$. Let $v, v' \in V$ and $\alpha \in F$. Let $(x_n)_n$ and $(y_n)_n$ be sequences in U with $(x_n)_n \rightarrow v$ and $(y_n)_n \rightarrow v'$. Observe that:

$$\begin{aligned} T(v + \alpha v') &= \lim_{n \rightarrow \infty} T_0(x_n + \alpha y_n) \\ &= \lim_{n \rightarrow \infty} T_0(x_n) + \alpha \lim_{n \rightarrow \infty} T_0(y_n) \\ &= T(v) + \alpha T(v'). \end{aligned}$$

Note that the composition $V \xrightarrow{T} W \xrightarrow{\|\cdot\|_W} F$ will be continuous and bounded, so by Proposition ?? we have:

$$\begin{aligned} \|T\|_{\text{op}} &= \sup_{v \in B_V} \|T(v)\| \\ &= \sup_{v \in B_U} \|T(v)\| \\ &= \sup_{v \in B_U} \|T_0(v)\| \\ &= \|T_0\|_{\text{op}}. \end{aligned}$$

Thus $T \in B(V, W)$. $\square$

Example 2.6.2. *** (Something about Hilbert spaces...)

Proposition 2.6.17. *Let $T : V \rightarrow W$ be a continuous linear map between normed spaces which is bounded below, that is, there is a $C > 0$ with $\|Tv\| \geq C\|v\|$ for all $v \in V$. If V is complete, then $\text{im}(T) \subseteq W$ is a closed subspace, and $V \cong \text{im}(T)$ are uniformly isomorphic.*

Proof. Let $(T(v_n))_n$ be a sequence in $\text{im}(T)$ converging to $w \in W$. Given ϵ , find N large so that $n \geq N$ implies $\|T(v_n) - w\| < \frac{C\epsilon}{2}$. For $n, m \geq N$, observe that:

$$\begin{aligned} \|v_n - v_m\| &\leq \frac{1}{C} \|T(v_n - v_m)\| \\ &= \frac{1}{C} \|T(v_n) - T(v_m)\| \\ &\leq \frac{1}{C} \|T(v_n) - w\| + \frac{1}{C} \|w - T(v_m)\| \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

Thus $(v_n)_n$ is Cauchy. Since V is complete, let $v_0 := \lim_{n \rightarrow \infty} v_n$. Since T is continuous, we can see $(T(v_n))_n \rightarrow T(v_0)$. It must be the case that $T(v_0) = w$; i.e., $w \in \text{im}(T)$. Thus $\text{im}(T)$ is a closed subspace.

Since T is continuous, there exists some $\alpha > 0$ such that $\|Tv\| \leq \alpha \|v\|$. Clearly if $v = 0$, then $Tv = 0$, implying that T is injective. Whence $V \cong \text{im}(T)$ as vector spaces. Since T is continuous, it is uniformly continuous, so it remains to show that $T^{-1} : \text{im}(T) \rightarrow V$ (which exists) is also continuous. Let $w \in \text{im}(T)$, then there exists $v \in V$ with $T(v) = w$. Observe that:

$$\begin{aligned} \|T^{-1}w\| &= \|T^{-1}(T(v))\| \\ &= \|v\| \\ &\leq \frac{1}{C} \|Tv\| \\ &= \frac{1}{C} \|w\|. \end{aligned}$$

Thus T is a uniformism. □

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Definition 2.6.3. Let (X, d) be a metric space. A *completion* of (X, d) is a pair $((Z, \rho), \iota)$ where:

- (1) (Z, ρ) is a complete metric space;
- (2) $\iota : X \hookrightarrow Z$ is an isometry;
- (3) $\overline{\iota(X)}^\rho = Z$.

Example 2.6.3. $(([0, 1], \|\cdot\|), \iota(t) = t)$ is a completion of $(0, 1)$.

Lemma 2.6.18. Let $f : (X, d) \rightarrow (Y, \rho)$ be an isometry between metric spaces. If X is complete, then $f(X) \subseteq Y$ is closed. In particular, if $f(X)$ is dense, then f is onto.

Proof. If $(f(x_n))_n \rightarrow y$ in Y , then $(f(x_n))_n$ is ρ -Cauchy. Since f is an isometry, $d(x_n, x_m) = \rho(f(x_n), f(x_m))$, so $(x_n)_n$ is d -Cauchy. Let $x \in X$ such that $(x_n)_n \rightarrow x$. Since f is continuous, we have $(f(x_n))_n \rightarrow f(x)$. It must be that $y = f(x) \in f(X)$.

Since we've just shown that $f(X)$ is closed, if it were also dense then $f(X) = \overline{f(X)}^\rho = Y$, whence f is surjective. □

Theorem 2.6.19. Let (X, d) be a metric space. If $((Z, \rho), \iota)$ and $((Z', \rho'), \jmath)$ are completions of X , there exists a unique isometric isomorphism $\varphi : Z \rightarrow Z'$ with $\varphi \circ \iota = \jmath$; that is, the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{\iota} & Z \\ & \searrow \jmath & \downarrow \varphi \\ & & Z' \end{array}$$

Proof. Let $z \in Z$. Since $\iota(X)$ is dense in Z , there exists a sequence $(\iota(x_n))_n$ in $\iota(X)$ such that $(\iota(x_n))_n \rightarrow z$. Then $(\iota(x_n))_n$ is ρ -Cauchy since it is convergent, and furthermore we have:

$$\begin{aligned} \rho'(\jmath(x_n), \jmath(x_m)) &= d(x_n, x_m) \\ &= \rho(\iota(x_n), \iota(x_m)). \end{aligned}$$

So if $(\iota(x_n))_n$ is ρ -Cauchy in Z , then $(j(x_n))_n$ is ρ' -Cauchy in Z' . Since $j(X)$ is dense, the limit of this sequence is in Z' . Define $\varphi(z) := \lim_{n \rightarrow \infty} j(x_n)$.

We will show that $\varphi : Z \rightarrow Z'$ is well-defined, that is, it does not depend on our particular choice of sequence. Let (y_n) be another sequence in X with $(y_n)_n \rightarrow z$. We can see that:

$$\begin{aligned} d(x_n, y_n) &= \rho(\iota(x_n), \iota(y_n)) \\ &\leq \rho(\iota(x_n), z) + \rho(z, \iota(y_n)) \\ &\rightarrow 0, \end{aligned}$$

which gives:

$$\begin{aligned} \rho'(j(y_n), \varphi(z)) &\leq \rho'(j(y_n), j(x_n)) + \rho'(j(x_n), \varphi(z)) \\ &= d(y_n, x_n) + \rho'(j(x_n), \varphi(z)) \\ &\rightarrow 0. \end{aligned}$$

Thus $(j(y_n))_n \rightarrow \varphi(z)$. Since $\lim_{n \rightarrow \infty} j(x_n) = \lim_{n \rightarrow \infty} j(y_n)$, φ is well-defined.

We will now show that φ is an isometric isomorphism. Given $z_1, z_2 \in Z$, let $(\iota(x_n))_n \rightarrow z_1$ and $(\iota(y_n))_n \rightarrow z_2$, corresponding to $(j(x_n))_n \rightarrow \varphi(z_1)$ and $(j(y_n))_n \rightarrow \varphi(z_2)$. Observe that:

$$\begin{aligned} \rho'(\varphi(z_1), \varphi(z_2)) &= \lim_{n \rightarrow \infty} \rho'(j(x_n), j(y_n)) \\ &= \lim_{n \rightarrow \infty} d(x_n, y_n) \\ &= \lim_{n \rightarrow \infty} \rho(\iota(x_n), \iota(y_n)) \\ &= \rho(z_1, z_2). \end{aligned}$$

Since φ is an isometry, it is injective. Chasing the above diagram will yield $\varphi \circ \iota = j$, so we have $j(X) = \varphi(\iota(X)) \subseteq \varphi(Z)$. Then $Z' = \overline{j(X)}^{\rho'} \subseteq \overline{\varphi(Z)}^{\rho'}$. Since Z is complete, by Lemma 2.6.18 the set $\varphi(Z)$ is closed, whence $Z' \subseteq \overline{\varphi(Z)}^{\rho'} = \varphi(Z)$. Thus φ is onto.

It remains to show that φ is unique. Let $\psi : Z \rightarrow Z'$ be another isometric isomorphism satisfying $\psi \circ \iota = j$. Given $z \in Z$, we can find a sequence $(\iota(x_n))_n$ in $\iota(X)$ such that $(\iota(x_n))_n \rightarrow z$. Observe that:

$$\begin{aligned} \varphi(z) &= \lim_{n \rightarrow \infty} \varphi(\iota(x_n)) \\ &= \lim_{n \rightarrow \infty} j(x_n) \\ &= \lim_{n \rightarrow \infty} \psi(\iota(x_n)) \\ &= \psi(z). \end{aligned}$$

Since $z \in Z$ was arbitrary, this proves $\varphi = \psi$. □

Lemma 2.6.20. *If (X, d) is a metric space and $i : (X, d) \rightarrow (Y, \rho)$ is an isometry into a complete metric space (Y, ρ) , then $\left(\overline{i(X)}^\rho, \rho, i\right)$ is a completion of X .*

Proof. The space $(\overline{i(X)}^\rho, \rho)$ is complete by Proposition 2.6.10. Clearly $i : X \rightarrow \overline{i(X)}^\rho$ is an isometry because $i(X)^\rho \subseteq Y$. □

Theorem 2.6.21. *Every metric space admits a unique completion up to isometric isomorphism.*

Proof. Uniqueness was shown in Theorem 2.6.19. Given (X, d) , consider the Banach space $(C_b(X), \|\cdot\|_u)$. By the previous lemma we only need to construct an isometry $X \xrightarrow{i} C_b(X)$.

Fix any $x_0 \in X$. Define $f_x : X \rightarrow F$ by $f_x(t) = d(t, x) - d(t, x_0)$. Clearly f_x is continuous, and it is bounded because $|f_x(t)| = |d(t, x) - d(t, x_0)| \leq d(x, x_0)$. So $f_x \in C_b(X)$. Now define $i : X \rightarrow C_b(X)$ by $x \mapsto f_x$. Observe that:

$$\begin{aligned} \|f_x - f_y\|_u &= \sup_{t \in X} |f_x(t) - f_y(t)| \\ &= \sup_{t \in X} |d(t, x) - d(t, x_0) - d(t, y) + d(t, x_0)| \\ &= \sup_{t \in X} |d(t, x) - d(t, y)| \\ &= d(x, y). \end{aligned}$$

Thus i is an isometry, making $(\overline{i(X)}^{\|\cdot\|_u}, \|\cdot\|_u, i)$ a completion of X . $\square$

Theorem 2.6.22. *Let (X, d) be a metric space with completion $((Z, \rho), i)$. If $f : X \rightarrow Y$ is a uniformly continuous function into a complete metric space Y , then there exists a unique uniformly continuous function $\tilde{f} : Z \rightarrow Y$ such that $\tilde{f} \circ i = f$; i.e., the following diagram commutes:*

$$\begin{array}{ccc} X & \xrightarrow{i} & Z \\ & \searrow f & \downarrow \tilde{f} \\ & & Y \end{array}$$

Proof. Define $g : i(X) \rightarrow Z$ by $g = f \circ i^{-1}$. Since g is uniformly continuous and $i(X) \subseteq Z$ is dense, Theorem 2.6.14 says there exists a unique uniformly continuous map $\tilde{f} : Z \rightarrow Y$. Clearly $\tilde{f} \circ i = f$. $\square$

Theorem 2.6.23. **** Let $(V, \|\cdot\|)$ be a normed space. The completion of V is a Banach space.*

Proof. Let $((W, \rho), i)$ be the completion of V . We must first verify that W is a vector space. In doing so, we have to define what the vector space operations of W are. Let $w \in W$ and $\alpha \in F$. We know there exists a sequence $(i(v_n))_n$ in $i(V)$ converging to w . Since this sequence is convergent, it is ρ -Cauchy. From the following:

$$\begin{aligned} \rho(i(\alpha v_n), i(\alpha v_m)) &= \|\alpha v_n - \alpha v_m\| \\ &= |\alpha| \|v_n - v_m\| \\ &= |\alpha| \rho(i(v_n), i(v_m)), \end{aligned}$$

we can see $(i(\alpha v_n))_n$ is also ρ -Cauchy, whence it is convergent. Define $s : F \times W \rightarrow W$ by $s(\alpha, w) = \lim_{n \rightarrow \infty} i(\alpha v_n) := \alpha w$. We will first show that this action is well-defined. Let $(i(u_n))_n$ be another sequence in $i(V)$ converging to w . As above, we have:

$$\begin{aligned} \rho(i(\alpha v_n), i(\alpha u_n)) &= |\alpha| \rho(i(v_n), i(u_n)) \\ &\leq \alpha(\rho(i(v_n), w) + \rho(w, i(u_n))) \\ &\rightarrow 0. \end{aligned}$$

This gives:

$$\begin{aligned}\rho(i(\alpha u_n), \alpha w) &\leq \rho(i(\alpha u_n), i(\alpha v_n)) + \rho(i(\alpha v_n), \alpha w) \\ &\rightarrow 0.\end{aligned}$$

Thus $(i(\alpha u_n))_n \rightarrow \alpha w$, meaning scalar multiplication is well-defined. Now let $w_1, w_2 \in W$. Let $(i(v_n))_n$ and $(i(u_n))_n$ be sequences in $i(V)$ converging respectively to w_1 and w_2 . Since these sequences are convergent, they are ρ -Cauchy. From the fact that:

$$\begin{aligned}\rho(i(v_n + u_n), i(v_m + u_m)) &= \|v_n + u_n - v_m - u_m\| \\ &\leq \|v_n - v_m\| + \|u_n - u_m\| \\ &= \rho(i(v_n), i(v_m)) + \rho(i(u_n), i(u_m)),\end{aligned}$$

we can see $(i(v_n + u_n))_n$ is also ρ -Cauchy, hence it is convergent. Define $\alpha : W \times W \rightarrow W$ by $\alpha(w_1, w_2) = \lim_{n \rightarrow \infty} i(v_n + u_n) := w_1 + w_2$. We will show this binary operation is well-defined. Let $(i(x_n))_n$ and $(i(y_n))_n$ be sequences in $i(V)$ also converging to w_1 and w_2 respectively. Note that:

$$\begin{aligned}\rho(i(v_n + u_n), i(x_n + y_n)) &= \|v_n + u_n - x_n - y_n\| \\ &\leq \|v_n - x_n\| + \|u_n - y_n\| \\ &= \rho(i(v_n), i(x_n)) + \rho(i(u_n), i(y_n)) \\ &\leq \rho(i(v_n), w_1) + \rho(w_1, i(x_n)) + \rho(i(u_n), w_2) + \rho(w_2, i(y_n)) \\ &\rightarrow 0.\end{aligned}$$

This gives:

$$\begin{aligned}\rho(i(x_n + y_n), w_1 + w_2) &\leq \rho(i(x_n + y_n), i(v_n + u_n)) + \rho(i(v_n + u_n), w_1 + w_2) \\ &\rightarrow 0.\end{aligned}$$

Thus $(i(x_n + y_n))_n \rightarrow w_1 + w_2$, meaning vector addition is well-defined.

Before showing W paired with the above operations is a vector space, we need to verify that the isometry $i : V \rightarrow W$ is linear. If $v, v' \in V$ and $\alpha \in F$, by taking $v_n = v$ and $u_n = v'$ for all $n \geq 1$, we can see that:

$$\begin{aligned}i(v + \alpha v') &= \lim_{n \rightarrow \infty} i(v + \alpha v') \\ &= i(v) + \alpha i(v').\end{aligned}$$

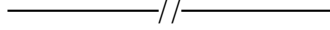
With this fact, we can show that W satisfies all of the vector-space axioms —this is left as an exercise. Moreover, we have that $i(V) \subseteq W$ is a ρ -dense linear subspace².

It remains to show that W is a normed space. Exactly as before, if $w \in W$, we can find a sequence $(i(v_n))_n$ in $i(V)$ converging to w . Since this sequence is Cauchy, observe that:

$$\begin{aligned}\left| \|v_n\| - \|v_m\| \right| &\leq \|v_n - v_m\| \\ &= \rho(i(v_n), i(v_m)).\end{aligned}$$

So $(\|v_n\|)_n$ is Cauchy, hence it is convergent. Define $\|w\|_W := \lim_{n \rightarrow \infty} \|v_n\|$. Showing this definition is well-defined, and that it satisfies the properties of a norm are left as an exercise. $\square$

²If $T : V \rightarrow W$ is injective, then $V \cong T(V)$



There is a softer proof of Theorem 2.6.23, but it requires heavier machinery. If V is a vector space over F , recall that

$$V' = \{ \varphi \mid \varphi : V \rightarrow F \text{ linear} \}$$

is the linear space of all linear functionals on V . By Zorn's Lemma, $V' \neq \emptyset$. If V is a normed space, then

$$V^* = \{ \varphi \in V' \mid \varphi \text{ continuous} \}$$

is called the *continuous dual space*. This is in fact a Banach space with norm $\|\varphi\|_{\text{op}} = \sup_{v \in B_V} |\varphi(v)|$. However, is $V^* \neq \emptyset$?

Theorem 2.6.24 (Hahn-Banach). *Let V be a normed space. For any nonzero $v_0 \in V$, there is a $\varphi_{v_0} \in V^*$ with $\varphi_{v_0}(v_0) = \|v_0\|$. Such a φ_{v_0} is called a *norming functional*.*

Corollary 2.6.25. *Let V be a normed space and $v \in V$. Then $\|v\| = \sup_{\varphi \in B_{V^*}} |\varphi(v)|$.*

Proof. If $\varphi \in B_{V^*}$, then $|\varphi(v)| \leq \|\varphi\|_{\text{op}} \|v\| \leq \|v\|$. Choose $\varphi_v \in V^*$. Since $\varphi_v(v) = \|v\|$, we have $\sup_{\varphi \in B_{V^*}} |\varphi(v)| \geq |\varphi_v(v)| = \|v\|$. $\square$

Corollary 2.6.26. *There is a linear isometry $V \hookrightarrow (V^*)^*$.*

Proof. Let $v \in V$. Define $\widehat{v} : V^* \rightarrow F$ by $\varphi \mapsto \varphi(v)$. We can easily verify that $\widehat{v} \in (V^*)'$. By the above corollary:

$$\begin{aligned} \|\widehat{v}\| &= \sup_{\varphi \in B_{V^*}} |\widehat{v}(\varphi)| \\ &= \sup_{\varphi \in B_{V^*}} |\varphi(v)| \\ &= \|v\|. \end{aligned}$$

Whence $\widehat{v} \in (V^*)^*$. Define $i_V : V \rightarrow (V^*)^*$ by $v \mapsto \widehat{v}$. Note that this is an isometry since $\|i_V(v)\| = \|\widehat{v}\| = \|v\|$. Given $v_1, v_2 \in V$, $c \in F$, and $\varphi \in V^*$ we can see:

$$\begin{aligned} i_V(v_1 + cv_2)(\varphi) &= \widehat{v_1 + cv_2}(\varphi) \\ &= \varphi(v_1 + cv_2) \\ &= \varphi(v_1) + c\varphi(v_2) \\ &= \widehat{v_1}(\varphi) + c\widehat{v_2}(\varphi) \\ &= i_V(v_1)(\varphi) + ci_V(v_2)(\varphi). \end{aligned}$$

Thus i_V is linear. $\square$

Using the tools above, we can now demonstrate in a cleaner way that the completion of any normed space forms a Banach space.

Alternative proof of Theorem 2.6.23. If V is a normed space, then $(V^*)^* = B(V^*, F)$ is complete by Proposition 2.6.12. Then $\overline{i_V(V)}^{\|\cdot\|_{\text{op}}} \subseteq (V^*)^*$ is complete by Proposition 2.6.10. Clearly $i_V : V \rightarrow \overline{i_V(V)}^{\|\cdot\|_{\text{op}}}$ is an isometry because $\overline{i_V(V)}^{\|\cdot\|_{\text{op}}} \subseteq (V^*)^*$. Thus the completion of V is $\left(\overline{i_V(V)}^{\|\cdot\|_{\text{op}}}, \|\cdot\|_{\text{op}}, i_V\right)$ $\square$

Lemma 2.6.27. *Let $T : V \rightarrow W$ be continuous and linear. There is an induced continuous and linear map $T^* : W^* \rightarrow V^*$ with $T^*(\psi) = \psi \circ T$; i.e., the following diagram commutes:*

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ & \searrow & \downarrow \psi \\ & T^*(\psi) & F \end{array}$$

Proof. Let $\psi_1, \psi_2 \in W^*$ and $\alpha \in F$. We have:

$$\begin{aligned} T^*(\psi_1 + \alpha\psi_2) &= (\psi_1 + \alpha\psi_2) \circ T \\ &= \psi_1 \circ T + \alpha(\psi_2 \circ T) \\ &= T^*(\psi_1) + \alpha T^*(\psi_2). \end{aligned}$$

Thus T^* is linear. Moreover:

$$\begin{aligned} \|T^*\|_{\text{op}} &= \sup_{\|\psi\|_{\text{op}} \leq 1} \|T^*(\psi)\|_{\text{op}} \\ &= \sup_{\|\psi\|_{\text{op}} \leq 1} \|\psi \circ T\|_{\text{op}} \\ &\leq \sup_{\|\psi\|_{\text{op}} \leq 1} (\|\psi\|_{\text{op}} \|T\|_{\text{op}}) \\ &= \|T\|_{\text{op}} \cdot \sup_{\|\psi\|_{\text{op}} \leq 1} \|\psi\|_{\text{op}} \\ &\leq \|T\|_{\text{op}}. \end{aligned}$$

Thus T^* is continuous. $\square$

Theorem 2.6.28. *Let V and W be normed spaces with completions $\tilde{V} = \overline{i_V(V)}$ and $\tilde{W} = \overline{i_W(W)}$. If $T : V \rightarrow W$ is a continuous and linear map, there exists a unique continuous and linear map $\tilde{T} : \tilde{V} \rightarrow \tilde{W}$ such that $\tilde{T} \circ i_V = i_W \circ T$; i.e., the following diagram commutes:*

$$\begin{array}{ccc} \tilde{V} & \xrightarrow{\tilde{T}} & \tilde{W} \\ i_V \uparrow & & \uparrow i_W \\ V & \xrightarrow{T} & W \end{array}$$

Proof. Since T is continuous, we can induce the map $T^* : W^* \rightarrow V^*$ where $T^*(\psi) = \psi \circ T$. Since T^* is continuous, we can induce the map $T^{**} : V^{**} \rightarrow W^{**}$ where $T^{**}(\xi) = \xi \circ T^*$. If

$v \in V$ and $\psi \in V^*$, note that:

$$\begin{aligned} (\widehat{v} \circ T^*)(\psi) &= (T^* \circ \psi)(v) \\ &= (\psi \circ T)(v) \\ &= \widehat{T(v)}(\psi). \end{aligned}$$

This allows us to show:

$$\begin{aligned} T^{**} \circ i_V(v) &= \widehat{v} \circ T^* \\ &= \widehat{T(v)} \\ &= i_W(T(v)). \end{aligned}$$

Whence $T^{**}(i_V(V)) \subseteq i_W(W)$. Since T^{**} is continuous:

$$\begin{aligned} T^{**}(\widetilde{V}) &= T^{**}(\overline{i_V(V)}) \\ &\subseteq \overline{T^{**}(i_V(V))} \\ &\subseteq \overline{i_W(W)} \\ &= \widetilde{W}. \end{aligned}$$

Thus $T^{**}|_{\widetilde{V}} : \widetilde{V} \rightarrow \widetilde{W}$ is the desired linear extension. We obtain the following diagram:

$$\begin{array}{ccc} V^{**} & \xrightarrow{T^{**}} & W^{**} \\ \cup & & \cup \\ \widetilde{V} & \xrightarrow{T^{**}|_{\widetilde{V}}} & \widetilde{W} \\ \cup & & \cup \\ i_V(V) & \xrightarrow{T^*|_{i_V(V)}} & i_W(W) \\ i_V \uparrow & & \uparrow i_W \\ V & \xrightarrow{T} & W \end{array}$$

□

Exercise 2.6.1. Show that $\|\widetilde{T}\|_{\text{op}} = \|T\|_{\text{op}}$.

Definition 2.6.4. If V is a normed space and $i_V : V \rightarrow (V^*)^*$ is surjective, then V is called *reflexive*.

Example 2.6.4. Hilbert spaces are reflexive by the Riesz representation theorem.

—————//—————

Recall that $A \subseteq X$ is *nowhere dense* if $\overline{A}^{\circ} = \emptyset$. For example, if $f : \mathbb{R} \rightarrow \mathbb{R}$ is any map, the set $\{(x, y) \in \mathbb{R}^2 \mid y = f(x)\}$ is nowhere dense.

Proposition 2.6.29. For a metric space (X, d) and $A \subseteq X$, the following are equivalent:

- (1) A is nowhere dense;
- (2) There exists a closed subset $F \subseteq X$ such that $F^\circ = \emptyset$ and $F \supseteq A$;
- (3) There exists an open and dense subset $U \subseteq X$ such that $U \subseteq A^c$.

Proof. (1) $\Rightarrow$ (2) Take $F = \overline{A}$.

(2) $\Rightarrow$ (1) Let F be such a set. Then $\overline{A} \subseteq \overline{F}$. So $\overline{A}^\circ \subseteq \overline{F}^\circ = \emptyset$.

(2) $\Rightarrow$ (3) Let F be such a set. Let $U = F^c$. Then $\overline{U} = \overline{F^c} = (F^\circ)^c = \emptyset^c = X$. We also have $U = F^c \subseteq A^c$.

(3) $\Rightarrow$ (2) Let U be such a set. Take $F = U^c$. Then $F^\circ = (U^c)^\circ = (\overline{U})^c = X^c = \emptyset$. Also $F = U^c \supseteq (A^c)^c = A$. $\square$

Definition 2.6.5. A point $x \in X$ is *isolated* if there exists an $\epsilon > 0$ such that $U(x, \epsilon) = \{x\}$.

Proposition 2.6.30. Let (X, d) be a metric space.

- (1) If $A \subseteq X$ is nowhere dense and $B \subseteq A$, then B is nowhere dense.
- (2) If $A \subseteq X$ is nowhere dense, then $\overline{A}$ is nowhere dense.
- (3) If $A_1, A_2, \dots, A_n$ are nowhere dense, then $\bigcup_{k=1}^n A_k$ is nowhere dense.
- (4) If X has no isolated points, then every finite subset is nowhere dense.

Proof. (1) If $B \subseteq A$ then $\overline{B} \subseteq \overline{A}$, so $(\overline{B})^\circ \subseteq (\overline{A})^\circ = \emptyset$.

(2) Note that $\overline{A} = \overline{\overline{A}}$. So $(\overline{A})^\circ = (\overline{\overline{A}})^\circ = \emptyset$.

(3) We show this for $n = 2$. Let A_1 and A_2 be nowhere dense. By Proposition 2.6.29, $A_1^c \supseteq U_1$, where U_1 is open and dense. Similarly, $A_2^c \supseteq U_2$, where U_2 is open and dense. Then:

$$\begin{aligned} (A_1 \cup A_2)^c &= A_1^c \cap A_2^c \\ &\supseteq U_1 \cap U_2. \end{aligned}$$

Clearly $U_1 \cap U_2$ is open by Proposition 2.2.1. Claim: $U_1 \cap U_2$ is dense. Let $x \in X$ and $\epsilon > 0$. We'd like to show, by Proposition 2.2.8, that $(U_1 \cap U_2) \cap U(x, \epsilon) \neq \emptyset$. Since U_1 is dense, we know $U_1 \cap U(x, \epsilon) \neq \emptyset$. Let $z \in U_1 \cap U(x, \epsilon)$. Since $U_1 \cap U(x, \epsilon)$ is open, by Definition 2.2.2 there exists a $\delta > 0$ such that $U(z, \delta) \subseteq U_1 \cap U(x, \epsilon)$. Now since U_2 is dense, $U(z, \delta) \cap U_2 \neq \emptyset$. Therefore $U(x, \epsilon) \cap (U_1 \cap U_2) \neq \emptyset$. Thus $U_1 \cap U_2$ is dense.

(4) Since X has no isolated points, $\{x\}$ is closed but not open. Then $(\overline{\{x\}})^\circ = \{x\}^\circ = \emptyset$. So $\{x\}$ is nowhere dense. By (3), any finite set $\{x_1, x_2, \dots, x_n\} = \{x_1\} \cup \{x_2\} \cup \dots \cup \{x_n\}$ is nowhere dense. $\square$

Note that $\mathbb{Q}$ is not nowhere dense, but it is the *countable* union of nowhere dense sets.

Definition 2.6.6. Let (X, d) be a metric space.

- (1) A set $A \subseteq X$ is *meager* if $A = \bigcup_{k=1}^\infty A_k$, where each A_k is nowhere dense.
- (2) A set $B \subseteq X$ is *residual* if B^c is meager.

Example 2.6.5. A meager set is "topologically small", whereas a residual set is "topologically big". Consider $\mathbb{Q} \subseteq \mathbb{R}$ meager. Then $\mathbb{R} \setminus \mathbb{Q} \subseteq \mathbb{R}$ is residual.

Note that this definition only applies to *subsets* of a specified metric space. For instance, $\mathbb{Z} \subseteq \mathbb{R}$ is meager, but $\mathbb{Z} \subseteq \mathbb{Z}$ is not meager. Indeed, $\mathbb{Z}$ with the discrete topology means that every subset of $\mathbb{Z}$ is open, and every subset of $\mathbb{Z}$ is closed. So given $A \subseteq \mathbb{Z}$ nowhere dense, we have $A = (\overline{A})^c = \emptyset$. So if $A \subseteq \mathbb{Z}$ is meager, then $A \neq \emptyset$.

Proposition 2.6.31. *Let (X, d) be a metric space.*

- (1) *If $A \subseteq X$ is meager and $B \subseteq A$, then B is meager.*
- (2) *If $A_1, A_2, \dots, A_k$ are meager, then $\bigcup_{k=1}^{\infty} A_k$ is meager.*
- (3) *If X has no isolated points, every countable set is meager.*

Proof. (1) If A is meager, then $A = \bigcup_{k=1}^{\infty} A_k$, where each A_k is nowhere dense. Then:

$$\begin{aligned} B &= B \cap A \\ &= B \cap \bigcup_{k=1}^{\infty} A_k \\ &= \bigcup_{k=1}^{\infty} (B \cap A_k) \end{aligned}$$

Since $B \cap A_k$ is nowhere dense for all k , B is meager.

(2) If A_k is meager, then $A_k = \bigcup_{j=1}^{\infty} A_{kj}$, where A_{kj} is nowhere dense for each j . We have:

$$\bigcup_{k=1}^{\infty} A_k = \bigcup_{k=1}^{\infty} \bigcup_{j=1}^{\infty} A_{kj}.$$

Since the countable union of countable sets is countable, we can see $\bigcup_{k=1}^{\infty} A_k$ is meager.

(3) We can write $\{x_k \mid k \geq 1\}$ as $\bigcup_{k=1}^{\infty} \{x_k\}$. Since singletons are nowhere dense, every countable set is meager. $\square$

Proposition 2.6.32 (Cantor's Intersection Theorem). *Let (X, d) be a complete metric space and $F_1 \supseteq F_2 \supseteq F_3 \supseteq \dots$ be a nested sequence of closed nonempty sets with $(\text{diam}(F_n))_n \rightarrow 0$. Then $\bigcap_{n \geq 1} F_n = \{x\}$ for some $x \in X$.*

Proof. Let $x_n \in F_n$ for all $n \geq 1$. Note that $(x_n)_n$ is Cauchy —given $\epsilon > 0$, let N be large so that $n \geq N$ implies $\text{diam}(F_n) < \epsilon$. For $m, n \geq N$, we have $x_m, x_n \in F_N$ (because $F_N \supseteq F_{N+1} \supseteq \dots$), which gives:

$$\begin{aligned} d(x_n, x_m) &\leq \sup_{x_n, x_m \in F_N} d(x_n, x_m) \\ &= \text{diam}(F_N) \\ &< \epsilon. \end{aligned}$$

Since X is complete, $(x_n)_n \rightarrow x_0$ for some $x_0 \in X$.

Claim: $\bigcap_{n \geq 1} F_n = \{x_0\}$. To see this, fix $m \in \mathbb{N}$ and consider the sequence $(x_{m+k})_k$ in F_m . Since $(x_n)_n$ converges to x_0 , we know the tail $(x_{m+k})_k \xrightarrow{k \rightarrow \infty} x_0$. Since F_m is closed, we know $x_0 \in F_m$. Since m was arbitrary, $x_0 \in \bigcap_{n \geq 1} F_n$. Now suppose $x_0, x \in \bigcap_{n \geq 1} F_n$ with

$d(x_0, x) > 0$. We can find N large so that $\text{diam}(F_n) < d(x_0, x)$. But $x_0, x \in \bigcap_{n \geq 1} F_n$ implies $x_0, x \in F_n$, giving $d(x_0, x) < d(x_0, x)$, which is a contradiction. Thus $\bigcap_{n \geq 1} F_n = \{x_0\}$. $\square$

Example 2.6.6. Note that $\bigcap_{n \geq 1} (0, \frac{1}{n}) = \emptyset$ because $(0, \frac{1}{n})$ is not closed, and $\bigcap_{n \geq 1} [n, \infty) = \emptyset$ because $(\text{diam}([n, \infty)))_n \rightarrow 0$.

Theorem 2.6.33 (Baire Category Theorem). *Let (X, d) be a complete metric space.*

- (1) *If $\{V_k\}_{k=1}^\infty$ is a countable family of open and dense subsets, then $\bigcap_{k=1}^\infty V_k$ is also dense (but not necessarily open).*
- (2) *X is not meager.*

Proof. (1) Let U_0 be any open ball. Since V_1 is open and dense, $U_0 \cap V_1$ is open and nonempty. So there exists an open ball $U(x, \delta) \subseteq U_0 \cap V_1$. Notice we also have $U_1 := U(x, \frac{\delta}{2}) \subseteq U_0 \cap V_1$. Define $B_1 = \overline{U_1} \subseteq U_0 \cap V_1$. Since δ was arbitrary, we arbitrarily shrink out open ball to ensure $\text{diam}(B_1) < 1$. Now $U_1 \cap V_2$ is open and nonempty. So there exists some open ball contained in $U_1 \cap V_2$. Exactly as we just did, find $B_2 := \overline{U_2} \subseteq U_1 \cap V_2$ such that $\text{diam}(B_2) < \frac{1}{2}$. Inductively, having constructed $U_1, B_1, U_2, B_2, \dots$, we note that $U_{n-1} \cap V_n$ is open and nonempty, so it contains an open ball. Find $B_n := \overline{U_n} \subseteq U_{n-1} \cap V_n$ such that $\text{diam}(B_n) < \frac{1}{n}$. Observe that:

$$B_1 \supseteq U_1 \supseteq B_2 \supseteq U_2 \supseteq B_3 \supseteq U_3 \supseteq \dots$$

So $\{B_n\}_{n \geq 1}$ is a nested sequence of closed sets with $(\text{diam}(B_n))_n \rightarrow 0$. By Cantor's Intersection Theorem, $\bigcap_{n \geq 1} B_n = \{x\}$. Claim: $x \in U_0 \cap (\bigcap_{k=1}^\infty V_k)$. Note that $B_n \subseteq U_{n-1} \cap V_n \subseteq V_n$. So $x \in \bigcap_{n \geq 1} B_n \subseteq \bigcap_{n \geq 1} V_n$. Also, $x \in B_1 = \overline{U_1} \subseteq U_0 \cap V_1 \subseteq U_0$. This means $x \in U_0 \cap (\bigcap_{k=1}^\infty V_k)$. Since U_0 was an arbitrary open set, and since its intersection with $\bigcap_{k=1}^\infty V_k$ is nonempty, we can conclude that $\bigcap_{k=1}^\infty V_k$ is dense.

(2) Suppose towards contradiction that X is meager, that is, $X = \bigcup_{k=1}^\infty A_k$, where each A_k is nowhere dense. Fixing k , by Proposition 2.6.29 we can find an open and dense set V_k with $V_k \subseteq A_k^c$. But then:

$$\begin{aligned} \emptyset &= X^c \\ &= \left(\bigcup_{k=1}^\infty A_k \right)^c \\ &= \bigcap_{k=1}^\infty A_k^c \\ &\supseteq \bigcap_{k=1}^\infty V_k. \end{aligned}$$

This is a contradiction. Thus, if X is complete, then X is not meager. $\square$

Exercise 2.6.2. Show that $\mathbb{R} \setminus \mathbb{Q}$ is not meager.

Skipped all of the "application" stuff

Example 2.6.7. Let $X = (C([0, 1]), \|\cdot\|_u)$. Recall that $f(x) = |x|$ is continuous everywhere, but not differentiable at $x = 0$. Does there exist a function $f : X \rightarrow \mathbb{R}$ which is continuous everywhere, but differentiable nowhere. The answer is yes, and in fact, the set:

$$\{f \in X \mid f \text{ differentiable nowhere}\}$$

is *residual*. This is surprising, as the complement of this set—the set containing every continuous function which is differentiable at one or more points, is "topologically small."

§ 2.7. Compactness

Compactness is a generalization of "closed and bounded in $\mathbb{R}^n$ for metric spaces. It is the algebraic analogue of "finite-dimensional".

Definition 2.7.1. Let (X, d) be a metric space and $K \subseteq X$ a subset.

- (1) A *cover* for K is a family $\mathcal{U} = \{U_i\}_{i \in I}$ with $U_i \subseteq X$ for all i and $K \subseteq \bigcup_{i \in I} U_i$.
- (2) The family $\mathcal{U}$ is an *open cover* for K if each U_i is open.
- (3) The family $\mathcal{U}$ is a *finite cover* for K if I is finite.
- (4) A *subcover* of $\mathcal{U}$ is a subfamily $\{U_i\}_{i \in J}$, where $J \subseteq I$ and $K \subseteq \bigcup_{i \in J} U_i$.
- (5) K is called *compact* if every open cover $\mathcal{U}$ of K admits a finite subcover. That is, if $K \subseteq \bigcup_{i \in I} U_i$ with each $U_i \subseteq X$ open, there exists a finite subset $F \subseteq I$ with $K \subseteq \bigcup_{i \in F} U_i$.

Since K is not necessarily proper, a metric space X is compact if it can be written as the union of finitely many open sets.

Example 2.7.1. The set $(0, 1]$ is not compact in $\mathbb{R}$. The family $\{(\frac{1}{n}, 2)\}_{n \geq 1}$ is an open cover of $(0, 1]$, but admits no finite subcover.

Exercise 2.7.1. Let (X, d) be a metric space with $Y \subseteq X$ and $K \subseteq Y$. $K \subseteq X$ is compact if and only if $K \subseteq Y$ is compact.

Proposition 2.7.1. Let (X, d) be a metric space and $K \subseteq X$.

- (1) If K is compact, then K is closed and bounded,
- (2) If X is compact and K is closed, then K is compact.

Proof. (1) We will first show that K is bounded. Pick $x_0 \in K$. Note that $\{U(x_0, n)\}_{n=1}^\infty$ is a cover for K , whence $K \subseteq \bigcup_{n \geq 1} U(x_0, n)$. By the compactness of K , $K \subseteq \bigcup_{n=1}^N U(x_0, n)$. For $x, x' \in K$:

$$\begin{aligned} d(x, x') &\leq d(x, x_0) + d(x_0, x') \\ &\leq 2N. \end{aligned}$$

Thus $\text{diam}(K) \leq 2N$, so K is bounded. We will now show that K^c is open. Let $x_0 \in K^c$ be arbitrary. For each $x \in K$, we can find a $\delta_x > 0$ such that $U(x_0, \delta_x) \cap U(x, \delta_x) = \emptyset$. Note that $K \subseteq \bigcup_{x \in K} U(x, \delta_x)$. Since K is compact, there exists $x_1, \dots, x_n \in K$ with $K \subseteq \bigcup_{j=1}^n U(x_j, \delta_{x_j})$.

Let $\delta = \min_{j=1}^n \delta_{x_j}$. Claim: $U(x_0, \delta) \subseteq K^c$. If not, we can find some $z \in U(x_0, \delta) \cap K$, which implies $z \in U(x_0, \delta) \cap U(x_j, \delta_{x_j})$ for some j . But this means $\emptyset \neq U(x_0, \delta) \cap U(x_j, \delta_{x_j}) \subseteq U(x_0, \delta_{x_j}) \cap U(x_j, \delta_{x_j}) = \emptyset$. Thus K is closed.

(2) Let $K \subseteq \bigcup_{i \in I} U_i$, where $U_i \subseteq X$ is open. Then $X = K^c \cup (\bigcup_{i \in I} U_i)$. Since $K^c \cup (\bigcup_{i \in I} U_i)$ is an open cover of X , there exists a finite subset $F \subseteq I$ with $X = K^c \cup (\bigcup_{i \in F} U_i)$. It follows that $K \subseteq \bigcup_{i \in F} U_i$. $\square$

Definition 2.7.2. Let (X, d) be a metric space. A non-empty family of subsets $\{A_i\}_{i \in I}$ is said to have the *finite intersection property* if for any finite subset $F \subseteq I$, we have $\bigcap_{i \in F} A_i \neq \emptyset$.

Proposition 2.7.2. Let (X, d) be a metric space. The following are equivalent:

- (1) X is compact;
- (2) For any family of closed subsets $\{C_i\}_{i \in I}$ satisfying the finite intersection property, we have $\bigcap_{i \in I} C_i \neq \emptyset$.

Proof. Suppose that X is compact and let $\{C_i\}_{i \in I}$ be a family of closed sets satisfying the finite intersection property. Assume towards contradiction that $\bigcap_{i \in I} C_i = \emptyset$. Then $\bigcup_{i \in I} C_i^c = X$. Since X is compact, there exists some finite subset $F \subseteq I$ with $\bigcup_{i \in F} C_i^c = X$. But this implies $\bigcap_{i \in F} C_i = \emptyset$, contradicting our family of closed subsets satisfying the finite intersection property.

We will assume the contrapositive of (2). Let $X = \bigcup_{i \in I} V_i$, where each V_i is open. Then $\emptyset = \bigcap_{i \in I} V_i^c$. Since $\{V_i^c\}_{i \in I}$ is a family of closed subsets, it must be the case that $\bigcap_{i \in F} V_i^c = \emptyset$ for some finite subset $F \subseteq I$. Whence $\bigcup_{i \in F} V_i = X$. $\square$

Proposition 2.7.3. *** Every compact metric space is separable.

Proof. Fix $n \geq 1$. We have an open cover $X = \bigcup_{x \in X} U(x, \frac{1}{n})$, which admits a subcover $X = \bigcup_{j=1}^{J_n} U(x_{n,j}, \frac{1}{n})$. Let $S = \{x_{n,j} \mid n \geq 1, 1 \leq j \leq J_n\}$.

Let $x \in X$ and $\epsilon > 0$. Let m be such that $\epsilon > \frac{1}{m}$. Find $j \in \{1, \dots, J_m\}$ with $x \in U(x_{m,j}, \frac{1}{m})$. Thus $d(x, x_{m,j}) < \frac{1}{m}$. Whence $S \cap U(x, \epsilon) \neq \emptyset$. So S is dense and certainly countable. $\square$

Definition 2.7.3. Let (X, d) be a metric space and let $K \subseteq X$, K is called *sequentially compact* if every sequence $(x_n)_n$ in K admits a convergent subsequence in K .

Proposition 2.7.4. Let (X, d) be a metric space. If $K \subseteq X$ is compact, then K is sequentially compact.

Proof. Let $(x_k)_k$ be a sequence in K . Consider $C_n = \overline{\{x_k \mid k > n\}}$. Note that:

$$C_1 \supseteq C_2 \supseteq C_3 \supseteq \dots$$

In particular, $\bigcap_{j=1}^n C_j \neq \emptyset$ for any $n > 1$. So $\{C_n\}_{n \geq 1}$ is a family of closed sets which satisfies the finite intersection property. By Proposition 2.7.2 we have $\bigcap_{n \geq 1} C_n \neq \emptyset$. Let $x \in \bigcap_{n \geq 1} C_n$.

Clearly $x \in C_1$. This means $U(x, 1) \cap \{x_k \mid k > 1\} \neq \emptyset$. Let $x_{k_1} \in U(x, 1) \cap C_1$. Since $x_{k_1} \in C_1$, we know that $k_1 > 1$. Moreover, $d(x, x_{k_1}) < 1$.

Similarly, $x \in C_{k_1}$. This means $U(x, \frac{1}{2}) \cap \{x_k \mid k > k_1\} \neq \emptyset$. Let $x_{k_2} \in U(x, \frac{1}{2}) \cap \{x_k \mid k > k_1\}$. Since $x_{k_2} \in \{x_k \mid k > k_1\}$, we know $k_2 > k_1$. Moreover, $d(x, x_{k_2}) < \frac{1}{2}$.

Inductively, $x \in C_{k_{j-1}}$. This means $U(x, \frac{1}{j}) \cap \{x_k \mid k > k_{j-1}\} \neq \emptyset$. Let $x_{k_j} \in U(x, \frac{1}{j}) \cap \{x_k \mid k > k_{j-1}\}$. Since $x_{k_j} \in \{x_k \mid k > k_{j-1}\}$, we know $k_j > k_{j-1}$. Moreover, $d(x, x_{k_j}) < \frac{1}{j}$. We've obtained a sequence $(x_{k_j})_j$ which converges to x . Thus (X, d) is sequentially compact. $\square$

Proposition 2.7.5. *If (X, d) is sequentially compact, then (X, d) is complete.*

Proof. Let $(x_n)_n$ be Cauchy. We know there exists a subsequence $(x_{n_k})_k$ which converges to some $x \in X$. Let $\epsilon > 0$. Find N large so $p, q \geq N$ implies $d(x_p, x_q) < \frac{\epsilon}{2}$. Find K large so $k \geq K$ implies $d(x_{n_k}, x) < \frac{\epsilon}{2}$. For $n, k \geq \max\{N, K\}$:

$$\begin{aligned} d(x_n, x) &\leq d(x_n, x_{n_k}) + d(x_{n_k}, x) \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

Thus $(x_n)_n \rightarrow x$. Whence (X, d) is complete. $\square$

Definition 2.7.4. Let (X, d) be a metric space. We say $Y \subseteq X$ is *totally bounded* if for all $\epsilon > 0$, there exists a finite subset $F \subseteq Y$ such that $Y \subseteq \bigcup_{z \in F} U(z, \epsilon)$.

Proposition 2.7.6. *Let (X, d) be a metric space. If $K \subseteq X$ is sequentially compact, then K is totally bounded.*

Proof. Suppose towards contradiction that K is not totally bounded. Then there exists some $\epsilon_0 > 0$ such that, for any finite subset $F \subseteq K$, we have $K \not\subseteq \bigcup_{z \in F} U(z, \epsilon_0)$. Let $x_1 \in K$ be arbitrary. Then $K \not\subseteq U(x_1, \epsilon_0)$, since K cannot be covered by finitely many open balls. Find $x_2 \in K \setminus U(x_1, \epsilon_0)$. Then $K \not\subseteq U(x_1, \epsilon_0) \cup U(x_2, \epsilon_0)$. Find $x_3 \in K \setminus (U(x_1, \epsilon_0) \cup U(x_2, \epsilon_0))$. Then $K \not\subseteq U(x_1, \epsilon_0) \cup U(x_2, \epsilon_0) \cup U(x_3, \epsilon_0)$. Inductively, we obtain a sequence $(x_n)_n$, where $x_n \in K \setminus \bigcup_{j=1}^{n-1} U(x_j, \epsilon_0)$. Since $x_n \notin \bigcup_{j=1}^{n-1} U(x_j, \epsilon_0)$, we have that $d(x_n, x_j) \geq \epsilon_0$ for all $j \neq n$. Now since X is sequentially compact, $(x_n)_n$ admits a convergent subsequence, call it $(x_{n_k})_k$. Since $(x_{n_k})_k$ converges, it is Cauchy. But this is a contradiction, because $d(x_{n_l}, x_{n_k}) \geq \epsilon_0$ for all $l \neq k$. It must be the case that K is totally bounded. $\square$

Proposition 2.7.7. *If (X, d) is a metric space and $K \subseteq X$ is totally bounded, then K is bounded.*

Proof. Let $\epsilon = 1$. We have that $K \subseteq \bigcup_{j=1}^n U(x_j, 1)$. Define $C = \max_{1 \leq i, j \leq n} d(x_i, x_j)$. If $x, y \in K$, then $x \in U(x_i, 1)$ and $y \in U(x_j, 1)$ for some i, j . This gives:

$$\begin{aligned} d(x, y) &\leq d(x, x_i) + d(x_i, x_j) + d(x_j, y) \\ &\leq 2 + C. \end{aligned}$$

Thus $\text{diam}(K) \leq 2 + C$. $\square$

Corollary 2.7.8. *If $K \subseteq \mathbb{R}$ is compact and nonempty, then $\sup(K) \in K$.*

Proof. Define $u := \sup(K)$. Let $(x_n)_n$ be a sequence in K converging to u . Note that $u \in \mathbb{R}$ since K is bounded. Since K is closed, $u \in K$. $\square$

Theorem 2.7.9. *Let (X, d) be a metric space. The following are equivalent:*

- (1) X is compact;

- (2) X is sequentially compact;
 (3) X is complete and totally bounded.

Proof. Propositions 2.7.4, 2.7.5, and 2.7.6 established (1) $\Rightarrow$ (2) and (2) $\Rightarrow$ (3). It remains to prove (3) $\Rightarrow$ (1). Suppose towards contradiction $\mathcal{V}$ is an open cover for X which fails to admit a finite subcover. Let $\epsilon = 1$. By total boundedness:

$$X = U(x_1, 1) \cup U(x_2, 1) \cup \dots \cup U(x_{m_1}, 1).$$

There must be at least one $U(x_j, 1)$ that can't be covered by finitely many members of $\mathcal{V}$. Label one of these balls at $B_1(x_1)$. Let $\epsilon = \frac{1}{2}$. By total boundedness,

$$X = U(x_1, \frac{1}{2}) \cup U(x_2, \frac{1}{2}) \cup \dots \cup U(x_{m_2}, \frac{1}{2}).$$

Note that:

$$\begin{aligned} \bigcup_{j=1}^{m_2} (B_1(x_1) \cap U(x_j, \frac{1}{2})) &= B_1(x_1) \cap \left(\bigcup_{j=1}^{m_2} U(x_j, \frac{1}{2}) \right) \\ &= B_1(x_1) \cap X \\ &= B_1(x_1). \end{aligned}$$

There must be some j such that $B_{x_1}(1) \cap U(x_j, \frac{1}{2})$ cannot be covered by finitely many members of $\mathcal{V}$. Label $U(x_j, \frac{1}{2})$ as $B_{1/2}(x_2)$. We continue this process inductively, along which we obtain a sequence $(x_n)_n$ such that

$$F_n := B_1(x_1) \cap B_{1/2}(x_2) \cap B_{1/3}(x_3) \cap \dots \cap B_{1/n}(x_n),$$

cannot be covered by finitely many members of $\mathcal{V}$. Define $C_n = \overline{F_n}$. Note that $C_1 \supseteq C_2 \supseteq C_3 \supseteq \dots$ and $\text{diam}(C_n) \leq \frac{2}{n}$. Since X is complete, by Cantor's Intersection Theorem $\bigcap_{n \geq 1} C_n = \{x\}$. Locate this x in $V \in \mathcal{V}$. Since V is open, we can find some $\epsilon > 0$ with $U(x, \epsilon) \subseteq V$. Find N large so that $\frac{2}{N} < \epsilon$. We can see that $C_N \subseteq U(x, \epsilon)$ since $d(z, x) \leq \frac{2}{N} < \epsilon$ for all $z \in C_N$. But this means $F_N \subseteq C_N \subseteq U(x, \epsilon) \subseteq V$. This is a contradiction, as we've asserted F_n cannot be covered by finitely many members of $\mathcal{V}$. Thus X is compact. $\square$

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We're interested in studying compact subsets of $\mathbb{R}^d$.
 (Heine-Borel stuff here...)

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Compactness and Continuity

Proposition 2.7.10. *Let $f : (X, d) \rightarrow (Y, d)$ be continuous and $K \subseteq X$ compact. Then $f(K) \subseteq Y$ is compact.*

Proof. Let $f(K) \subseteq \bigcup_{i \in I} V_i$, where each $V_i \subseteq Y$ is open. Then:

$$\begin{aligned} K &\subseteq f^{-1}(f(K)) \\ &\subseteq f^{-1}\left(\bigcup_{i \in I} V_i\right) \\ &= \bigcup_{i \in I} f^{-1}(V_i). \end{aligned}$$

Since f is continuous, $f^{-1}(V_i) \subseteq X$ is open. So there exists a finite subset $F \subseteq I$ with $K \subseteq \bigcup_{i \in F} f^{-1}(V_i)$. Whence:

$$\begin{aligned} f(K) &\subseteq f\left(\bigcup_{i \in F} f^{-1}(V_i)\right) \\ &= \bigcup_{i \in F} f(f^{-1}(V_i)) \\ &= \bigcup_{i \in F} V_i. \end{aligned}$$

□

Corollary 2.7.11. *If d_1 and d_2 are topologically equivalent metrics on X , then $K \subseteq X$ is d_1 -compact if and only if $K \subseteq X$ is d_2 -compact.*

Proof. By the definition of topological equivalence $\text{id} : (X, d_1) \rightarrow (X, d_2)$ is continuous. Let $K \subseteq X$ be d_1 -compact. Then $K = \text{id}(K) \subseteq (X, d_2)$. The previous proposition says K is d_2 -compact. The converse direction is identical. □

Theorem 2.7.12 (Extreme Value Theorem). *Let $f : (X, d) \rightarrow \mathbb{R}$ be continuous and $K \subseteq X$ compact. Then there exists $x_m, x_M \in K$ with $f(x_m) = \inf_{x \in K} f(x)$ and $f(x_M) = \sup_{x \in K} f(x)$.*

Proof. If f is continuous, then $f(K) \subseteq \mathbb{R}$ is compact. Then Proposition 2.7.8 gives $\inf(f(K)), \sup(f(K)) \in f(K)$. Hence there exists x_m, x_M with $f(x_m) = \inf(K)$ and $f(x_M) = \sup(K)$. □

Proposition 2.7.13. *Let V be a finite dimensional vector space over F . All norms on V are equivalent.*

Proof. Let $\{v_1, \dots, v_n\}$ be a basis for V . Define $\|\cdot\|_1 : V \rightarrow [0, \infty)$ by:

$$\left\| \sum_{i=1}^n t_i v_i \right\|_1 = \sum_{i=1}^n |t_i|.$$

Then $\varphi : \ell_1^n \rightarrow (V, \|\cdot\|_1)$ given by $(t_1, \dots, t_n) \mapsto \sum_{i=1}^n t_i v_i$ is a linear isomorphism and

isometry. Indeed:

$$\begin{aligned}
 \|\varphi((t_1, \dots, t_n)) - \varphi((t'_1, \dots, t'_n))\|_1 &= \left\| \sum_{i=1}^n t_i v_i - \sum_{i=1}^n t'_i v_i \right\|_1 \\
 &= \left\| \sum_{i=1}^n (t_i - t'_i) v_i \right\|_1 \\
 &= \sum_{i=1}^n |t_i - t'_i| \\
 &= \|(t_1, \dots, t_n) - (t'_1, \dots, t'_n)\|_{\ell_2^n}.
 \end{aligned}$$

By the Heine-Borel Theorem $B_{\ell_1^n} = \{t \in \ell_1^n \mid \|t\|_{\ell_1^n} \leq 1\}$ is compact as it is a closed and bounded subset of ℓ_1^n . Moreover, $B_1 = \{v \in V \mid \|v\| \leq 1\}$ is compact by Proposition 2.7.10 since $B_1 = \varphi(B_{\ell_1^n})$. Now $S_1 = \{v \in V \mid \|v\|_1 = 1\} \subseteq B_1$ is closed, hence compact. Let $\|\cdot\|$ be any norm on V . We can see:

$$\begin{aligned}
 \left\| \sum_{i=1}^n t_i v_i \right\| &= \sum_{i=1}^n |t_i| \|v_i\| \\
 &\leq c \sum_{i=1}^n |t_i| \\
 &= c \left\| \sum_{i=1}^n t_i v_i \right\|_1,
 \end{aligned}$$

where $c = \max_{i=1}^n \|v_i\|$.

Now consider $g : (V, \|\cdot\|_1) \rightarrow \mathbb{R}$ given by $g(v) = \|v\|$. We have:

$$\begin{aligned}
 |g(v) - g(v')| &= |\|v\| - \|v'\|| \\
 &\leq \|v - v'\| \\
 &\leq c \|v - v'\|_1.
 \end{aligned}$$

Thus g is continuous. Since $S_1 \subseteq (V, \|\cdot\|_1)$ is compact, the Extreme Value Theorem says there exists $w \in S_1$ with $g(w) = \inf_{v \in S_1} g(v)$. If $g(w) = \|w\| = 0$, then $w = 0$. This is a contradiction since $w \in S_1$, so it must be the case that $g(w) > 0$. Define $c' := g(w)$, then $c' \leq g(v) = \|v\|$ for every $v \in V$. Using the fact that $\frac{v}{\|v\|_1} \in S_1$ for any nonzero $v \in V$, we can see $c' \leq \left\| \frac{v}{\|v\|_1} \right\|$ implies $\|v\|_1 \leq \frac{1}{c'} \|v\|$. By transitivity, all norms on V are equivalent. $\square$

Corollary 2.7.14. *Let V be a normed space and $W \subseteq V$ a finite dimensional subspace. Then $W \subseteq V$ is closed.*

Proof. We know that all norms on V are equivalent, so there exists a uniformism and linear isomorphism $\varphi : W \rightarrow \ell_1^n$. Let $(w_n)_n$ be a sequence in W converging to some $v \in V$. Since $(w_n)_n$ is convergent, it is $\|\cdot\|$ -Cauchy. Since φ is uniformly continuous, $(\varphi(w_n))_n$ is $\|\cdot\|_{\ell_1^n}$ -Cauchy, whence it converges to some $x \in \ell_1^n$. Since φ^{-1} is uniformly continuous, we have that $(w_n)_n \rightarrow \varphi^{-1}(x) \in W$. Thus $v = \varphi^{-1}(x) \in W$; i.e., W is closed. $\square$

Lemma 2.7.15 (Riesz). *** Let V be a normed space and W a closed, proper subspace of V . For each $t \in (0, 1)$, there is a $v_t \in S_V$ with $\text{dist}_W(v_t) \geq t$.

Proof. Recall that $\text{dist}_W : V \rightarrow [0, \infty)$ is given by $v \mapsto \inf_{w \in W} \|v - w\|$. Let $v_0 \in V \setminus W$ and define $\delta := \text{dist}_W(v_0)$. Clearly $\delta > 0$ —if not the closedness of W implies $v_0 \in W$, which is a contradiction. Since $t \in (0, 1)$, we have $\delta < \frac{\delta}{t}$. Choose $w_t \in W$ with $\delta < \|v_0 - w_t\| < \frac{\delta}{t}$. Define $v_t := \frac{w_t - v_0}{\|w_t - v_0\|}$. $\square$

Theorem 2.7.16. Let V be a normed space and $B_V = \{v \in V \mid \|v\| \leq 1\}$. The following are equivalent:

- (1) B_V is compact;
- (2) $\dim(V) < \infty$.

Proof. Let B_V be compact and suppose towards contradiction $\dim(V) = \infty$. Choose $v_1 \in S_V$. Since V is infinite dimensional, $\text{span}\{v_1\} \neq V$. By Riesz' Lemma, there exists a $v_2 \in S_V$ so that $\text{dist}_{\text{span}\{v_1\}}(v_2) \geq \frac{1}{2}$. In particular, $\|v_2 - v_1\| \geq \frac{1}{2}$. Again, since V is infinite dimensional, $\text{span}\{v_1, v_2\} \neq V$. By Riesz' Lemma, there exists a $v_3 \in S_V$ so that $\text{dist}_{\text{span}\{v_1, v_2\}}(v_3) \geq \frac{1}{2}$. Then $\|v_3 - v_2\| \geq \frac{1}{2}$ and $\|v_3 - v_1\| \geq \frac{1}{2}$. Inductively, we obtain a sequence $(v_n)_n$ in S_V with $\|v_n - v_j\| \geq \frac{1}{2}$ for all $1 \leq j \leq n-1$. Since $(v_n)_n$ is not Cauchy, by Proposition 2.7.5 $(v_n)_n$ does not admit a convergent subsequence. So S_V is not compact by Theorem 2.7.9. But this contradicts Proposition 2.7.1, as we've assumed $B_V \supseteq S_V$ is compact.

Suppose V is finite dimensional. By Proposition 2.5.7 there exists a continuous linear isomorphism $f : \mathbb{C}^n \rightarrow V$. Note that $B_V = \{v \in V \mid \|v\| \leq 1\}$ is closed and bounded. Since f is continuous, by definition $f^{-1}(B_V)$ is closed. Since f^{-1} is continuous, Proposition 2.5.6 says f^{-1} is Lipschitz. So $f^{-1}(B_V)$ is bounded. By the Heine-Borel Theorem, $f^{-1}(B_V)$ is compact. Thus $f(f^{-1}(B_V)) = B_V$ is also compact by Proposition 2.7.10. $\square$

Proposition 2.7.17. Let (X, d) be compact and $f : (X, d) \rightarrow (Y, \rho)$ continuous. Then f is uniformly continuous

Proof. Let $\epsilon > 0$. Since f is continuous, for every $x \in X$ there exists a $\delta_x > 0$ such that, for any $z \in X$, we have $d(x, z) < \delta_x$ implies $\rho(f(x), f(z)) < \frac{\epsilon}{2}$. Note that $X = \bigcup_{x \in X} U(x, \frac{\delta_x}{2})$. Since X is compact, there exists $x_1, \dots, x_n \in X$ with $X = \bigcup_{i=1}^n U(x_i, \frac{\delta_{x_i}}{2})$. Let $x, x' \in X$ be arbitrary. Define $\delta = \min_{i=1}^n \delta_{x_i}$. Locate $U(x_i, \frac{\delta_{x_i}}{2}) \ni x$. If $d(x, x') < \delta$, then

$$\begin{aligned} d(x', x_i) &\leq d(x', x) + d(x, x_i) \\ &< \delta + \frac{\delta_{x_i}}{2} \\ &\leq \delta_{x_i}, \end{aligned}$$

which gives:

$$\begin{aligned} \rho(f(x), f(x')) &\leq \rho(f(x), f(x_i)) + \rho(f(x_i), f(x')) \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

$\square$

Alternative proof of Proposition 2.7.17. Suppose towards contradiction f is not uniformly continuous. Then there exists an $\epsilon_0 > 0$ and sequences $(u_n)_n, (v_n)_n$ in X such that $d(u_n, v_n) < \frac{1}{n}$ and $\rho(f(u_n), f(v_n)) \geq \epsilon_0$. Since X is compact there exists a subsequence $(u_{n_k})_k$ converging to $u \in X$. Then:

$$\begin{aligned} d(v_{n_k}, u) &\leq d(v_{n_k}, u_{n_k}) + d(u_{n_k}, u) \\ &< \frac{1}{n_k} + d(u_{n_k}, u). \end{aligned}$$

Taking the limit as $k \rightarrow \infty$ gives $(d(v_{n_k}, u))_k \rightarrow 0$; i.e., $(v_{n_k})_k$ converges to u . Since f is continuous, both $(f(u_{n_k}))_k$ and $(f(v_{n_k}))_k$ converge to $f(u)$. But this is a contradiction, as $\rho(f(u_n), f(v_n)) \rightarrow 0 \not\geq \epsilon_0$. $\square$

Lemma 2.7.18. *Let (X, d) be compact. Suppose $(f_n)_n$ is a monotonically decreasing sequence of continuous real-valued functions on X which converges pointwise to 0. Then $(f_n)_n$ converges uniformly to 0.*

Proof. Let $\epsilon > 0$. Since $(f_n)_n$ converges pointwise to 0, for each $x \in X$ there exists $N_x \in \mathbb{N}$ such that $n \geq N_x$ implies $f_n(x) < \frac{\epsilon}{2}$. Because f_{N_x} is continuous at x , there exists $\delta_x > 0$ such that, for every $z \in X$, $d(x, z) < \delta_x$ implies $|f_{N_x}(x) - f_{N_x}(z)| < \frac{\epsilon}{2}$. The collection $\{U(x, \delta_x)\}_{x \in X}$ covers X , so by compactness there is a finite set $F \subseteq X$ with $X = \bigcup_{x \in F} U(x, \delta_x)$. Set $N = \max_{x \in F} N_x$. Let $z \in X$ be arbitrary and locate $x \in F$ such that $z \in U(x, \delta_x)$. Notice that our choice of N does not depend on z . For $n \geq N$:

$$\begin{aligned} f_n(z) &\leq f_{N_x}(z) \\ &= f_{N_x}(z) - f_{N_x}(x) + f_{N_x}(x) \\ &\leq |f_{N_x}(z) - f_{N_x}(x)| + f_{N_x}(x) \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

Thus for $n \geq N$, we have $\|f_n\|_u \leq \epsilon$. $\square$

Alternative proof of Lemma 2.7.18. Let $\epsilon > 0$. For $n \geq 1$, define:

$$\begin{aligned} U_n &:= \{x \mid f_n(x) < \frac{\epsilon}{2}\} \\ &= f_n^{-1}\left((-\infty, \frac{\epsilon}{2})\right) \end{aligned}$$

Note $U_1 \subseteq U_2 \subseteq U_3 \subseteq \dots$, and in particular $\bigcup_{n \geq 1} U_n = X$. Since X is compact, $X = U_N$ for some N . That is, $f_N < \frac{\epsilon}{2}$ for all x , so $\|f_n\|_u < \epsilon$. $\square$

Theorem 2.7.19 (Dini's Theorem). *Let (X, d) be compact. Suppose $(f_n)_n$ is a monotone sequence of continuous real-valued functions on X which converges pointwise to f . Then $(f_n)_n$ converges uniformly to f .*

Proof. If $(f_n)_n$ is decreasing, apply Lemma 2.7.18 to $f_n - f$. If $(f_n)_n$ is increasing, apply Lemma 2.7.18 to $f - f_n$. $\square$

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The goal of this subsection is to develop the structure of $C(X)$. If (X, d) is compact, by the Extreme Value Theorem $C(X) = C_b(X)$; i.e., $C(X)$ is a Banach algebra. We know that compact subsets of $C(X)$ must be closed and bounded. Conversely however, we'd like to know which closed and bounded subsets of $C(X)$ are compact. This leads us to the following definition.

Definition 2.7.5. Let (X, d) be a metric space and $\mathcal{F} \subseteq C(X)$.

(1) $\mathcal{F}$ is *pointwise equicontinuous* if:

$$(\forall x_0 \in X)(\forall \epsilon > 0)(\exists \delta > 0) : (\forall x \in X)(\forall f \in \mathcal{F})(d(x, x_0) < \delta \implies |f(x) - f(x_0)| < \epsilon)$$

(2) $\mathcal{F}$ is *uniformly equicontinuous* if:

$$(\forall \epsilon > 0)(\exists \delta > 0) : (\forall x, x' \in X)(\forall f \in \mathcal{F})(d(x, x') < \delta \implies |f(x) - f(x')| < \epsilon)$$

Proposition 2.7.20. Let $\mathcal{F} \subseteq C(X)$ be finite. Then $\mathcal{F}$ is pointwise equicontinuous.

Proof. Suppose $\mathcal{F} = \{f_1, f_2, \dots, f_n\}$. Let $x_0 \in X$ and $\epsilon > 0$. For each $k \in \{1, 2, \dots, n\}$, there exists $\delta_k > 0$ such that, for any $x \in X$, $d(x, x_0) < \delta_k$ implies $|f_k(x) - f_k(x_0)| < \epsilon$. Set $\delta = \min_{k=1}^n \delta_k$. If $x \in X$ and $d(x, x_0) < \delta$, then by our choice of δ we have $|f(x) - f(x_0)| < \epsilon$ for any $f \in \mathcal{F}$. Hence $\mathcal{F}$ is pointwise equicontinuous. $\square$

Theorem 2.7.21 (Arzela-Ascoli). Let (X, d) be compact and $\mathcal{F} \subseteq C(X)$. The following are equivalent:

- (1) $\mathcal{F}$ is compact;
- (2) $\mathcal{F}$ is closed, bounded, and uniformly equicontinuous.

Proof. Suppose $\mathcal{F}$ is compact. Then $\mathcal{F}$ is closed and (totally) bounded. We only need to show that $\mathcal{F}$ is uniformly equicontinuous. Let $\epsilon > 0$. Since $\mathcal{F}$ is totally bounded, there exists $f_1, \dots, f_n \in \mathcal{F}$ with $\mathcal{F} \subseteq \bigcup_{j=1}^n \mathcal{U}(f_j, \frac{\epsilon}{3})$. Now since X is compact, each $f_1, \dots, f_n$ are uniformly continuous. So there exists $\delta_j > 0$ such that, for all $x, y \in X$, $d(x, y) < \delta_j$ implies $|f_j(x) - f_j(y)| < \frac{\epsilon}{3}$. Let $\delta = \min_{j=1}^n \delta_j$. Given $f \in \mathcal{F}$, locate f_j such that $f \in \mathcal{U}(f_j, \frac{\epsilon}{3})$. Then given any $x, y \in X$ satisfying $d(x, y) < \delta$, we have:

$$\begin{aligned} |f(x) - f(y)| &\leq |f(x) - f_j(x)| + |f_j(x) - f_j(y)| + |f_j(y) - f(y)| \\ &\leq 2\|f - f_j\|_u + \frac{\epsilon}{3} \\ &< \frac{2\epsilon}{3} + \frac{\epsilon}{3} \\ &= \epsilon. \end{aligned}$$

Thus $\mathcal{F}$ is uniformly equicontinuous.

Now suppose $\mathcal{F}$ is closed, bounded, and uniformly equicontinuous. Since X is compact, $C(X)$ is complete. Since $\mathcal{F}$ is closed, it is complete. We only need to show $\mathcal{F}$ is totally bounded. Let $\epsilon > 0$. By the uniform equicontinuity of $\mathcal{F}$, there exists $\delta > 0$ such that, for any $f \in \mathcal{F}$ and $x, y \in X$ satisfying $d(x, y) < \delta$, we have $|f(x) - f(y)| < \frac{\epsilon}{4}$. Note that since X is compact, it is totally bounded. So we can find $x_1, \dots, x_n \in X$ with $X = \bigcup_{i=1}^n \mathcal{U}(x_i, \delta)$. Note that any $f \in \mathcal{F}$ is uniformly continuous, hence bounded. So it must be the case that

$C_{\mathcal{F}} = \{(f(x_1), \dots, f(x_n)) \mid f \in \mathcal{F}\} \subseteq \mathbb{R}^n$ is bounded, hence totally bounded. As a result, there exists $f_1, \dots, f_m \in \mathcal{F}$ such that $C_{\mathcal{F}} \subseteq \bigcup_{i=1}^m U((f_i(x_1), \dots, f_i(x_n)), \frac{\epsilon}{4})$. This means, given any $f \in \mathcal{F}$, there exists $i = 1, \dots, m$ such that $(f(x_1), \dots, f(x_n)) \in \bigcup_{i=1}^m U((f_i(x_1), \dots, f_i(x_n)), \frac{\epsilon}{4})$. Claim: $\mathcal{F} \subseteq \bigcup_{i=1}^m U(f_i, \epsilon)$. Let $f \in \mathcal{F}$ and $x \in X$ be arbitrary. Locate x_j so that $x \in U(x_j, \delta)$. Locate i so that $(f(x_1), \dots, f(x_n)) \in U((f_i(x_1), \dots, f_i(x_n)), \frac{\epsilon}{4})$. Then $\sum_{j=1}^n |f(x_j) - f_i(x_j)| < \frac{\epsilon}{4}$. We finally have:

$$\begin{aligned} |f(x) - f_i(x)| &\leq |f(x) - f(x_j)| + |f(x_j) - f_i(x_j)| + |f_i(x_j) - f_i(x)| \\ &< \frac{3\epsilon}{4}. \end{aligned}$$

Thus $\|f - f_i\| < \epsilon$, so our claim holds. Since $\mathcal{F}$ is both complete and totally bounded, it is compact. $\square$

Just as we were interested in compact subsets of $C(X)$, we'd like to know which subalgebras of $C(X)$ are dense. It turns out that the crucial property that a subalgebra must satisfy is that it separates points: a set A of functions defined on X is said to *separate* points if, for every two different points x and y in X , there exists a function f in A with $f(x) \neq f(y)$. We restate this in one of the most vital results in mathematical analysis.

Theorem 2.7.22 (Stone-Weierstrass). *Let (X, d) be a compact metric space. Suppose $A \subseteq C(X)$ is a separating unital subalgebra. Then A is $\|\cdot\|_u$ -dense in $C(X)$.*

The Stone-Weierstrass Theorem for continuous complex-valued functions requires $A \subseteq C(X, \mathbb{C})$ to be separating, unital, and self-adjoint.

Example 2.7.2.

- (1) The polynomials $\{\sum_{k=0}^n a_k x^k \mid a_k \in \mathbb{F}, n \geq 0\}$ are dense in $C([a, b])$.
- (2) By Theorem 2.7.16 and Proposition 2.7.1, the set $S_{\mathbb{C}} = \{z \in \mathbb{C} \mid |z| = 1\}$ is compact. Consider the set of all trigonometric polynomials $\mathcal{T} = \{z \mapsto \sum_{k=-n}^n a_k z^k \mid a_k \in \mathbb{C}\}$. Note that $\bar{z} = z^{-1}$ for $z \in S_{\mathbb{C}}$. Then $\mathcal{T} \subseteq C(S_{\mathbb{C}})$ is dense.

§ 2.8. Connectedness

Definition 2.8.1. Let (X, d) be a metric space and $Y \subseteq X$.

- (1) A *splitting* for Y in X is a pair of open subsets $U, V \subseteq X$ with $Y \subseteq U \cup V$ and $U \cap V \cap Y = \emptyset$.
- (2) Such a splitting for Y is *trivial* if $U \cap Y = \emptyset$ or $V \cap Y = \emptyset$.
- (3) Y is *connected* in X if every splitting is trivial. Otherwise, Y is said to be *disconnected*.

Lemma 2.8.1.

Proposition 2.8.2. $[a, b] \subseteq \mathbb{R}$ is connected.

Proof. Let $[a, b] \subseteq U \cup V$ be a splitting. Suppose towards contradiction that $V \cap [a, b] \neq \emptyset$; i.e., our splitting is not trivial. Set $c = \inf(V \cap [a, b])$. Since U is open, there exists $\epsilon > 0$ such that $[a, a + \epsilon) \subseteq U$.

Claim: $V \cap [a, b] \subseteq [a + \epsilon, b]$. If not, then there exists so $x \in V \cap [a, b]$ with $x \notin [a + \epsilon, b]$. It must be the case that $x \in [a, a + \epsilon) \subseteq U$. But this gives $x \in U \cap V \cap [a, b]$, which contradicts U, V being a splitting. From this, it follows that $c \geq a + \epsilon > a$.

Claim: $[a, c) \subseteq U$. If not, there must be some $x \in [a, c)$ with $x \notin U$. Since U, V splits $[a, b]$, it must be that $x \in V$. But $x < c$, and $x \in V \cap [a, b]$, contradicting our definition of c .

Claim: $c \in V$. Suppose towards contradiction $c \in U$. We proceed by cases. Case 1: $c < b$. Since U is open, there exists some $\delta > 0$ with $(c - \delta, c + \delta) \subseteq U$. Then $[a, c + \delta) \subseteq U \cap [a, b]$. Consider $x \in (c, c + \delta)$. If $x \in V$, then $x \in U \cap V \cap [a, b]$, contradicting the fact that U, V is a splitting. If $x \in U$, then this contradicts c being the least upper bound of $V \cap [a, b]$. Case 2: $c = b$. Then $[a, b) \subseteq U$ and $\{b\} \subseteq U$. But this means $[a, b] \subseteq U$. In particular, $U \cap [a, b] = [a, b]$. Thus $U \cap V \cap [a, b] \neq \emptyset$, contradicting again that U, V is a splitting.

Since V is open, there exists $\gamma > 0$ such that $(c - \gamma, c + \gamma) \subseteq V$. This clearly contradicts c being the least upper bound, as we can find an element less than c still contained in $V \cap [a, b]$. $\square$